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Nuclear Decay Data for Dosimetric
Calculations



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Nuclear Decay Data for Dosimetric Calculations

ICRP Publication 107

Approved by ICRP Committee 2 in October 2007

Abstract—In this report, the Commission provides an electronic database of the physical data needed in calculations of radionuclide-specific protection and operational quantities. This database supersedes the data of *Publication 38* (ICRP, 1983), and will be used in future ICRP publications of dose coefficients for the intake of or exposure to radionuclides in the workplace and the environment.

The database contains information on the half-lives, decay chains, and yields and energies of radiations emitted in nuclear transformations of 1252 radionuclides of 97 elements. The CD accompanying the publication provides electronic access to complete tables of the emitted radiations, as well as the beta and neutron spectra. The database has been constructed such that user-developed software can extract the data needed for further calculations of a radionuclide of interest. A Windows-based application is provided to display summary information on a user-specified radionuclide, as well as the general characterisation of the nuclides contained in the database. In addition, the application provides a means by which the user can export the emissions of a specified radionuclide for use in subsequent calculations.

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Keywords: Radionuclide transformation; Half-life; Beta spectra; Fission neutron spectra; Auger-electron spectra

Guest Editorial

IN THE BOOK OF LIFE, THE ANSWERS AREN'T IN THE BACK

– Charlie Brown, fictional character of
the *Peanuts* comic strip created by Charles Schulz

This publication tabulates the energies and intensities of radiations emitted in spontaneous nuclear transformation (decay) of 1252 radionuclides, and thus supersedes *Publication 38* (ICRP, 1983). The tabulations, collectively referred to as ‘nuclear decay data’, are provided electronically on a compact disk (CD) accompanying the publication rather than in printed tables as was the case for *Publication 38*. Unlike the oversize format of *Publication 38*, this issue of the *Annals of the ICRP* can reside on your bookshelf with the other issues of the journal.

For the past 30 years, all nuclide-specific dose quantities issued by ICRP have been based on the nuclear decay data of *Publication 38*. *Publication 38* was prepared by the Task Group on Dose Calculations (DOCAL), formed in June 1974 to assist Committee 2 with the preparation of *Publication 30* (ICRP, 1979, 1980, 1981). For that effort, DOCAL reviewed metabolic data, evaluated the radiations emitted in nuclear transformations of radionuclides, and implemented computational methods to derive the annual limit on intake (ALI) that appeared in *Publication 30*. *Publication 38* presented the nuclear decay data that DOCAL used in its computation of the ALIs.

Reliable information on physical characteristics of a radionuclide (half-life, modes of decay, energies and intensities of the emitted radiations, etc.) is the starting point in assessing the radiological significance of a radionuclide’s presence in the workplace or in the environment. As the various radiations differ in their range in tissues, it is particularly important to account for the fraction of the available decay energy given to radiations of discrete energy (alpha particles, gamma rays, conversion electrons, Auger electrons, and characteristic x rays) as well as the continuous energy spectra of beta particles. Accounting for such details requires very specific expertise and is a laborious task. Thus, as part of a schema for absorbed dose calculations of biologically distributed radionuclides, the Medical Internal Radiation Dose (MIRD) Committee of the Society of Nuclear Medicine began an effort to compile the physical data needed by dosimetrists. In the early 1970s, Dillman prepared a series of MIRD pamphlets (Dillman, 1969, 1970; Dillman and von der Lage, 1975) on the decay schemes and nuclear parameters of radionuclides of interest in nuclear medicine. As a member of DOCAL, Dillman developed the EDISTR software (Dillman, 1980) to translate the information contained in the evaluated nuclear structure data files

(ENSDF) into the required data. This computer-based file is maintained by the National Nuclear Data Center at Brookhaven National Laboratory (Tuli, 2001). An updated version of the EDISTR software was used to derive the data contained on the CD accompanying this publication. The nature of the updates is briefly discussed in this publication and is detailed in technical reports included on the CD.

The DOCAL Task Group, as formed in 1974, was centred at Oak Ridge National Laboratory (ORNL), and with the completion of *Publication 30* and issuance of *Publication 38*, the Task Group expected to be disbanded. However, Committee 2 was concerned about its reliance on a single organisation for computational support, ORNL had previously played a major role in the preparation of *Publication 2* (ICRP, 1959) and other publications, and thus requested that DOCAL become international in its membership, and established within several organisations the capability to serve the needs of Committee 2 in computational dosimetry of incorporated radionuclides. This capability was achieved by inclusion of members from the United Kingdom Health Protection Agency (formerly the National Radiation Protection Board), the German Federal Office for Radiation Protection (BfS), and the Ukraine Radiation Protection Institute. With this structure, DOCAL served the Committee's needs in development of age-specific dose coefficients for members of the public (ICRP, 1989, 1993, 1995a, 1995b, 1996, 2004) and dose coefficients for the workers (ICRP, 1994) following the issuance of the Commission's 1990 recommendations (ICRP, 1991). In the course of these efforts, DOCAL worked closely with the Task Group on Internal Dosimetry (INDOS), chaired by J. Stather (1994–2007) and J. Harrison (from 2007). In 2000, DOCAL was further restructured to be responsible for the computational dosimetry associated with external radiation fields. During its existence, DOCAL has been chaired by W.S. Snyder (1974–1977), M.R. Ford (1977–1984), K.F. Eckerman (1984–2004), and W.E. Bolch (from 2004).

With the issuance of the Commission's recommendations in *Publication 103* (ICRP, 2007), DOCAL now begins its efforts to derive dose coefficients for exposure to radionuclides in the workplace following the definition of the effective dose in the new recommendations. This publication sets forth the nuclear decay data for that effort by replacing DOCAL's first publication, *Publication 38*, issued in 1983. The publication provides comprehensive nuclear decay data for dose calculations in occupational, environmental, and medical exposures. The data are also useful for understanding the properties of radionuclides that are used in the calibration and testing of radiation protection instrumentation and spectrometers. While some readers may miss the graphic decay schemes and printed tables of emitted radiations given for 820 radionuclides in *Publication 38*, it must be realised that format is prohibitive when addressing the 1252 radionuclides of this publication. For these readers, it is noted that data for 333 radionuclides of interest in nuclear medicine have been tabulated with accompanying decay scheme graphics in a monograph entitled 'MIRD: Radionuclide Data and Decay Schemes' published by the Society of Nuclear Medicine (Eckerman and Endo, 2008). Furthermore, software distributed with the enclosed CD was developed to provide the user with insight into the decay characteristics of the radionuclides, and to tabulate summary and detailed information for a user-specified radionuclide.

This publication is an essential source of information for those working or studying in the field of nuclear medicine, radiation protection, medical physics, and health physics. Contrary to Charlie Brown's book of life, the answers are included on the enclosed CD.

KEITH F. ECKERMAN
HANS-G. MENZEL

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Ⓢ This article does not appear in the printed product, it is available in the online version of this issue.

PREFACE

Committee 2 of the International Commission on Radiological Protection (ICRP) has the responsibility for providing dose coefficients for intakes of and exposure to radionuclides by workers who are occupationally exposed and by members of the public exposed to radionuclides in the environment. Over the past 30 years, the dosimetric coefficients calculated by the Task Group on Dose Calculations (formed in June 1974) were based on the yields and energies of radiations emitted in nuclear transformations as tabulated in *Publication 38* (ICRP, 1983). *Publication 38* provided data for the radionuclides addressed in *Publication 30* (ICRP, 1979) and for a few additional radionuclides of interest in nuclear medicine. Although *Publication 30* and subsequent publications of the Commission only considered the intake of radionuclides with half-lives of more than 10 min, the emissions of short-lived members of their decay chain had to be considered when deriving the parent's dose coefficient. Thus, *Publication 38* included data for 820 radionuclides, 764 with half-lives of more than 10 min and 56 with half-lives of less than 10 min, as either a member of the decay chain of a radionuclide with a half-life of more than 10 min or of potential interest in nuclear medicine.

This publication provides a comprehensive database of nuclear decay data for 1252 radionuclides. The database provides the physical data needed for calculation of absorbed doses in organs and tissues of the body, and the evaluation of localised depth dose distributions. The database includes all nuclides with a half-life of more than 1 min for which the nuclear structure information was judged sufficient for a meaningful assessment of the emissions. Data are provided for 922 radionuclides with a half-life of more than 10 min and 330 radionuclides with a half-life of less than 10 min. This publication departs from the format of *Publication 38*, and provides the data in electronic form on a CD in a pocket on the inside of the back cover. The electronic format enables inclusion of the continuous energy distribution of beta emissions that are needed in calculations of the depth dose distributions for walled organs and the skin. Similarly, the spectra of neutrons accompanying spontaneous fission are also included on the CD. The electronic format enables presentation of the emissions without invoking a limitation on the number of radiations considered, which would be necessary if the data were provided in printed tables.

The membership of the Working Party on Nuclear Decay Data that contributed to the development of the report was as follows:

K. Eckerman

A. Endo

The work of the Working Party was greatly aided by significant technical contributions from T. Tamura and K. Umeda.

The membership of the Task Group on Dose Calculations during the period of preparation of the report was:

W. Bolch (Chair)	
D. Noßke (Vice-Chair Internal Dose)	A. Endo
N. Petoussi-Henß (Vice-Chair External Dose)	N. Hertel
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L. Bertelli	R.W. Leggett	
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J.G. Hunt	G. Miller	G. Xu
L. Johansson	M. Stabin	R. Richardson

The membership of Committee 2 during the period of preparation of this report (2005–2009) was:

C. Streffer (Chair, –2007)	G. Dietze	H-G. Menzel (Chair, 2007–)	Y. Zhou
M. Balonov	K.F. Eckerman	F. Paquet	V. Berkovski
J.D. Harrison	H. Paretzke	W.E. Bolch	N. Ishigure
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MAIN POINTS

- The yields and energies of radiations emitted in nuclear transformations of 1252 radionuclides have been assembled for use in calculating nuclide-specific protection and operational quantities.
- This publication supersedes *Publication 38* which was issued in 1983, and will be used in future calculations by the Task Group on Dose Calculations of Committee 2.
- Nuclides were selected for consideration if their half-life exceeded 1 min or if formed in nuclear transformation of a selected radionuclide. N-16 (half-life 7.13 s) was included largely for historical purposes. It is, of course, necessary that the basic nuclear physics information should be suitable for meaningful calculations of yields and energies of the emitted radiations.
- Data were compiled for 922 radionuclides with a half-life of more than 10 min, 330 radionuclides with a half-life of less than 10 min, and assembled in five data files.
- The data files contained on the CD accompanying this publication embody no restriction on the number of radiations, as was necessary in the tables of *Publication 38*. A Windows-based application is provided to enable tabular and graphical viewing of the emitted radiations of the radionuclides. Electronic access to the data is provided to assist those wanting to use the data in calculations.

GLOSSARY

Alpha decay

In alpha decay, the nucleus of an atom of atomic number Z and mass number A emits an alpha particle (${}^4\text{He}$ nucleus) to form an atom of atomic number $Z - 2$ and mass number $A - 4$.

Air-kerma rate constant (Γ_δ)

This constant, a characteristic of a radionuclide, is defined in terms of an ideal point source as $\Gamma_\delta = l^2 \dot{K}_\delta / A$ where $\dot{K}_\delta$ is the air-kerma rate due to photons (gamma and x rays) of energy greater than δ at a distance l in a vacuum from a point source of the radionuclide of activity A .

Annihilation photons

The electromagnetic radiation (photon) emitted as a positron–electron pair undergoes annihilation. Annihilation generally occurs after the positron has lost its initial kinetic energy to the absorbing medium, and two photons, each equivalent in energy to the rest mass of an electron ($m_0 c^2 = 0.511 \text{ MeV}$), are created in opposite directions. The number of annihilation photons is taken to be twice the number of positrons emitted in beta-plus decay.

Auger transition

The filling of a vacancy in an inner electron shell of an atom, e.g. as created in electron capture, is accompanied by the emission of an electron from the outer shell, an Auger electron.

Beta-minus decay

In beta-minus decay, the nucleus of an atom of atomic number Z and mass number A emits a negative electron (β^-) and an antineutrino ($\bar{\nu}$) to form an atom of atomic number $Z + 1$ and mass number A .

Beta-plus decay

In beta-plus decay, the nucleus of an atom of atomic number Z and mass number A emits a positive electron (β^+ , positron) and a neutrino (ν) to form an atom of atomic number $Z - 1$ and mass number A .

Branching fraction

Branching fraction is the fraction of the nuclear transformations of the radionuclide that forms a particular daughter nucleus.

Coster-Kronig (CK) transition

CK transition is an Auger transition in which the atomic shell vacancy is filled by an electron from a higher subshell of the same shell.

Inner bremsstrahlung

Inner bremsstrahlung is a continuous electromagnetic radiation produced by the sudden change of nuclear charge which occurs in beta-plus, beta-minus, or electron-capture decay. This radiation is not included here because of its low energy and low yield.

Electron-capture decay

In electron-capture decay, the nucleus of an atom of atomic number Z and mass number A captures an atomic electron and emits a neutrino (ν) to form an atom of atomic number $Z - 1$ and mass number A .

Forbiddenness

Beta transitions are classified as allowed transition or forbidden transition based on the spin and parity of the nuclear states involved in the transition. Forbiddenness indicates how 'fast' the transition occurs.

Half-life ($T_{1/2}$)

The half-life of a radionuclide is the interval required for its activity to decay to half of its initial value. The half-life is related to the decay constant λ as $T_{1/2} = \ln(2)/\lambda$ and to the mean life τ as $\tau = T_{1/2}/\ln(2)$.

Gamma ray

Gamma-rays are electromagnetic radiations (photons) emitted by the nucleus in transition from a high energy state to a lower energy state.

Isomeric transition decay

A nucleus in an excited energy state of an atom of atomic number Z and mass number A can transition to a lower energy state by emission of a gamma ray, or eject an orbital electron with the excess energy. No change in atomic and mass numbers occurs; the parent and daughter nuclei are isomers. Isomers of energy above the ground state are identified by appending 'm', 'n', 'p', or 'q' to the mass number.

Isomers

Nuclei of the same atomic number Z and mass number A but with a different energy state and different physical half-life. Isomers of energy above the ground state are identified by appending 'm', 'n', 'p', or 'q' to the mass number.

Internal conversion electron

An internal conversion electron is an orbital electron ejected from the atom as a result of the energy difference between states of the nucleus being transferred directly to a bound atomic electron. This process is a form of isomeric decay.

Nuclear transformation

Nuclear transformation is the occurrence of the decay process(es) that spontaneously transform(s) an atom from one state to another. It is often referred to as 'decay' and its SI unit is the becquerel.

Spontaneous fission

Spontaneous fission is a decay mechanism resulting in splitting of the nucleus into lighter nuclei, referred to as 'fission fragments', with the emission of neutrons.

Transition

The individual process by which a nucleus is spontaneously transformed to another energy state.

X ray

The filling of a vacancy in an inner electron shell of an atom, e.g. created in electron capture, may be accompanied by the emission of an x-ray photon.

1. INTRODUCTION

(1) This document provides information on the half-lives, decay chains, and energies and yields of radiations emitted by nuclear transformations and subsequent atomic processes that are needed in the computation of the radiation dose from internal or external exposure to radionuclides. These data were compiled as a replacement for the data of *Publication 38* (ICRP, 1983). Data are provided for 1252 radionuclides of 97 elements with an atomic number of less than 101 (see Fig. 1 and Table A.1). The data are presented here as an electronic database rather than the printed tables of *Publication 38*. Eckerman and Endo (2008) have published a monograph with printed tables and decay scheme drawings for 333 radionuclides of interest in nuclear medicine. The electronic database provided here is similar to the NUCDECAY database (Eckerman et al., 1994), and enables complete listings of the emissions including beta and neutron spectra. These data were assembled from the output of the EDISTR04 code (Endo et al., 2005); an updated version of EDISTR (Dillman, 1980) used in the production of *Publication 38*. EDISTR04 uses the basic nuclear structure data files [evaluated nuclear structure data files (ENSDF)] (Tuli, 2001) as input. The updating of EDISTR and the quality assurance efforts are discussed in detail by Endo et al. (2005). Below is a brief discussion of the methodology implemented in EDISTR04 and a detailed description of the electronic database. The electronic database, the software package DECDATA, and additional documentation, including the reports of Dillman (1980), Endo et al. (2005), and Endo and Eckerman (2007), are contained on the CD that accompanies this publication and as supplementary data to the online report at <http://www.icrp.info>.

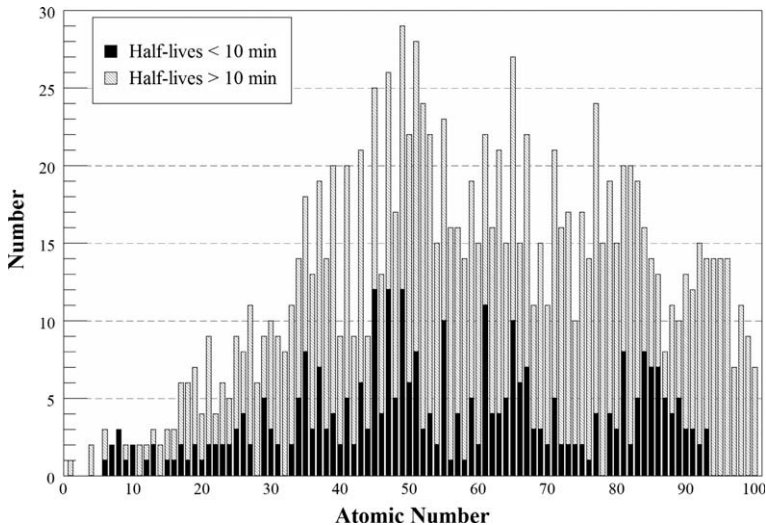


Fig. 1. Number of radioisotopes of the elements in this compilation.

2. METHODS EMPLOYED IN COMPILING THE NUCLEAR DECAY DATA

2.1. General

(2) The computer code EDISTR04 (Endo et al., 2005) was used to calculate the energies and yields of the radiations emitted by nuclear transformations and subsequent atomic processes. The code calculates the energy spectra of beta-minus and beta-plus emissions and, in the case of alpha transitions, the unique energies and yields of alpha particles and recoil nuclei. The energies and yields of various x ray and Auger-electron groups arising as a result of electron capture or internal conversion of gamma rays are also calculated. In the few cases in which spontaneous fission (SF) occurs (28 radionuclides in the collection), the yields of radiations concomitant with SF (fission fragments, neutron, prompt and delayed gamma, and delayed beta emissions) are computed.

(3) Information contained in the ENSDF is used as input data to EDISTR04. The ENSDF data are part of the computer database of the National Nuclear Data Center at Brookhaven National Laboratory, from which the radioactive decay data are published in the periodical ‘Nuclear Data Sheets’ (Academic Press, New York). The ENSDF were subject to review in terms of consistency with the latest nuclear parameters, by comparing the computed energies of emitted radiations with total decay energies. The fundamental decay properties in the ENSDF, such as total decay energy (Q values), branching fractions, excitation energies of isomers, physical half-life, and spin and parity of the initial and final states, were updated as necessary to those of NUBAS2003 (Audi et al., 2003) and AME2003 (Wapstra et al., 2003). The revised ENSDF were then processed by EDISTR04 in order to calculate the energies E (MeV) and yields Y (/Bq s) of alpha particles, beta particles, gamma rays including annihilation photons, internal conversion electrons, x rays, and Auger and Coster-Kronig (CK) electrons. For nuclides undergoing SF, the average energy and yields of fission fragments, prompt neutrons, prompt gamma rays, delayed gamma rays, and delayed beta particles were also computed.

(4) A comprehensive description of the theoretical and numerical techniques employed in the EDISTR and EDISTR04 codes has been published elsewhere (Dillman, 1980; Endo et al., 2005). These reports are included on the accompanying CD. A brief overview of these methods follows.

2.2. Alpha decay

(5) The emission of an alpha particle transforms the parent nucleus with an atomic number Z and mass number A to a daughter nucleus with atomic number $Z - 2$ and mass number $A - 4$. This is expressed symbolically as:



where X and Y represent the parent and daughter elements, respectively, and γ represents the gamma ray accompanying a transformation ending in an excited state of the daughter nucleus. The alpha particle and the recoiling nucleus share the available

kinetic energy. The input data consist of the ground-state Q value (reaction energy for a ground-state-to-ground-state transition); the excitation energy of levels in the daughter nucleus at which the alpha transitions end, together with the corresponding yields; and the excitation energy of the parent (zero except for decay from an isomeric-level parent). From these data, the kinetic energies of the alpha particles and associated recoil nuclei are computed using conservation of energy and momentum. The corresponding absolute yields are computed from relative yields using the normalisation data directly available in the ENSDF.

2.3. Beta decay

(6) Beta decay is a three-body process that gives rise to a continuous energy spectrum of beta particles. For dosimetric purposes, the average energy of the spectrum is often sufficient; however, for calculation of the dose at a depth into tissue, e.g. the walls of the airways and the alimentary tract, the energy spectrum must be considered. The neutrino emitted in beta decay carries off a portion of the available energy; however, this energy does not contribute to dose because of the negligible interaction of neutrinos with matter. The average kinetic energy, $\bar{E}$, of the beta particles is determined from the spectrum as:

$$\bar{E} = \int_0^{E_0} P(E)E \, (dE / \int_0^{E_0} P(E) \, dE) \quad (2)$$

where $P(E)$ is proportional to the probability that the beta particle will be emitted with kinetic energy between E and $E + dE$, and E_0 is the maximum possible kinetic energy of the beta transition; the so-called end-point energy. The end-point kinetic energy, E_0 , is determined from conservation of energy according to the equations:

$$\begin{aligned} E_0(\beta^-) &= Q + E_p - E_L \\ \text{and} \\ E_0(\beta^+) &= Q + E_p - E_L - 2m_0c^2. \end{aligned} \quad (3)$$

(7) In these equations, $E_0(\beta^-)$ and $E_0(\beta^+)$ are the end-point kinetic energies for beta-minus and beta-plus particles, respectively, Q is the energy difference between the ground states of parent and daughter nuclei, E_p is the excitation energy of the parent ($E_p = 0.0$ except for an isomeric-level parent nuclide), E_L is the excitation energy of the level in the daughter nucleus at which the transition ends, and m_0c^2 is the electron rest-mass energy (0.511 MeV). The values of Q , E_p , and E_L are all included in the ENSDF.

(8) The emission of a negatively charged beta particle (β^- or negatron) from the nucleus transforms the parent nucleus X with atomic number Z and mass number A to a daughter nucleus Y with atomic number $Z + 1$ and mass number A . The daughter nucleus has one more proton and one less neutron than the parent radio-nuclide. Symbolically, beta-minus decay is expressed as:



Table 2.1. Classification of beta transitions.

Type of transition	Forbiddenness	Change in spin	Change in parity
Super-allowed		0	No
Allowed		0, ± 1	No
Forbidden unique	First	± 2	Yes
	Second	± 3	No
	Third	± 4	Yes
	Fourth	± 5	No
Forbidden non-unique	First	0, ± 1	Yes
	Second	± 2	No
	Third	± 3	Yes
	Fourth	± 4	No

where $\bar{\nu}$ is an antineutrino and γ represents the gamma-ray emissions accompanying a transition ending in an excited state of the daughter nucleus. The emission of a positively charged beta particle (β^+ or positron) from the nucleus transforms the parent nucleus X with atomic number Z and mass number A to a daughter nucleus Y with atomic number Z – 1 and mass number A. The daughter nucleus has one less proton and one more neutron than the parent nucleus. The beta-plus transition is expressed as:



where ν is the neutrino.

(9) Beta spectra are classified into various degrees of forbiddenness depending on the spin and parity changes that occur between the parent and daughter nuclear levels. The forbiddenness classifications are illustrated in Table 2.1.

(10) The mathematical form of $P(E)$ becomes lengthy for the case of forbidden unique spectra and, in the interest of brevity, is not presented here. Even for the case of allowed transitions, Coulomb screening and finite nuclear size corrections to the theory complicate otherwise simple expressions for $P(E)$. The interested reader is referred to Dillman (1980) for further details, and Endo et al. (2005) for a comparison of beta spectra calculated by EDISTR04 with other published works.

(11) Positrons are unstable in matter and will annihilate with an electron of the absorbing media to create two annihilation photons. If the annihilation occurs after the beta-plus particle has come to rest, which is the usual circumstance, the energy of each annihilation photon corresponds with the rest-mass energy of an electron or 0.511 MeV. In the database, the yield of annihilation photons is taken to be twice the beta-plus yield.

2.4. Electron-capture decay

(12) Electron capture is a form of beta decay that results in vacancies in the atomic subshells of the daughter atom. Vacancies are most likely to be created in the K shell if energetically possible. The distribution of primary vacancies among the atomic

shells and subshells must be calculated because this distribution affects the relative yields of the x rays and Auger electrons that result from the initial vacancies. Given a specific electron-capture transition, the basic computations required are the K/L/M... capture ratios. Electron capture is expressed as:



where all variables are as defined previously.

(13) EDISTR04 calculates the capture ratios from 24 subshells from K up to O₈ using the comprehensive tables of Firestone et al. (1996) and Bambynek et al. (1977). The electron-binding energies are taken from the evaluated atomic data library (EADL) of Perkins et al. (1991). EADL provides the atomic data for $Z = 1 - 100^*$ and includes theoretical binding energies of 29 subshells from K up to Q₁ subshells. When the transition energy between the parent and daughter nuclei is greater than twice the rest mass energy of an electron (1.022 MeV), the emission of a positron can compete with electron capture. This competition has dosimetric implications as, if positron emission is possible, the energy of the transition is shared between the positron and the neutrino. The neutrino leaves the nucleus with approximately two-thirds of the transitional energy. The probability of electron capture compared with positron emission generally increases with decreasing transition energy and increasing atomic number. Only electron capture can occur when the transition energy is less than the 1.022-MeV threshold required for positron emission.

2.5. Isomeric transition and conversion of gamma rays

(14) The de-excitation of a nuclear state of more than transitory existence (greater than a nanosecond) by the release of gamma rays or conversion electrons is referred to as 'isomeric transition'. Whereas the previous discussion of other decay modes showed gamma-ray emission from nuclei formed in an excited state by alpha- or beta-particle decay, the term 'isomeric transition' describes the decay of longer-lived metastable states where the loss of energy is the only change in the nucleus. Radionuclides that decay by isomeric transition are identified with an 'm' immediately after the mass number[†], such as in ^{99m}Tc. Symbolically, this decay mode is expressed as:



(15) An alternative means of de-excitation of the nucleus is internal conversion. This is a process by which energy, equal to the energy of the gamma ray, is transferred to an orbital electron and the orbital electron is ejected from the atom. The energy of the internal conversion electron emitted from one of the atomic shells

* EDISTR04 is not applicable to mendelevium (Md) and heavier elements; two isotopes of Md were addressed in *Publication 38*.

[†] Isomers are identified in order of increasing excitation energy by appending 'm', 'n', 'p', ... to mass number A.

(K, L,...) is equal to the energy of the corresponding gamma ray minus the binding energy of the ejected orbital electron. As with electron capture, vacancies can be created in various electron shells, and the distribution of these vacancies must be calculated in order to determine the yields of the resulting x rays and Auger electrons. The ratio of the number of conversion electrons to the number of gamma radiations emitted per transition is called the ‘internal conversion coefficient’. The coefficient for each transition depends on transition energy, atomic number of the nucleus, and the multipolarity of the transition. The multipolarity depends on the total angular momentum and parity changes between the nuclear levels involved in the transition. Usually, the internal conversion coefficient for an atomic shell or subshell increases as the transition energy decreases, as the atomic number increases, and as the multipole order increases.

2.6. X-ray and Auger-electron yields and energies

(16) An atom excited by an atomic vacancy in one of its inner shells de-excites by x ray or Auger-electron emission or by CK transitions. The resulting cascade of atomic events leads to a myriad of x rays and Auger electrons being emitted, even in very simple nuclear decay schemes. In the x ray emission process, an electron in an outer shell, Y, makes a transition to the vacancy in an inner shell, X, and a photon is emitted. The energy of the emitted x ray is equal to E_x , the binding energy of an electron in shell X, minus E_y , the binding energy of an electron in shell Y. The original vacancy in shell X is transferred to shell Y, which will be subsequently filled.

(17) In Auger-electron emission, an electron in an outer shell, Y, makes a transition to the vacancy in an inner shell, X, and an electron in outer shell Y' is ejected from the atom. Shell Y' may or may not be the same as shell Y. The energy of the emitted electron (Auger electron) is approximately equal to $E_x - E_y - E_{y'}$. This process results in vacancies in both the Y and Y' shells from a single vacancy in the X shell. If X is the K shell, Y the L₁ subshell, and Y' the L₂ subshell, the electron is called a ‘KL₁L₂ Auger electron’. A KLL Auger electron is one in which the Y and Y' shells could be any of three L subshells.

(18) A CK transition is a non-radiative process involving an electron transition between two subshells of the same major shell. For example, if a vacancy is transferred from the L₁ subshell to the L₂ subshell, and the small energy difference between the subshells is transferred to one of the outermost electrons, ejecting it from the atom, the transition is said to be a ‘CK transition’. The essential difference between CK and Auger transitions is that in the case of CK transitions, one of the final vacancies is in the same major shell as the initial vacancy. In Auger transitions, both of the final vacancies are in a different major shell from that of the initial vacancy.

(19) A method, based on the RELAX computer code of Cullen (1992), for calculating detailed atomic radiations has been introduced into EDISTR04 in order to treat transitions from outer shells. The energies and yields of x rays, Auger electrons,

and CK electrons are calculated for each transition; the Auger and CK emissions can exceed 3000 in number. These emissions are collected into 15 composite groups to

Table 2.2. Assignment for Auger and Coster-Kronig (CK) electrons in the tables.

Notation	Definition
Auger KLL	K-shell Auger transitions where the two new vacancies are in the L shell
Auger KLX	K-shell Auger transitions where one of the two new vacancies is in the L shell
Auger KXY	K-shell Auger transitions where neither of the two new vacancies is in the L shell
CK LLX	All L-shell CK transitions
Auger LMM	L-shell Auger transitions where the two new vacancies are in the M shell
Auger LMX	L-shell Auger transitions where one of the two new vacancies is in the M shell
Auger LXY	L-shell Auger transitions where neither of the two new vacancies is in the M shell
CK MMX	All M-shell CK transitions
Auger MNN	M-shell Auger transitions where the two new vacancies are in the N shell
Auger MNX	M-shell Auger transitions where one of the two new vacancies is in the N shell
Auger MXY	M-shell Auger transitions where neither of the two new vacancies is in the N shell
CK NNX	All N-shell CK transitions
Auger NXY	All N-shell Auger transitions
CK OOX	All O-shell CK transitions
Auger OXY	All O-shell Auger transitions

Table 2.3. International Union of Pure and Applied Chemistry (IUPAC) and Siegbahn notations for x rays (Jenkins et al., 1991).

K x-ray series			
Siegbahn	IUPAC	Siegbahn	IUPAC
K_{x_2}	K – L ₂	$K_{\beta_3}^I$	K – M ₅
K_{x_1}	K – L ₃	$K_{\beta_2}^{II}$	K – N ₂
K_{β_3}	K – M ₂	$K_{\beta_2}^I$	K – N ₃
K_{β_1}	K – M ₃	$K_{\beta_4}^{II} K_{\beta_4x}$	K – N ₄
$K_{\beta_5}^{II}$	K – M ₄	$K_{\beta_3}^I$	K – N ₅
L x-ray series			
Siegbahn	IUPAC	Siegbahn	IUPAC
L_{β_4}	L ₁ – M ₂	L_{γ_6}	L ₂ – O ₄
L_{β_3}	L ₁ – M ₃	L_{ℓ}	L ₃ – M ₁
$L_{\beta_{10}}$	L ₁ – M ₄	L_t	L ₃ – M ₂
L_{β_9}	L ₁ – M ₅	L_{δ}	L ₃ – M ₃
L_{γ_2}	L ₁ – N ₂	L_{x_2}	L ₃ – M ₄
L_{γ_3}	L ₁ – N ₃	L_{x_1}	L ₃ – M ₅
L_{γ_4}	L ₁ – O ₂	L_{β_6}	L ₃ – N ₁
L_{γ_4}	L ₁ – O ₃	$L_{\beta_{15}}$	L ₃ – N ₄
L_{η}	L ₂ – M ₁	L_{β_2}	L ₃ – N ₅
$L_{\beta_{17}}$	L ₂ – M ₃	$L_{\beta_7'} L_u$	L ₃ – N ₆
L_{β_1}	L ₂ – M ₄	$L_{\beta_7'} L_u$	L ₃ – N ₇
L_{γ_5}	L ₂ – N ₁	L_{β_7}	L ₃ – O ₁
L_{γ_1}	L ₂ – N ₄	L_{β_5}	L ₃ – O ₄
L_{γ_8}, L_v	L ₂ – N ₆	L_{β_5}	L ₃ – O ₅
L_{γ_8}	L ₂ – O ₁		

keep the tabulated data within an appropriate size. The grouping of the Auger and CK electrons is noted in Table 2.2. Ungrouped Auger and CK electron data (e.g. KL_1L_2 , $M_1M_2M_5$) are contained on the accompanying CD for selected radionuclides. The x-ray notation of the International Union of Pure and Applied Chemistry (IUPAC) has been adopted for the data tables (Jenkins et al., 1991). The relationship of this notation with the conventional Siegbahn notation for the K and L series is given in Table 2.3. For example, the Siegbahn notation for the K_{α_2} x ray is denoted as 'K-L₂' under the IUPAC notation.

2.7. Spontaneous fission

(20) SF is nuclear fission that occurs without the addition of particles or energy to the nucleus. From a dosimetric viewpoint, SF is very important because substantial energy is associated with fission. In addition, fission gives rise to a variety of radiations including fission fragments, neutrons, beta particles, prompt gamma rays, and delayed gamma rays. Dillman and Jones (1975) developed a set of empirical formulae to compute the energies and yields of these radiations, which were used in EDISTR. These formulae have been updated as discussed by Endo et al. (2005) in EDISTR04. The empirical formulations express the yields and energies of the radiations in terms of the fission decay fraction BF_{SF} , the mass number A and the atomic number Z of the parent nuclide, and the average number of neutrons emitted per fission, $\bar{\nu}$. The pertinent formulae are summarised in Table 2.4.

(21) The neutron spectral shape is well expressed by the Watt spectrum and the approximation has been incorporated into EDISTR04. The spectrum is expressed as:

$$N(E) = BF_{SF} \bar{\nu} e^{-\frac{E}{a}} \sinh(\sqrt{bE}) \quad (8)$$

where the SF branching fraction and the parameters a and b are given in Table 2.5.

Table 2.4. Energies and yields of radiations accompanying spontaneous fission.

Radiation type	Average energy (MeV)	Yield (/Bq s)
Neutrons	$0.25a^2b + 1.5a^*$	$BF_{SF}\bar{\nu}$
Fission fragments	$0.05945 \frac{Z^2}{A^{1/3}} + 3.65$	$2 BF_{SF}$
Prompt gamma rays	$-1.33 + 119.6 \frac{Z^{1/3}}{A}$	$\frac{(2.51 - 1.13 \times 10^{-5} Z^2 A^{1/2}) \bar{\nu} + 4}{\bar{E}_{p\gamma}}$
Delayed gamma rays	0.9578	$0.2102 \left(5.98 + \frac{92}{236} A - Z \right)^2 BF_{SF}$
Delayed beta particles	$0.2058 \left(5.98 + \frac{92}{236} A - Z \right)$	$\left(5.98 + \frac{92}{236} A - Z \right) BF_{SF}$

* a and b are parameters of the Watt approximation of the neutron spectrum.

Table 2.5. Parameters of the Watt spectrum for spontaneous fission.

Nuclide	BF_{SF}	$\bar{\nu}$	Watt parameters	
			a	b
U-238	5.450E-07	2.010	0.648318	6.81057
Pu-236	1.370E-09	2.130	0.988270	3.10386
Pu-238	1.850E-09	2.220	0.847833	4.16933
Pu-240	5.750E-08	2.160	0.794930	4.68927
Pu-242	5.540E-06	2.150	0.819150	4.36668
Pu-244	1.210E-03	2.300	0.694716	6.00370
Cm-240	3.900E-08	2.390	1.071680	2.69829
Cm-242	6.370E-08	2.520	0.887353	3.89176
Cm-244	1.371E-06	2.690	0.902523	3.72033
Cm-245	6.100E-09	2.870	0.911919	3.62393
Cm-246	2.630E-04	3.180	0.878224	3.88585
Cm-246	2.630E-04	3.180	0.878224	3.88585
Cm-248	8.390E-02	3.110	0.808387	4.53623
Cm-250	7.400E-01	3.310	0.734482	5.43559
Cf-246	2.500E-06	3.100	1.02600	2.93000
Cf-248	2.900E-05	3.340	1.02772	2.93228
Cf-249	5.020E-09	3.410	1.02600	2.93000
Cf-250	7.700E-04	3.530	1.02600	2.93000
Cf-252	3.092E-02	3.765	1.02500	2.92600
Cf-254	9.969E-01	3.890	1.02600	2.93000
Es-253	8.900E-08	3.930	0.82000	4.60000
Es-254	3.000E-08	3.950	0.82000	4.60000
Es-254m	4.500E-04	3.950	0.82000	4.60000
Es-255	4.500E-05	3.970	0.82000	4.60000
Fm-252	2.300E-05	3.900	0.82000	4.60000
Fm-255	2.300E-07	3.730	0.82000	4.60000
Fm-256	9.190E-01	4.010	0.82000	4.60000
Fm-257	2.100E-03	3.850	0.82000	4.60000

3. CONTENT OF ACCOMPANYING CD AND THE DATA FILES

3.1. Content of the CD

(22) The CD in the pocket on the inside back cover of this publication provides electronic files of the unabridged nuclear decay data for 1252 radionuclides of 97 elements in the ICRP collection. Table A.1 in Annex A lists the radionuclides addressed in the CD. The CD contains the following:

- README.TXT. A file containing information post preparation of the publication.
- CONTENT.PDF. A file detailing the content of the CD.
- LICENSE.TXT. A file containing the licence agreement and a liability statement.
- USERGUIDE.PDF. The user guide to the DECDATA software.
- Nuclear decay data files:
 - ICRP-07.NDX
 - ICRP-07.RAD
 - ICRP-07.BET
 - ICRP-07.NSF
 - ICRP-07.ACK
- SETUP.EXE. Executable file for installing the DECDATA software.
- Archive folders:
 - INPUT folder. EDISTR04 input files for nuclides of the ICRP-07 collection.
 - OUTPUT folder. EDISTR04 output files for nuclides of the ICRP-07 collection.
 - SOURCE folder. The source code for DECDATA.

(23) The software package DECDATA[‡] provides access to the nuclear decay data for a user-requested radionuclide, and extracts the energies and yields of the emitted radiations into ASCII files for use in further computations. DECDATA enables the user to view the decay data in tabular and graphical form on a personal computer with Windows operating system. However, the five data files have been constructed in a manner such that the data can be accessed directly by user-developed software. Further guidance on this issue is provided in the DECDATA user guide.

3.2. Nuclear data files

(24) The nuclear decay data are embodied in five formatted (hence can be viewed with an ASCII editor) direct-access files, each with the root name ICRP-07. The file ICRP-07.RAD contains the data on the absolute yields and mean[§] or discrete

[‡] Portions © 2002 Perfect Sync, Inc.

[§] Mean values are reported for beta transitions, Auger and CK electron groups, and the radiations emitted during SF – fission fragments, neutrons, and delayed beta emission.

energies of the radiations emitted by the nuclides of the ICRP-07 collection. This file contains data for all emitted radiations, i.e. no cut-off on the number of radiations has been used in creating the data files.

(25) The file ICRP-07.BET contains the beta spectra for all beta emitters in the collection. EDISTR04 computes the spectrum for each beta transition to determine the mean beta energy of the transition, and tabulates the composite spectrum for all beta transitions of the radionuclide. The composite spectrum is included in this file.

(26) The file ICRP-07.ACK contains detailed Auger–CK electron spectra for 136 selected radionuclides. In the RAD file, these discrete emissions were collected in no more than 15 groups (see Table 2.2) for each decay mode, with the average energy and yield reported for the group.

(27) The spectra of neutrons emitted during SF are contained in the file ICRP-07.NSF. In the ICRP-07 collection, 28 radionuclides decay by SF.

(28) To facilitate access to the data in these files, an additional file (ICRP-07.NDX) was constructed. A brief description of each of these five files is presented below. For convenience, the files are referred to by their extensions, i.e. NDX, RAD, BET, ACK, and NSF.

3.2.1. Index file: ICRP-07.NDX

(29) The NDX file serves as the entrance into the radiation file (RAD) and spectral (BET, ACK, and NSF) data files. The NDX file contains one record for each nuclide of the collection. The fields of the nuclide record specify the location (a pointer) of the nuclide's record in the RAD, BET, ACK, and NSF files. In addition to these pointers, the record includes fields giving the nuclide's physical half-life, decay mode (e.g. alpha or beta), identity of progeny (daughters), fraction of the nuclear transformations forming each daughter (so-called branching fraction), the total energies emitted by alpha, electron, and photon emissions (including electrons and photons accompanying SF), and other supportive data. A description of the NDX records is given in Table 3.1.

(30) The records of the index file have been sorted by the field containing the nuclide name, making it possible for user-developed software to locate the record for a radionuclide of interest by a binary search. While the purpose of the NDX file is to provide entry into the other data files, it is of considerable utility in its own right. For example, the decay chain headed by a radionuclide from information in the NDX file includes identification of the stable nucleus terminating the chain. Tantalum-180m, formed in the beta decay of Hf-180m, is assigned here as stable due to its very long half-life as in NUBASE2003 (Audi et al., 2003).

(31) The nuclide record in the NDX file has a field containing the air-kerma rate constant. This constant, a characteristic of a radionuclide, is defined in terms of an ideal point source. The International Commission on Radiation Units and Measurements (ICRU) defined the constant as $\Gamma_{\delta} = l^2 \dot{K}_{\delta} / A$ where $\dot{K}_{\delta}$ is the air-kerma rate due to photons of energy greater than δ at a distance l in a vacuum from a point source of the radionuclide of activity A (ICRU, 1998). The photons referred to

Table 3.1 Structure of records in the ICRP-07.NDX file.

Field	Format*	Description
Record 1		
First	I4	Record number of first data record
Last	I4	Record number of last data record
Data records (first, ..., last)		
Nuclide	A7	Name of nuclide (parent); e.g. Am-241, Tc-99m
Half-life	A8	Half-life of nuclide
Units	A2	Half-life units: μ s, microsecond; ms, millisecond; s, second; m, minute; d, day; y, year
Decay mode	A8	A, alpha; B-, beta minus; B+, beta plus; EC, electron capture; IT, isomeric transition; SF, spontaneous fission
Pointer-1	I7	Location of nuclide in ICRP-07.RAD file
Pointer-2	I7	Location of nuclide in ICRP-07.BET file
Pointer-3	I7	Location of nuclide in ICRP-07.ACK file
Pointer-4	I6	Location of nuclide in ICRP-07.NSF file
The next block of three fields are repeated for radioactive daughter i , $i = 1 - 4$.		
Daughter- i	A7	Name of daughter nuclide i
Pointer- i	I6	Location of daughter i in ICRP-07.NDX file
Branch- i	E11.0	Branching fraction to daughter i
E-alpha	E7.0	Energy of alpha emissions (MeV/nuclear transformation)
E-electron	E8.0	Energy of electrons, including beta (MeV/nuclear transformation)
E-photon	E8.0	Energy of photon emission (MeV/nuclear transformation)
Number-1	I4	Number of photons of energy less than 10 keV
Number-2	I4	Number of photons of energy greater than 10 keV
Number-3	I4	Number of beta transitions
Number-4	I5	Number of mono-energetic electrons
Number-5	I4	Number of alpha transitions
AMU	E11.0	Atomic mass of radionuclide (Audi et al., 2003)
Γ_{10}	E10.0	Air-kerma rate constant ($\text{Gy}\cdot\text{m}^2/\text{Bq s}$)
K_{air}	A9	Point source air-kerma coefficient ($\text{Gy}\cdot\text{m}^2/\text{Bq s}$)

* The format is expressed in FORTRAN notation, e.g. A8 denotes an alphanumeric field of length 8, I5 denotes an integer field of length 5, and E11.0 denotes a real numeric field of length 11.

include gamma rays, characteristic x rays, and inner bremsstrahlung; the latter are not addressed here. It should also be noted that annihilation photons, included in the RAD file, are not considered. In a source of finite size, attenuation and scattering occur, and external bremsstrahlung can be produced. In addition, any medium between the source and the point of measurement will give rise to absorption and scattering, external bremsstrahlung, and annihilation photons. In many cases, these processes can influence the observed kerma rate substantially. The constant, as defined, is given by:

$$\Gamma_{\delta} = \frac{1}{4\pi} \sum_i (\mu_k/\rho)_i Y_i E_i \quad (9)$$

where $(\mu_k/\rho)_i$ is the mass energy-transfer coefficient for air for photons of energy E_i emitted by the nuclide with yield Y_i . The value of δ used in the calculations here is 10 keV.

Point source air-kerma coefficient

(32) To address the annihilation photons associated with positron emission and the neutrons and photons that accompany SF, the air-kerma coefficient, $K_{\text{air},\delta}$, for a hypothetical point source is defined as:

$$K_{\text{air},\delta} = \frac{1}{4\pi} \left[\sum_i (\mu_k/\rho)_i Y_i E_i + \sum_i Y(E_i, E_{i+1}) \bar{k}(E_i, E_{i+1}) \right] \quad (10)$$

This has been included on the nuclide record in the NDX file. The summation in the first term of Eq. (10) extends over all photons of energy greater than δ , including annihilation photons arising from position emission, and the delay and prompt gamma radiations accompanying SF. The second term of Eq. (10) is the contribution to kerma of neutrons accompanying SF with yield $Y(E_i, E_{i+1})$ per nuclear transformation, and $\bar{k}(E_i, E_{i+1})$ denotes the average value of the neutron air-kerma coefficient (Chadwick et al., 1999) over energy E_i and E_{i+1} . Only neutrons of energy greater than δ (10 keV) are considered. The constant and the coefficient are numerically equal except in cases of positron emission or SF decay. The cautionary statements above regarding the air-kerma rate constant and observed kerma rate for a real source in air are also applicable to the coefficient.

3.2.2. Radiation file: ICRP-07.RAD

(33) The records of the RAD file contain the data on the energy and yield of each radiation emitted in nuclear transformations of the radionuclide. The fields of the RAD records are described in Table 3.2. Briefly, the first record (header record) of a nuclide in the file contains the name of the nuclide, its half-life, and the number of data records for the emitted radiations that follow. The data record for each emitted radiation has the following fields: (1) an integer code (ICODE) that identifies the radiation type; (2) the absolute yield of the radiation (i.e. number per nuclear transformation); (3) the unique or average energy of the radiation (MeV); and (4) a two-character mnemonic denoting the radiation type. The ICODE code and the

Table 3.2 Structure of records in the ICRP-07.RAD file.

Field	Format	Description
Nuclide record		
Nuclide	A7	Nuclide name, e.g. Tc-99m
$T_{1/2}$	E11.0	Physical half-life of nuclide
Time unit	A2	Units of $T_{1/2}$
N	I9	Number of data records
Data record (1, ..., N)		
ICODE	A2	Radiation type (see Table 2.3)
Yield	E12.0	Yield of the radiation (/nuclear transformation)
Energy	E12.0	Energy of the radiation (MeV)
JCODE	A3	Mnemonic (see Table 2.3)

Table 3.3. Description of integer code (ICODE) variable.

ICODE	Mnemonic	Description
1	G	Gamma rays
	PG	Prompt gamma rays*
	DG	Delayed gamma rays*
2	X	X rays
3	AQ	Annihilation photons
4	B+	Beta-plus particles
5	B-	Beta-minus particles
	BD	Delayed beta particles*
6	IE	Internal conversion electrons
7	AE	Auger electrons
8	A	Alpha particles
9	AR	Alpha recoil nuclei
10	FF	Fission fragments
11	N	Neutrons

* Prompt and delayed radiations of spontaneous fission.

mnemonic are defined in Table 3.3. The pointer on the nuclide's records in the NDX file is the record number of the nuclide's header record in the RAD file.

(34) The radiation records are ordered by radiation type, in ICODE order, and in increasing energy. The radiation groups are as follows:

- Photons, ICODE 1–3: x ray, gamma ray, annihilation photons, and prompt and delayed photons of SF
- Beta particles, ICODE 4–5: mean energies of each beta transition
- Mono-energetic electrons, ICODE 6–7: internal conversion and Auger–CK electrons
- Alpha particles, ICODE 8
- Alpha recoil nuclei, ICODE 9
- Fission fragments, ICODE 10
- Neutrons, ICODE 11

(35) The records of each group are sorted by increasing energy. This sorting facilitates interpolation of energy-dependent quantities, e.g. energy-dependent specific absorbed fraction data for the radiation type. Radiations of a particular type within its group can be identified using the mnemonic (e.g. annihilation photons, delayed gamma rays, etc.). The prompt and delayed gamma spectra following SF are represented by 48 and 28 discrete photons, respectively, following the suggestion of Dillman and Jones (1975). These emissions can be identified by their mnemonic.

3.2.3. Beta spectra file: ICRP-07.BET

(36) The BET file contains the beta spectra for each beta emitter in the ICRP-07 collection. The spectral data are tabulated on a fixed logarithmic-type energy grid. For each nuclide, the header record gives the name of the nuclide and the number of data records that follow. The fields of the data records contain the electron energy E (MeV) and the number of beta particles per MeV per nuclear transformation

Table 3.4 Structure of records in the ICRP-07.BET file.

Field	Format	Description
Nuclide record		
Nuclide	A7	Nuclide name, e.g. Tc-99m
N	I10	Number of data records
Data record (1, ..., N)		
Energy	F7.0	Energy grid point (MeV)
Number	E10.0	Number of beta particles per MeV per nuclear transformation at this energy

emitted at this energy. A spectrum was synthesised for the delayed beta emissions following SF by first assuming forbidden unique transitions of atomic numbers 39 and 58 with the energy and yield indicated in Table 2.4. The structure of the records in the BET file is given in Table 3.4.

3.2.4. Auger–CK electron spectra file: ICRP-07.ACK

(37) For 136 selected nuclides, detailed Auger–CK electron spectra are contained in the ACK file. These emissions were collapsed into no more than 15 groups for each decay mode in the RAD file. The maximum number of discrete electrons in a spectrum is 3015 in the case of Hg-195m and Hg-197m. The fields of the header record contain the name of the nuclide and the number of data records that follow. The data records contain the electron energy E (eV), the yield (number of electrons of that energy per nuclear transformation), and identification of the atomic electron transition. The records of the ACK file are described in Table 3.5. The group data in the RAD file are sufficient to compute the average absorbed dose in organs and tissues; however, the data in the ACK file may be of interest in microdosimetric calculations of radionuclides within the cell nucleus.

Table 3.5 Structure of records in the ICRP-07.ACK file.

Field	Format	Description
Nuclide record		
Nuclide	A8	Nuclide name, e.g. I-123
N	I8	Number of discrete electron lines
Data record (1, ..., N)		
Yield	E12.0	Number of electrons per nuclear transformation at this energy
Energy	E12.0	Electron energy (eV)
Transition	A9	Identification of atomic transition; e.g. L1 M2 M3

Table 3.6 Structure of records in the ICRP-07.NSF file.

Field	Format	Description
Nuclide record		
Nuclide	A8	Nuclide name, e.g. Cf-252
BF	E12.0	Spontaneous fission branching fraction
N	I3	Number of neutron energy bins
Data record (1, ..., N)		
Energy-1	E9.0	Lower energy (MeV)
Energy-2	E9.0	Upper energy (MeV)
Yield	E12.0	Number of neutrons per nuclear transformation in bin

3.2.5. Neutron spectra file: ICRP-07.NSF

(38) The NSF file contains the spectra of neutrons emitted by the 28 radionuclides of the ICRP-07 collection that undergo SF. For each nuclide, the header record gives the name of the nuclide, the branching fraction for SF, and the number of data records that follow. The data records define a neutron energy bin (or energy group) and specify the number of neutrons per nuclear transformation in that energy bin. The records in the NSF file are described in Table 3.6.

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ANNEX A. RADIONUCLIDES OF THE ICRP-07 COLLECTION

(A1) The 1252 radionuclides of the ICRP-07 collection are listed in [Table A.1](#) in order of atomic number. Indented on the line following the radionuclide is the identity of the daughter nuclide formed by the nuclear transformation (nt) of the parent, along with the fraction of the parent's transformation creating the daughter. If the daughter nucleus is stable then it is shown in italics. The half-life, decay mode, and energy emitted per nuclear transformation are also shown.

(A2) The units of the half-life are abbreviated as: μ s - microsecond, ms - millisecond, s - second, m - minute, d - day, and y - year. The decay modes are abbreviated as: A - alpha, B- - beta minus, B+ - beta plus, EC - electron capture, IT - isomeric transition, SF-spontaneous fission. The contribution of fission fragments and neutrons associated with spontaneous fission are included in the total energy.

Table A.1 Properties of the radionuclides of the ICRP-07 collection.

Z	Element	Nuclide	Half-life	Decay mode	Emitted energy (MeV/nt)			
					Alpha	Electron	Photon	Total
1	Hydrogen	H-3	12.32 y	B-	—	0.005	—	0.0057
		<i>He-3</i>	1.00					
4	Beryllium	Be-7	53.22 d	EC	—	<E-04	0.0499	0.0499
		<i>Li-7</i>	1.00					
		Be-10	1.51E+6 y	B-	—	0.2525	—	0.2525
		<i>B-10</i>	1.00					
6	Carbon	C-10	19.255 s	ECB+	—	0.8087	1.7443	2.5531
		<i>B-10</i>	1.00					
		C-11	20.39 m	ECB+	—	0.3847	1.0196	1.4043
		<i>B-11</i>	1.00					
		C-14	5.70E+3 y	B-	—	0.0495	—	0.0495
		<i>N-14</i>	1.00					
7	Nitrogen	N-13	9.965 m	ECB+	—	0.4909	1.0200	1.5109
		<i>C-13</i>	1.00					
		N-16	7.13 s	B-	—	2.7646	4.4932	7.2579
		<i>O-16</i>	1.00					
8	Oxygen	O-14	70.606 s	ECB+	—	0.7763	3.3201	4.0965
		<i>N-14</i>	1.00					
		O-15	122.24 s	ECB+	—	0.7347	1.0210	1.7557
		<i>N-15</i>	1.00					
		O-19	26.464 s	B-	—	1.7608	0.9397	2.7006
		<i>F-19</i>	1.00					
9	Fluorine	F-17	64.49 s	ECB+	—	0.7385	1.0208	1.7593
		<i>O-17</i>	1.00					
		F-18	109.77 m	ECB+	—	0.2416	0.9886	1.2302
		<i>O-18</i>	1.00					
10	Neon	Ne-19	17.22 s	ECB+	—	0.9624	1.0210	1.9834
		<i>F-19</i>	1.00					
		Ne-24	3.38 m	B-	—	0.8035	0.5419	1.3455
		<i>Na-24</i>	1.00					

Table A.1 continued

Z	Element	Nuclide	Half-life	Decay mode	Emitted energy (MeV/nt)			
					Alpha	Electron	Photon	Total
11	Sodium	Na-22	2.6019 y	ECB+	–	0.1941	2.1926	2.3866
			<i>Ne-22</i>	1.00				
		Na-24	14.9590 h	B-	–	0.5538	4.1232	4.6770
			<i>Mg-24</i>	1.00				
12	Magnesium	Mg-27	9.458 m	B-	–	0.7021	0.8912	1.5933
			<i>Al-27</i>	1.00				
		Mg-28	20.915 h	B-	–	0.1610	1.3700	1.5310
			<i>Al-28</i>	1.00				
13	Aluminium	Al-26	7.17E+5 y	ECB+	–	0.4444	2.6751	3.1195
			<i>Mg-26</i>	1.00				
		Al-28	2.2414 m	B-	–	1.2417	1.7789	3.0205
			<i>Si-28</i>	1.00				
		Al-29	6.56 m	B-	–	0.9764	1.3794	2.3558
			<i>Si-29</i>	1.00				
14	Silicon	Si-31	157.3 m	B-	–	0.5949	0.0009	0.5957
			<i>P-31</i>	1.00				
		Si-32	132 y	B-	–	0.0686	–	0.0686
			<i>P-32</i>	1.00				
15	Phosphorus	P-30	2.498 m	ECB+	–	1.4389	1.0222	2.4610
			<i>Si-30</i>	1.00				
		P-32	14.263 d	B-	–	0.6948	–	0.6948
			<i>S-32</i>	1.00				
		P-33	25.34 d	B-	–	0.0764	–	0.0764
			<i>S-33</i>	1.00				
16	Sulphur	S-35	87.51 d	B-	–	0.0487	–	0.0487
			<i>Cl-35</i>	1.00				
		S-37	5.05 m	B-	–	0.7998	2.9311	3.7309
			<i>Cl-37</i>	1.00				
		S-38	170.3 m	B-	–	0.4898	1.6953	2.1851
			<i>Cl-38</i>	1.00				
17	Chlorine	Cl-34	1.5264 s	ECB+	–	2.0508	1.0212	3.0720
			<i>S-34</i>	1.00				
		Cl-34m	32.00 m	ECB+IT	–	0.4596	2.1127	2.5722
			<i>Cl-34</i>	4.4600E-01				
			<i>S-34</i>	5.5400E-01				
		Cl-36	3.01E+5 y	B-ECB+	–	0.2732	0.0001	0.2733
			<i>Ar-36</i>	9.8100E-01				
			<i>S-36</i>	1.9000E-02				
		Cl-38	37.24 m	B-	–	1.5504	1.4430	2.9934
			<i>Ar-38</i>	1.00				
		Cl-39	55.6 m	B-	–	0.8246	1.4509	2.2755
			<i>Ar-39</i>	1.00				
		Cl-40	1.35 m	B-	–	1.5265	4.0821	5.6085
			<i>Ar-40</i>	1.00				
18	Argon	Ar-37	35.04 d	EC	–	0.0023	0.0003	0.0025
			<i>Cl-37</i>	1.00				
		Ar-39	269 y	B-	–	0.2188	–	0.2188
			<i>K-39</i>	1.00				

Table A.1 continued

Z	Element	Nuclide	Half-life	Decay mode	Emitted energy (MeV/nt)			
					Alpha	Electron	Photon	Total
19	Potassium	Ar-41	109.61 m <i>K-41</i>	B- 1.00	—	0.4637	1.2836	1.7474
		Ar-42	32.9 y <i>K-42</i>	B- 1.00	—	0.2325	—	0.2325
		Ar-43	5.37 m <i>K-43</i>	B- 1.00	—	1.3570	1.5352	2.8921
		Ar-44	11.87 m <i>K-44</i>	B- 1.00	—	0.5245	1.9376	2.4621
		K-38	7.636 m <i>Ar-38</i>	ECB+ 1.00	—	1.2107	3.1867	4.3974
		K-40	1.251E+9 y <i>Ca-40</i> <i>Ar-40</i>	B-ECB+ 8.9140E-01 1.0860E-01	—	0.5218	0.1567	0.6785
		K-42	12.360 h <i>Ca-42</i>	B- 1.00	—	1.4303	0.2787	1.7090
		K-43	22.3 h <i>Ca-43</i>	B- 1.00	—	0.3097	0.9641	1.2739
		K-44	22.13 m <i>Ca-44</i>	B- 1.00	—	1.4567	2.3846	3.8413
		K-45	17.3 m <i>Ca-45</i>	B- 1.00	—	0.9959	1.8360	2.8319
		K-46	105 s <i>Ca-46</i>	B- 1.00	—	2.3161	2.8710	5.1871
20	Calcium	Ca-41	1.02E+5 y <i>K-41</i>	EC 1.00	—	0.0027	0.0005	0.0032
		Ca-45	162.67 d <i>Sc-45</i>	B- 1.00	—	0.0772	<E-04	0.0772
		Ca-47	4.536 d <i>Sc-47</i>	B- 1.00	—	0.3521	1.0521	1.4042
		Ca-49	8.718 m <i>Sc-49</i>	B- 1.00	—	0.8693	3.1675	4.0369
21	Scandium	Sc-42m	62.0 s <i>Ca-42</i>	ECB+ 1.00	—	1.2565	4.2042	5.4607
		Sc-43	3.891 h <i>Ca-43</i>	ECB+ 1.00	—	0.4195	0.9841	1.4035
		Sc-44	3.97 h <i>Ca-44</i>	ECB+ 1.00	—	0.5961	2.1369	2.7330
		Sc-44m	58.61 h <i>Sc-44</i> <i>Ca-44</i>	ITEC 9.8800E-01 1.2000E-02	—	0.0328	0.2743	0.3071
		Sc-46	83.79 d <i>Ti-46</i>	B- 1.00	—	0.1121	2.0096	2.1217
		Sc-47	3.3492 d <i>Ti-47</i>	B- 1.00	—	0.1624	0.1089	0.2713
		Sc-48	43.67 h <i>Ti-48</i>	B- 1.00	—	0.2216	3.3529	3.5744

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Table A.1 continued

Z	Element	Nuclide	Half-life	Decay mode	Emitted energy (MeV/nt)			
					Alpha	Electron	Photon	Total
22	Titanium	Sc-49	57.2 m <i>Ti-49</i>	B- 1.00	—	0.8177	0.0010	0.8188
		Sc-50	102.5 s <i>Ti-50</i>	B- 1.00	—	1.6362	3.2022	4.8384
		Ti-44	60.0 y Sc-44	EC 1.00	—	0.0108	0.1396	0.1504
		Ti-45	184.8 m <i>Sc-45</i>	ECB+ 1.00	—	0.3728	0.8703	1.2431
		Ti-51	5.76 m <i>V-51</i>	B- 1.00	—	0.8692	0.3692	1.2383
		Ti-52	1.7 m V-52	B- 1.00	—	0.7530	0.1282	0.8812
23	Vanadium	V-47	32.6 m <i>Ti-47</i>	ECB+ 1.00	—	0.8027	0.9951	1.7978
		V-48	15.9735 d <i>Ti-48</i>	ECB+ 1.00	—	0.1526	2.9140	3.0666
		V-49	330 d <i>Ti-49</i>	EC 1.00	—	0.0035	0.0009	0.0045
		V-50	1.50E+17 y <i>Ti-50</i> <i>Cr-50</i>	ECB- 8.3000E-01 1.7000E-01	—	0.0158	1.4235	1.4393
		V-52	3.743 m <i>Cr-52</i>	B- 1.00	—	1.0684	1.4449	2.5133
		V-53	1.61 m <i>Cr-53</i>	B- 1.00	—	1.0048	1.0382	2.0430
24	Chromium	Cr-48	21.56 h V-48	ECB+ 1.00	—	0.0086	0.4363	0.4449
		Cr-49	42.3 m V-49	ECB+ 1.00	—	0.6047	1.0538	1.6584
		Cr-51	27.7025 d <i>V-51</i>	EC 1.00	—	0.0038	0.0329	0.0367
		Cr-55	3.497 m <i>Mn-55</i>	B- 1.00	—	1.1007	0.0007	1.1013
		Cr-56	5.94 m Mn-56	B- 1.00	—	0.6083	0.0918	0.7000
25	Manganese	Mn-50m	1.75 m <i>Cr-50</i>	ECB+ 1.00	—	1.5243	4.6397	6.1640
		Mn-51	46.2 m Cr-51	ECB+ 1.00	—	0.9344	0.9977	1.9321
		Mn-52m	21.1 m Mn-52 <i>Cr-52</i>	ECB+IT 1.7500E-02 9.8250E-01	—	1.1321	2.4086	3.5407
		Mn-52	5.591 d <i>Cr-52</i>	ECB+ 1.00	—	0.0750	3.4585	3.5335
		Mn-53	3.7E+6 y <i>Cr-53</i>	EC 1.00	—	0.0040	0.0014	0.0053

Table A.1 continued

Z	Element	Nuclide	Half-life	Decay mode	Emitted energy (MeV/nt)			
					Alpha	Electron	Photon	Total
26	Iron	Mn-54	312.12 d	ECB+B- <i>Cr-54</i> <i>Fe-54</i>	—	0.0042	0.8360	0.8402
				1.00 2.9000E-06				
		Mn-56	2.5789 h	B- <i>Fe-56</i>	—	0.8299	1.6916	2.5215
				1.00				
		Mn-57	85.4 s	B- <i>Fe-57</i>	—	1.1013	0.0996	1.2009
				1.00				
		Mn-58m	65.2 s	B- <i>Fe-58</i>	—	1.7443	2.3854	4.1297
				1.00				
		Fe-52	8.275 h	ECB+	—	0.1917	0.7405	0.9322
		Mn-52m		1.00				
		Fe-53	8.51 m	ECB+	—	1.1013	1.1782	2.2795
				Mn-53 1.00				
		Fe-53m	2.526 m	IT Fe-53	—	0.0023	3.0547	3.0570
				1.00				
27	Cobalt	Fe-55	2.737 y	EC <i>Mn-55</i>	—	0.0042	0.0017	0.0058
				1.00				
		Fe-59	44.495 d	B- <i>Co-59</i>	—	0.1179	1.1883	1.3062
				1.00				
		Fe-60	1.5E+6 y	B- Co-60m	—	0.0647	—	0.0647
				1.00				
		Fe-61	5.98 m	B- Co-61	—	1.0934	1.3910	2.4844
				1.00				
		Fe-62	68 s	B- Co-62	—	0.8258	0.5061	1.3319
				1.00				
		Co-54m	1.48 m	ECB+ <i>Fe-54</i>	—	2.0484	3.9306	5.9790
				1.00				
		Co-55	17.53 h	ECB+ Fe-55	—	0.4312	1.9960	2.4272
				1.00				
		Co-56	77.23 d	ECB+ <i>Fe-56</i>	—	0.1198	3.6404	3.7602
				1.00				
		Co-57	271.74 d	EC <i>Fe-57</i>	—	0.0186	0.1252	0.1439
				1.00				
		Co-58	70.86 d	ECB+ <i>Fe-58</i>	—	0.0340	0.9749	1.0089
				1.00				
		Co-58m	9.04 h	IT Co-58	—	0.0229	0.0020	0.0249
				1.00				
		Co-60	5.2713 y	B- <i>Ni-60</i>	—	0.0969	2.5038	2.6007
				1.00				
		Co-60m	10.467 m	ITB- Co-60 <i>Ni-60</i>	—	0.0565	0.0067	0.0632
				9.9760E-01 2.4000E-03				
		Co-61	1.650 h	B- <i>Ni-61</i>	—	0.4664	0.0970	0.5634
				1.00				
		Co-62	1.50 m	B- <i>Ni-62</i>	—	1.6336	1.6050	3.2386
				1.00				

(continued on next page)

Table A.1 continued

Z	Element	Nuclide	Half-life	Decay mode	Emitted energy (MeV/nt)			
					Alpha	Electron	Photon	Total
28	Nickel	Co-62m	13.91 m	B-	–	1.0974	2.6923	3.7897
		<i>Ni-62</i>		1.00				
		Ni-56	6.075 d	ECB+	–	0.0073	1.7207	1.7280
		Co-56		1.00				
		Ni-57	35.60 h	ECB+	–	0.1571	1.9382	2.0953
		Co-57		1.00				
		Ni-59	1.01E+5 y	ECB+	–	0.0045	0.0024	0.0069
		<i>Co-59</i>		1.00				
		Ni-63	100.1 y	B-	–	0.0174	–	0.0174
		<i>Cu-63</i>		1.00				
		Ni-65	2.51719 h	B-	–	0.6277	0.5583	1.1860
		<i>Cu-65</i>		1.00				
		Ni-66	54.6 h	B-	–	0.0734	–	0.0734
		<i>Cu-66</i>		1.00				
29	Copper	Cu-57	0.1963 s	ECB+	–	3.5987	1.1407	4.7394
		<i>Ni-57</i>		1.00				
		Cu-59	81.5 s	ECB+	–	1.4892	1.4451	2.9342
		<i>Ni-59</i>		1.00				
		Cu-60	23.7 m	ECB+	–	0.8975	3.9111	4.8087
		<i>Ni-60</i>		1.00				
		Cu-61	3.333 h	ECB+	–	0.3090	0.8237	1.1327
		<i>Ni-61</i>		1.00				
		Cu-62	9.673 m	ECB+	–	1.2843	1.0069	2.2912
		<i>Ni-62</i>		1.00				
		Cu-64	12.700 h	ECB+B-	–	0.1248	0.1855	0.3102
		<i>Ni-64</i>		6.1000E-01				
		<i>Zn-64</i>		3.9000E-01				
		Cu-66	5.120 m	B-	–	1.0664	0.0978	1.1642
		<i>Zn-66</i>		1.00				
30	Zinc	Cu-67	61.83 h	B-	–	0.1504	0.1154	0.2657
		<i>Zn-67</i>		1.00				
		Cu-69	2.85 m	B-	–	0.8863	0.5284	1.4147
		<i>Zn-69</i>		1.00				
		Zn-60	2.38 m	ECB+	–	1.1301	1.5282	2.6584
		<i>Cu-60</i>		1.00				
		Zn-61	89.1 s	ECB+	–	1.8571	1.5327	3.3898
		<i>Cu-61</i>		1.00				
		Zn-62	9.186 h	ECB+	–	0.0326	0.4431	0.4757
		<i>Cu-62</i>		1.00				
		Zn-63	38.47 m	ECB+	–	0.9204	1.0967	2.0171
		<i>Cu-63</i>		1.00				
		Zn-65	244.06 d	ECB+	–	0.0069	0.5819	0.5888
		<i>Cu-65</i>		1.00				
		Zn-69m	13.76 h	ITB-	–	0.0226	0.4162	0.4388
		<i>Zn-69</i>		9.9967E-01				
		<i>Ga-69</i>		3.3000E-04				
		Zn-69	56.4 m	B-	–	0.3216	<E-04	0.3216
		<i>Ga-69</i>		1.00				

Table A.1 continued

Z	Element	Nuclide	Half-life	Decay mode	Emitted energy (MeV/nt)			
					Alpha	Electron	Photon	Total
31	Gallium	Zn-71m	3.96 h	B- <i>Ga-71</i>	—	0.5430	1.5606	2.1037
		Zn-71	2.45 m	B- <i>Ga-71</i>	—	1.0474	0.3150	1.3624
		Zn-72	46.5 h	B- <i>Ga-72</i>	—	0.1021	0.1519	0.2541
		Ga-64	2.627 m	ECB+ <i>Zn-64</i>	—	1.7031	3.3726	5.0756
		Ga-65	15.2 m	ECB+ <i>Zn-65</i>	—	0.8158	1.1650	1.9808
		Ga-66	9.49 h	ECB+ <i>Zn-66</i>	—	0.9634	2.4944	3.4577
		Ga-67	3.2612 d	EC <i>Zn-67</i>	—	0.0363	0.1595	0.1959
		Ga-68	67.71 m	ECB+ <i>Zn-68</i>	—	0.7379	0.9487	1.6866
		Ga-70	21.14 m	B-EC <i>Ge-70</i>	—	0.6441	0.0073	0.6514
				<i>Zn-70</i>				
		Ga-72	14.10 h	B- <i>Ge-72</i>	—	0.5060	2.7071	3.2131
		Ga-73	4.86 h	B- <i>Ge-73</i>	—	0.4999	0.3520	0.8519
		Ga-74	8.12 m	B- <i>Ge-74</i>	—	0.9945	3.1540	4.1486
32	Germanium	Ge-66	2.26 h	ECB+ <i>Ga-66</i>	—	0.0984	0.6780	0.7764
		Ge-67	18.9 m	ECB+ <i>Ga-67</i>	—	1.1688	1.4254	2.5942
		Ge-68	270.95 d	EC <i>Ga-68</i>	—	0.0050	0.0041	0.0091
		Ge-69	39.05 h	ECB+ <i>Ga-69</i>	—	0.1203	0.9505	1.0708
		Ge-71	11.43 d	EC <i>Ga-71</i>	—	0.0050	0.0042	0.0092
		Ge-75	82.78 m	B- <i>As-75</i>	—	0.4206	0.0352	0.4558
		Ge-77	11.30 h	B- <i>As-77</i>	—	0.6493	1.0787	1.7280
		Ge-78	88 m	B- <i>As-78</i>	—	0.2273	0.2781	0.5053
33	Arsenic	As-68	151.6 s	ECB+ <i>Ge-68</i>	—	1.9985	3.7113	5.7097
		As-69	15.23 m	ECB+ <i>Ge-69</i>	—	1.2182	1.1431	2.3613
		As-70	52.6 m	ECB+ <i>Ge-70</i>	—	0.8799	4.2528	5.1327

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Table A.1 continued

Z	Element	Nuclide	Half-life	Decay mode	Emitted energy (MeV/nt)			
					Alpha	Electron	Photon	Total
34	Selenium	As-71	65.28 h	ECB+	–	0.1167	0.5765	0.6932
		Ge-71	1.00					
		As-72	26.0 h	ECB+	–	1.0409	1.7832	2.8240
		Ge-72	1.00					
		As-73	80.30 d	EC	–	0.0606	0.0159	0.0765
		Ge-73	1.00					
		As-74	17.77 d	ECB+B-	–	0.2658	0.7582	1.0241
		Ge-74	6.6000E-01					
		Se-74	3.4000E-01					
		As-76	1.0778 d	B-	–	1.0670	0.4166	1.4836
		Se-76	1.00					
		As-77	38.83 h	B-	–	0.2258	0.0083	0.2342
		Se-77	1.00					
		As-78	90.7 m	B-	–	1.2460	1.3066	2.5525
		Se-78	1.00					
		As-79	9.01 m	B-	–	0.8771	0.0338	0.9109
		Se-79m	9.7188E-01					
		Se-79	2.8121E-02					
		Se-70	41.1 m	ECB+	–	0.2375	0.7211	0.9586
		As-70	1.00					
		Se-71	4.74 m	ECB+	–	1.3844	1.6056	2.9900
		As-71	1.00					
		Se-72	8.40 d	EC	–	0.0228	0.0343	0.0571
		As-72	1.00					
		Se-73	7.15 h	ECB+	–	0.3871	1.0924	1.4795
		As-73	1.00					
		Se-73m	39.8 m	ITECB+	–	0.1642	0.2633	0.4275
		Se-73	7.2600E-01					
		As-73	2.7400E-01					
		Se-75	119.779 d	EC	–	0.0144	0.3890	0.4034
		As-75	1.00					
		Se-77m	17.36 s	IT	–	0.0737	0.0888	0.1625
		Se-77	1.00					
		Se-79	2.95E+5 y	B-	–	0.0529	–	0.0529
		Br-79	1.00					
		Se-79m	3.92 m	ITB-	–	0.0820	0.0138	0.0957
		Se-79	9.9944E-01					
		Br-79	5.6000E-04					
		Se-81	18.45 m	B-	–	0.6108	0.0080	0.6188
		Br-81	1.00					
		Se-81m	57.28 m	ITB-	–	0.0871	0.0183	0.1054
		Se-81	9.9948E-01					
		Br-81	5.2000E-04					
		Se-83	22.3 m	B-	–	0.4528	2.6254	3.0782
		Br-83	1.00					
		Se-83m	70.1 s	B-	–	1.2570	0.9849	2.2419
		Br-83	1.00					

Table A.1 continued

Z	Element	Nuclide	Half-life	Decay mode	Emitted energy (MeV/nt)			
					Alpha	Electron	Photon	Total
35	Bromine	Se-84	3.1 m	B-	—	0.5499	0.4202	0.9701
		Br-84		1.00				
		Br-72	78.6 s	ECB+	—	2.7880	2.9624	5.7504
		Se-72		1.00				
		Br-73	3.4 m	ECB+	—	1.3434	1.4331	2.7765
		Se-73m		9.9881E-01				
		Se-73		1.1920E-03				
		Br-74	25.4 m	ECB+	—	1.0616	4.6130	5.6746
		Se-74		1.00				
		Br-74m	46 m	ECB+	—	1.2747	4.1467	5.4214
		Se-74		1.00				
		Br-75	96.7 m	ECB+	—	0.5283	1.1981	1.7263
		Se-75		1.00				
		Br-76	16.2 h	ECB+	—	0.6497	2.7933	3.4430
		Se-76		1.00				
		Br-76m	1.31 s	ITECB+	—	0.0690	0.0433	0.1123
		Br-76		9.9700E-01				
		Se-76		3.0000E-03				
		Br-77	57.036 h	ECB+	—	0.0094	0.3209	0.3303
		Se-77		1.00				
		Br-77m	4.28 m	IT	—	0.0876	0.0197	0.1073
		Br-77		1.00				
		Br-78	6.46 m	ECB+B-	—	1.0235	1.0336	2.0571
		Se-78		9.9990E-01				
		Kr-78		1.0000E-04				
		Br-80	17.68 m	B-ECB+	—	0.7247	0.0761	0.8007
		Kr-80		9.1700E-01				
		Se-80		8.3000E-02				
		Br-80m	4.4205 h	IT	—	0.0617	0.0242	0.0859
		Br-80		1.00				
		Br-82	35.30 h	B-	—	0.1454	2.6389	2.7844
		Kr-82		1.00				
		Br-82m	6.13 m	ITB-	—	0.0704	0.0081	0.0785
		Br-82		9.7600E-01				
		Kr-82		2.4000E-02				
		Br-83	2.40 h	B-	—	0.3258	0.0069	0.3326
		Kr-83m		9.9845E-01				
		Kr-83		1.5519E-03				
		Br-84	31.80 m	B-	—	1.2364	1.7595	2.9959
		Kr-84		1.00				
		Br-84m	6.0 m	B-	—	0.8983	2.7684	3.6667
		Kr-84		1.00				
		Br-85	2.90 m	B-	—	1.0384	0.0660	1.1045
		Kr-85m		9.9779E-01				
		Kr-85		2.2112E-03				

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Table A.1 continued

Z	Element	Nuclide	Half-life	Decay mode	Emitted energy (MeV/nt)			
					Alpha	Electron	Photon	Total
36	Krypton	Kr-74	11.50 m	ECB+	—	0.6006	1.0579	1.6585
			Br-74	1.00				
		Kr-75	4.29 m	ECB+	—	1.5430	1.2800	2.8230
			Br-75	1.00				
		Kr-76	14.8 h	EC	—	0.0154	0.4276	0.4430
			Br-76	9.9189E-01				
		Kr-77	74.4 m	ECB+	—	0.6744	1.0384	1.7128
			Br-77m	9.0386E-01				
		Kr-79	35.04 h	ECB+	—	0.0237	0.2549	0.2786
			Br-79	1.00				
		Kr-81	2.29E+5 y	EC	—	0.0052	0.0072	0.0124
			Br-81	1.00				
		Kr-81m	13.10 s	ITEC	—	0.0596	0.1309	0.1905
			Kr-81	9.9998E-01				
		Kr-83m	1.83 h	IT	—	0.0388	0.0028	0.0416
			Kr-83	1.00				
		Kr-85	10.756 y	B-	—	0.2507	0.0022	0.2529
			Rb-85	1.00				
		Kr-85m	4.480 h	B-IT	—	0.2549	0.1574	0.4123
			Kr-85	2.1400E-01				
		Kr-87	76.3 m	B-	—	1.3281	0.7919	2.1201
			Rb-87	1.00				
		Kr-88	2.84 h	B-	—	0.3689	1.9538	2.3227
			Rb-88	1.00				
		Kr-89	3.15 m	B-	—	1.3705	1.9313	3.3018
			Rb-89	1.00				
37	Rubidium	Rb-77	3.77 m	ECB+	—	1.6865	1.5452	3.2318
			Kr-77	1.00				
		Rb-78	17.66 m	ECB+	—	1.2889	4.0918	5.3808
			Kr-78	1.00				
		Rb-78m	5.74 m	ECB+IT	—	1.4992	3.2149	4.7141
			Rb-78	1.0000E-01				
		Rb-79	22.9 m	ECB+	—	0.8099	1.4493	2.2592
			Kr-79	1.00				
		Rb-80	33.4 s	ECB+	—	2.0455	1.1900	3.2355
			Kr-80	1.00				
		Rb-81	4.576 h	ECB+	—	0.1222	0.5081	0.6304
			Kr-81m	9.5691E-01				
		Rb-81m	30.5 m	ITECB+	—	0.0817	0.0303	0.1120
			Kr-81	4.3091E-02				
			Rb-81	9.7600E-01				
			Kr-81	2.3786E-02				
			Kr-81m	2.1355E-04				

Table A.1 continued

Z	Element	Nuclide	Half-life	Decay mode	Emitted energy (MeV/nt)			
					Alpha	Electron	Photon	Total
38	Strontium	Rb-82	1.273 m	ECB+	—	1.4112	1.1083	2.5195
		<i>Kr-82</i>		1.00				
		Rb-82m	6.472 h	ECB+	—	0.0935	2.9212	3.0147
		<i>Kr-82</i>		1.00				
		Rb-83	86.2 d	EC	—	0.0083	0.4914	0.4997
		<i>Kr-83m</i>		7.4292E-01				
		<i>Kr-83</i>		2.5708E-01				
		Rb-84	32.77 d	ECB+B-	—	0.1633	0.9077	1.0710
		<i>Kr-84</i>		9.6200E-01				
		<i>Sr-84</i>		3.8000E-02				
		Rb-84m	20.26 m	IT	—	0.0813	0.3831	0.4644
		Rb-84		1.00				
		Rb-86	18.642 d	B-EC	—	0.6680	0.0930	0.7610
		<i>Sr-86</i>		9.9995E-01				
		<i>Kr-86</i>		5.2000E-05				
		Rb-86m	1.017 m	IT	—	0.0100	0.5461	0.5561
		Rb-86		1.00				
		Rb-87	4.923E10 y	B-	—	0.1154	—	0.1154
		<i>Sr-87</i>		1.00				
		Rb-88	17.78 m	B-	—	2.0720	0.6370	2.7089
		<i>Sr-88</i>		1.00				
		Rb-89	15.15 m	B-	—	0.9528	2.2430	3.1959
		Sr-89		1.00				
		Rb-90	158 s	B-	—	2.0443	2.0286	4.0729
		Sr-90		1.00				
		Rb-90m	258 s	B-IT	—	1.4104	3.2406	4.6510
		Sr-90		9.7400E-01				
		Rb-90		2.6000E-02				
		Sr-79	2.25 m	ECB+	—	1.8572	1.1801	3.0373
		Rb-79		1.00				
		Sr-80	106.3 m	ECB+	—	0.0418	0.4371	0.4789
		Rb-80		1.00				
		Sr-81	22.3 m	ECB+	—	0.9743	1.3866	2.3609
		Rb-81		9.9856E-01				
		Rb-81m		1.4422E-03				
		Sr-82	25.36 d	EC	—	0.0054	0.0079	0.0132
		Rb-82		1.00				
		Sr-83	32.41 h	ECB+	—	0.1604	0.8213	0.9817
		Rb-83		1.00				
		Sr-85	64.84 d	EC	—	0.0089	0.5001	0.5090
		<i>Rb-85</i>		1.00				
		Sr-85m	67.63 m	ITECB+	—	0.0130	0.2177	0.2307
		Sr-85		8.6600E-01				
		<i>Rb-85</i>		1.3400E-01				
		Sr-87m	2.815 h	ITEC	—	0.0672	0.3202	0.3874
		Rb-87		3.0000E-03				
		<i>Sr-87</i>		9.9700E-01				

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Table A.1 continued

Z	Element	Nuclide	Half-life	Decay mode	Emitted energy (MeV/nt)			
					Alpha	Electron	Photon	Total
39	Yttrium	Sr-89	50.53 d	B- <i>Y-89</i>	–	0.5845	<E-04	0.5846
		Sr-90	28.79 y	B- <i>Y-90</i>	–	0.1957	–	0.1957
		Sr-91	9.63 h	B- <i>Y-91m</i>	–	0.6549	0.7072	1.3621
			<i>Y-91</i>	5.8247E-01 4.1753E-01				
		Sr-92	2.66 h	B- <i>Y-92</i>	–	0.2025	1.3368	1.5393
		Sr-93	7.423 m	B- <i>Y-93</i>	–	0.8094	2.2637	3.0731
		Sr-94	75.3 s	B- <i>Y-94</i>	–	0.8381	1.4270	2.2651
		Y-81	70.4 s	ECB+ <i>Sr-81</i>	–	1.9658	1.1702	3.1360
		Y-83	7.08 m	ECB+ <i>Sr-83</i>	–	1.3138	1.3458	2.6596
		Y-83m	2.85 m	ECB+IT <i>Sr-83</i>	–	0.8177	0.8370	1.6546
			<i>Y-83</i>	6.0000E-01 4.0000E-01				
		Y-84m	39.5 m	ECB+ <i>Sr-84</i>	–	1.2253	3.9751	5.2003
		Y-85	2.68 h	ECB+ <i>Sr-85m</i>	–	0.4881	1.0795	1.5676
		Y-85m	4.86 h	ECB+ <i>Sr-85</i>	–	0.5765	1.3242	1.9007
			<i>Sr-85m</i>	9.6000E-01 3.9998E-02				
		Y-86	14.74 h	ECB+ <i>Sr-86</i>	–	0.2179	3.5777	3.7956
		Y-86m	48 m	ITECB+ <i>Y-86</i>	–	0.0243	0.2203	0.2446
			<i>Sr-86</i>	9.9310E-01 6.9000E-03				
		Y-87	79.8 h	ECB+ <i>Sr-87m</i>	–	0.0072	0.4462	0.4534
		Y-87m	13.37 h	ITECB+ <i>Y-87</i>	–	0.0794	0.3072	0.3866
			<i>Sr-87</i>	9.8430E-01 1.5700E-02				
		Y-88	106.65 d	ECB+ <i>Sr-88</i>	–	0.0067	2.6950	2.7017
		Y-89m	15.663 s	IT <i>Y-89</i>	–	0.0077	0.9014	0.9091
		Y-90	64.10 h	B- <i>Zr-90</i>	–	0.9331	<E-04	0.9331
		Y-90m	3.19 h	ITB- <i>Y-90</i>	–	0.0470	0.6354	0.6823
			<i>Zr-90</i>	9.9998E-01 1.8000E-05				
		Y-91	58.51 d	B- <i>Zr-91</i>	–	0.6032	0.0031	0.6063
			<i>Zr-91</i>	1.00				

Table A.1 continued

Z	Element	Nuclide	Half-life	Decay mode	Emitted energy (MeV/nt)			
					Alpha	Electron	Photon	Total
40	Zirconium	Y-91m	49.71 m	IT	—	0.0279	0.5283	0.5562
		Y-91	1.00	1.00	—	—	—	—
		Y-92	3.54 h	B-	—	1.4494	0.2517	1.7011
		Zr-92	1.00	1.00	—	—	—	—
		Y-93	10.18 h	B-	—	1.1721	0.0961	1.2682
		Zr-93	1.00	1.00	—	—	—	—
		Y-94	18.7 m	B-	—	1.8133	0.7725	2.5858
		Zr-94	1.00	1.00	—	—	—	—
		Y-95	10.3 m	B-	—	1.4288	1.1080	2.5368
		Zr-95	1.00	1.00	—	—	—	—
		Zr-85	7.86 m	ECB+	—	1.3252	1.4718	2.7970
		Y-85m	9.6841E-01	9.6841E-01	—	—	—	—
		Y-85	3.1588E-02	3.1588E-02	—	—	—	—
		Zr-86	16.5 h	ECB+	—	0.0310	0.2952	0.3263
		Y-86	1.00	1.00	—	—	—	—
		Zr-87	1.68 h	ECB+	—	0.8218	0.9271	1.7488
		Y-87m	9.9704E-01	9.9704E-01	—	—	—	—
		Y-87	2.9636E-03	2.9636E-03	—	—	—	—
		Zr-88	83.4 d	EC	—	0.0160	0.3918	0.4078
		Y-88	1.00	1.00	—	—	—	—
		Zr-89	78.41 h	ECB+	—	0.1019	1.1581	1.2600
		Y-89	1.00	1.00	—	—	—	—
		Zr-89m	4.161 m	ITECB+	—	0.0318	0.6344	0.6662
		Zr-89	9.3770E-01	9.3770E-01	—	—	—	—
		Y-89	6.2300E-02	6.2300E-02	—	—	—	—
41	Niobium	Zr-93	1.53E+6 y	B-	—	0.0194	—	0.0194
		Nb-93m	9.7500E-01	9.7500E-01	—	—	—	—
		Nb-93	2.5000E-02	2.5000E-02	—	—	—	—
		Zr-95	64.032 d	B-	—	0.1185	0.7321	0.8506
		Nb-95	9.8920E-01	9.8920E-01	—	—	—	—
		Nb-95m	1.0802E-02	1.0802E-02	—	—	—	—
		Zr-97	16.744 h	B-	—	0.7212	0.8792	1.6004
		Nb-97	1.00	1.00	—	—	—	—
		Nb-87	3.75 m	ECB+	—	1.7709	1.2202	2.9911
		Zr-87	1.00	1.00	—	—	—	—
		Nb-88	14.5 m	ECB+	—	1.4555	4.2188	5.6743
		Zr-88	1.00	1.00	—	—	—	—
		Nb-88m	7.78 m	ECB+	—	1.4589	4.1063	5.5652
		Zr-88	1.00	1.00	—	—	—	—
		Nb-89	2.03 h	ECB+	—	1.0859	1.3668	2.4527
		Zr-89	9.8772E-01	9.8772E-01	—	—	—	—
		Zr-89m	1.2278E-02	1.2278E-02	—	—	—	—
		Nb-89m	66 m	ECB+	—	0.7855	1.3058	2.0913
		Zr-89m	1.00	1.00	—	—	—	—
		Nb-90	14.60 h	ECB+	—	0.4032	4.2135	4.6167
		Zr-90	1.00	1.00	—	—	—	—

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Table A.1 continued

Z	Element	Nuclide	Half-life	Decay mode	Emitted energy (MeV/nt)			
					Alpha	Electron	Photon	Total
42	Molybdenum	Nb-91	680 y	ECB+	–	0.0058	0.0118	0.0176
		<i>Zr-91</i>		1.00				
		Nb-91m	60.86 d	ITECB+	–	0.0963	0.0340	0.1303
		Nb-91		9.6600E-01				
		<i>Zr-91</i>		3.4000E-02				
		Nb-92	3.47E+7 y	EC	–	0.0079	1.5055	1.5133
		<i>Zr-92</i>		1.00				
		Nb-92m	10.15 d	ECB+	–	0.0065	0.9689	0.9753
		<i>Zr-92</i>		1.00				
		Nb-93m	16.13 y	IT	–	0.0294	0.0020	0.0314
		<i>Nb-93</i>		1.00				
		Nb-94	2.03E+4 y	B-	–	0.1684	1.5581	1.7265
		<i>Mo-94</i>		1.00				
		Nb-94m	6.263 m	ITB-	–	0.0356	0.0117	0.0473
		Nb-94		9.9500E-01				
		<i>Mo-94</i>		5.0000E-03				
		Nb-95	34.991 d	B-	–	0.0446	0.7645	0.8091
		<i>Mo-95</i>		1.00				
		Nb-95m	3.61 d	ITB-	–	0.1800	0.0697	0.2497
		Nb-95		9.4400E-01				
		<i>Mo-95</i>		5.6000E-02				
		Nb-96	23.35 h	B-	–	0.2534	2.4614	2.7149
		<i>Mo-96</i>		1.00				
		Nb-97	72.1 m	B-	–	0.4683	0.6650	1.1333
		<i>Mo-97</i>		1.00				
		Nb-98m	51.3 m	B-	–	0.7636	2.8177	3.5813
		<i>Mo-98</i>		1.00				
		Nb-99	15.0 s	B-	–	1.5132	0.1743	1.6875
		<i>Mo-99</i>		1.00				
		Nb-99m	2.6 m	B-IT	–	1.4148	0.7567	2.1714
		<i>Mo-99</i>		9.8000E-01				
		Nb-99		2.0000E-02				
		Mo-89	2.11 m	ECB+	–	1.9620	1.2173	3.1793
		Nb-89		1.00				
		Mo-90	5.56 h	ECB+	–	0.2107	0.8331	1.0438
		Nb-90		1.00				
		Mo-91	15.49 m	ECB+	–	1.4510	0.9773	2.4283
		Nb-91		9.9966E-01				
		Nb-91m		3.4232E-04				
		Mo-91m	64.6 s	ECB+IT	–	0.5559	1.3857	1.9416
		Nb-91m		5.0000E-01				
		<i>Mo-91</i>		5.0000E-01				
		Mo-93	4.0E+3 y	EC	–	0.0056	0.0107	0.0163
		Nb-93m		8.8000E-01				
		<i>Nb-93</i>		1.2000E-01				

Table A.1 continued

Z	Element	Nuclide	Half-life	Decay mode	Emitted energy (MeV/nt)			
					Alpha	Electron	Photon	Total
43	Technetium	Mo-93m	6.85 h	ITEC	—	0.1045	2.3183	2.4228
		Mo-93		9.9880E-01				
		Nb-93		1.2000E-03				
		Mo-99	65.94 h	B-	—	0.3929	0.1484	0.5413
		Tc-99m		8.7730E-01				
		Tc-99		1.2270E-01				
		Mo-101	14.61 m	B-	—	0.5524	1.4706	2.0230
		Tc-101		1.00				
		Mo-102	11.3 m	B-	—	0.3509	0.0185	0.3694
		Tc-102		1.00				
		Tc-91	3.14 m	ECB+	—	1.6985	2.4847	4.1832
		Mo-91		9.9302E-01				
		Mo-91m		6.9789E-03				
		Tc-91m	3.3 m	ECB+	—	1.8871	1.4231	3.3102
		Mo-91m		9.7976E-01				
		Mo-91		2.0245E-02				
		Tc-92	4.25 m	ECB+	—	1.7984	3.8278	5.6263
		Mo-92		1.00				
		Tc-93	2.75 h	ECB+	—	0.0436	1.5686	1.6122
		Mo-93		1.00				
		Tc-93m	43.5 m	ITECB+	—	0.1015	0.9522	1.0537
		Tc-93		7.6600E-01				
		Mo-93		2.3400E-01				
		Tc-94	293 m	ECB+	—	0.0475	2.6608	2.7083
		Mo-94		1.00				
		Tc-94m	52.0 m	ECB+	—	0.7543	1.9567	2.7110
		Mo-94		1.00				
		Tc-95	20.0 h	EC	—	0.0069	0.7965	0.8033
		Mo-95		1.00				
		Tc-95m	61 d	ECB+IT	—	0.0154	0.6887	0.7042
		Tc-95		3.8800E-02				
		Mo-95		9.6120E-01				
		Tc-96	4.28 d	EC	—	0.0089	2.5032	2.5121
		Mo-96		1.00				
		Tc-96m	51.5 m	ITECB+	—	0.0269	0.0480	0.0749
		Tc-96		9.8000E-01				
		Mo-96		2.0000E-02				
		Tc-97	2.6E+6 y	EC	—	0.0055	0.0114	0.0169
		Mo-97		1.00				
		Tc-97m	90.1 d	IT	—	0.0869	0.0096	0.0964
		Tc-97		1.00				
		Tc-98	4.2E+6 y	B-	—	0.1415	1.4127	1.5542
		Ru-98		1.00				
		Tc-99	2.111E+5 y	B-	—	0.1013	<E-04	0.1013
		Ru-99		1.00				

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Table A.1 continued

Z	Element	Nuclide	Half-life	Decay mode	Emitted energy (MeV/nt)			
					Alpha	Electron	Photon	Total
44	Ruthenium	Tc-99m	6.015 h	ITB-	—	0.0162	0.1266	0.1428
		Tc-99		9.9996E-01				
		<i>Ru-99</i>		3.7000E-05				
		Tc-101	14.2 m	B-	—	0.4725	0.3367	0.8092
		<i>Ru-101</i>		1.00				
		Tc-102	5.28 s	B-	—	1.9441	0.0808	2.0249
		<i>Ru-102</i>		1.00				
		Tc-102m	4.35 m	B-IT	—	0.7902	2.4744	3.2646
		Tc-102		2.0000E-02				
		<i>Ru-102</i>		9.8000E-01				
		Tc-104	18.3 m	B-	—	1.6014	2.2453	3.8467
		<i>Ru-104</i>		1.00				
		Tc-105	7.6 m	B-	—	1.2708	0.7981	2.0690
		Ru-105		1.00				
		Ru-92	3.65 m	ECB+	—	0.7939	2.0844	2.8783
		Tc-92		1.00				
		Ru-94	51.8 m	ECB+	—	0.0085	0.5197	0.5282
		Tc-94m		1.00				
		Ru-95	1.643 h	ECB+	—	0.0831	1.2419	1.3249
		Tc-95		9.7387E-01				
		Tc-95m		2.6131E-02				
		Ru-97	2.9 d	EC	—	0.0132	0.2408	0.2540
		Tc-97		9.9958E-01				
		Tc-97m		4.2179E-04				
45	Rhodium	Ru-103	39.26 d	B-	—	0.0660	0.4962	0.5622
		Rh-103m		9.8755E-01				
		<i>Rh-103</i>		1.2453E-02				
		Ru-105	4.44 h	B-	—	0.4406	0.7480	1.1886
		Rh-105		1.00				
		Ru-106	373.59 d	B-	—	0.0100	—	0.0100
		Rh-106		1.00				
		Ru-107	3.75 m	B-	—	1.0704	0.3453	1.4157
		Rh-107		1.00				
		Ru-108	4.55 m	B-	—	0.4803	0.0626	0.5429
		Rh-108		1.00				
		Rh-94	70.6 s	ECB+	—	2.9012	3.7634	6.6645
		Ru-94		1.00				
		Rh-95	5.02 m	ECB+	—	0.8973	2.5603	3.4576
		Ru-95		1.00				
		Rh-95m	1.96 m	ITECB+	—	0.1822	0.8973	1.0794
		Rh-95		8.8000E-01				
		Ru-95		1.2000E-01				
		Rh-96	9.90 m	ECB+	—	0.7419	3.9266	4.6685
		<i>Ru-96</i>		1.00				
		Rh-96m	1.51 m	ITECB+	—	0.5673	1.2819	1.8492
		Rh-96		6.0000E-01				
		<i>Ru-96</i>		4.0000E-01				

Table A.1 continued

Z	Element	Nuclide	Half-life	Decay mode	Emitted energy (MeV/nt)			
					Alpha	Electron	Photon	Total
		Rh-97	30.7 m	ECB+	–	0.5211	1.4448	1.9660
		Ru-97	1.00					
		Rh-97m	46.2 m	ECB+IT	–	0.1992	2.2017	2.4009
		Ru-97	9.4400E-01					
		Rh-97	5.6000E-02					
		Rh-98	8.7 m	ECB+	–	1.3371	1.8115	3.1486
		Ru-98	1.00					
		Rh-99	16.1 d	ECB+	–	0.0611	0.5627	0.6238
		Ru-99	1.00					
		Rh-99m	4.7 h	ECB+	–	0.0367	0.6511	0.6878
		Ru-99	1.00					
		Rh-100	20.8 h	ECB+	–	0.0494	2.7425	2.7920
		Ru-100	1.00					
		Rh-100m	4.6 m	ITECB+	–	0.0802	0.0637	0.1439
		Rh-100	9.8300E-01					
		Ru-100	1.7000E-02					
		Rh-101	3.3 y	EC	–	0.0267	0.2878	0.3145
		Ru-101	1.00					
		Rh-101m	4.34 d	ECIT	–	0.0199	0.2877	0.3076
		Rh-101	6.4000E-02					
		Ru-101	9.3600E-01					
		Rh-102	207 d	ECB+B-	–	0.1717	0.5060	0.6777
		Ru-102	7.8000E-01					
		Pd-102	2.2000E-01					
		Rh-102m	3.742 y	ECB+IT	–	0.0125	2.1536	2.1661
		Rh-102	2.3300E-03					
		Ru-102	9.9767E-01					
		Rh-103m	56.114 m	IT	–	0.0377	0.0017	0.0394
		Rh-103	1.00					
		Rh-104	42.3 s	B-EC	–	0.9823	0.0124	0.9947
		Pd-104	9.9550E-01					
		Ru-104	4.5000E-03					
		Rh-104m	4.34 m	ITB-	–	0.0846	0.0441	0.1287
		Rh-104	9.9870E-01					
		Pd-104	1.3000E-03					
		Rh-105	35.36 h	B-	–	0.1533	0.0773	0.2306
		Pd-105	1.00					
		Rh-106	29.80 s	B-	–	1.4111	0.2061	1.6172
		Pd-106	1.00					
		Rh-106m	131 m	B-	–	0.3492	2.8526	3.2018
		Pd-106	1.00					
		Rh-107	21.7 m	B-	–	0.4407	0.3133	0.7539
		Pd-107	1.00					
		Rh-108	16.8 s	B-	–	1.8209	0.3172	2.1381
		Pd-108	1.00					
		Rh-109	80 s	B-	–	0.9320	0.2995	1.2315
		Pd-109	1.00					

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Table A.1 continued

Z	Element	Nuclide	Half-life	Decay mode	Emitted energy (MeV/nt)			
					Alpha	Electron	Photon	Total
46	Palladium	Pd-96	122 s	ECB+	—	0.2257	1.4403	1.6660
		Rh-96m		1.00				
		Pd-97	3.10 m	ECB+	—	0.7524	2.3813	3.1337
		Rh-97		9.8838E-01				
		Rh-97m		1.1622E-02				
		Pd-98	17.7 m	ECB+	—	0.0455	0.4154	0.4609
		Rh-98		1.00				
		Pd-99	21.4 m	ECB+	—	0.4499	1.2833	1.7331
		Rh-99m		9.6647E-01				
		Rh-99		3.3529E-02				
		Pd-100	3.63 d	EC	—	0.0455	0.1231	0.1686
		Rh-100		1.00				
		Pd-101	8.47 h	ECB+	—	0.0328	0.3518	0.3846
		Rh-101m		9.9730E-01				
		Rh-101		2.7000E-03				
		Pd-103	16.991 d	EC	—	0.0058	0.0146	0.0204
		Rh-103m		9.9875E-01				
		Rh-103		1.2512E-03				
		Pd-107	6.5E+6 y	B-	—	0.0096	—	0.0096
		Ag-107		1.00				
		Pd-109	13.7012 h	B-	—	0.4380	0.0118	0.4497
		Ag-109		1.00				
		Pd-109m	4.69 m	IT	—	0.0777	0.1112	0.1889
		Pd-109		1.00				
		Pd-111	23.4 m	B-	—	0.8409	0.0478	0.8887
		Ag-111m		9.9756E-01				
		Ag-111		2.4368E-03				
		Pd-112	21.03 h	B-	—	0.0900	0.0051	0.0951
		Ag-112		1.00				
		Pd-114	2.42 m	B-	—	0.5317	0.0259	0.5576
		Ag-114		1.00				
47	Silver	Ag-99	124 s	ECB+	—	1.3076	2.3092	3.6168
		Pd-99		1.00				
		Ag-100m	2.24 m	ECB+	—	1.9068	2.8248	4.7316
		Pd-100		1.00				
		Ag-101	11.1 m	ECB+	—	0.8396	1.5703	2.4099
		Pd-101		1.00				
		Ag-102	12.9 m	ECB+	—	0.8430	3.4092	4.2522
		Pd-102		1.00				
		Ag-102m	7.7 m	ECB+IT	—	0.3964	1.9933	2.3897
		Ag-102		4.9000E-01				
		Pd-102		5.1000E-01				
		Ag-103	65.7 m	ECB+	—	0.1973	0.8440	1.0412
		Pd-103		1.00				
		Ag-104	69.2 m	ECB+	—	0.0917	2.7066	2.7982
		Pd-104		1.00				

Table A.1 continued

Z	Element	Nuclide	Half-life	Decay mode	Emitted energy (MeV/nt)			
					Alpha	Electron	Photon	Total
		Ag-104m	33.5 m	ECB+IT	—	0.7331	1.8047	2.5377
		Ag-104		7.0000E-04				
		<i>Pd-104</i>		9.9930E-01				
		Ag-105	41.29 d	EC	—	0.0192	0.5138	0.5330
		<i>Pd-105</i>		1.00				
		Ag-105m	7.23 m	ITECB+	—	0.0252	0.0013	0.0265
		Ag-105		9.9660E-01				
		<i>Pd-105</i>		3.4000E-03				
		Ag-106	23.96 m	ECB+B-	—	0.4967	0.6996	1.1963
		<i>Pd-106</i>		9.9000E-01				
		<i>Cd-106</i>		1.0000E-02				
		Ag-106m	8.28 d	EC	—	0.0131	2.8091	2.8222
		<i>Pd-106</i>		1.00				
		Ag-108	2.37 m	B-ECB+	—	0.6071	0.0186	0.6256
		<i>Cd-108</i>		9.7150E-01				
		<i>Pd-108</i>		2.8500E-02				
		Ag-108m	418 y	ECIT	—	0.0159	1.6209	1.6368
		Ag-108		8.7000E-02				
		<i>Pd-108</i>		9.1300E-01				
		Ag-109m	39.6 s	IT	—	0.0770	0.0111	0.0880
		<i>Ag-109</i>		1.00				
		Ag-110	24.6 s	B-EC	—	1.1812	0.0307	1.2119
		<i>Cd-110</i>		9.9700E-01				
		<i>Pd-110</i>		3.0000E-03				
		Ag-110m	249.76 d	B-IT	—	0.0758	2.7606	2.8363
		Ag-110		1.3600E-02				
		<i>Cd-110</i>		9.8640E-01				
		Ag-111	7.45 d	B-	—	0.3539	0.0265	0.3804
		<i>Cd-111</i>		1.00				
		Ag-111m	64.8 s	ITB-	—	0.0568	0.0079	0.0646
		Ag-111		9.9300E-01				
		<i>Cd-111</i>		7.0000E-03				
		Ag-112	3.130 h	B-	—	1.3540	0.6905	2.0445
		<i>Cd-112</i>		1.00				
		Ag-113	5.37 h	B-	—	0.7614	0.0719	0.8333
		Cd-113		9.8261E-01				
		Cd-113m		1.7388E-02				
		Ag-113m	68.7 s	ITB-	—	0.2278	0.2134	0.4412
		Ag-113		6.4000E-01				
		Cd-113		3.6000E-01				
		Ag-114	4.6 s	B-	—	2.1091	0.2589	2.3681
		<i>Cd-114</i>		1.00				
		Ag-115	20.0 m	B-	—	1.0928	0.4831	1.5760
		Cd-115		9.4210E-01				
		Cd-115m		5.7870E-02				
		Ag-116	2.68 m	B-	—	1.7540	2.1473	3.9014
		<i>Cd-116</i>		1.00				

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Table A.1 continued

Z	Element	Nuclide	Half-life	Decay mode	Emitted energy (MeV/nt)			
					Alpha	Electron	Photon	Total
48	Cadmium	Ag-117	73.6 s	B-	–	1.2813	1.3028	2.5841
		Cd-117		8.4703E-01				
		Cd-117m		1.5297E-01				
		Cd-101	1.36 m	ECB+	–	1.0685	2.4853	3.5538
		Ag-101		1.00				
		Cd-102	5.5 m	ECB+	–	0.0756	0.8380	0.9136
		Ag-102m		9.4624E-01				
		Ag-102		5.3762E-02				
		Cd-103	7.3 m	ECB+	–	0.3636	2.0891	2.4527
		Ag-103		1.00				
		Cd-104	57.7 m	EC	–	0.0306	0.2513	0.2819
		Ag-104m		1.00				
		Cd-105	55.5 m	ECB+	–	0.2167	1.2995	1.5161
		Ag-105m		8.2963E-01				
		Ag-105		1.7037E-01				
		Cd-107	6.50 h	ECB+	–	0.0870	0.0337	0.1207
		Ag-107		1.00				
		Cd-109	461.4 d	EC	–	0.0827	0.0265	0.1092
		Ag-109		1.00				
		Cd-111m	48.50 m	IT	–	0.1066	0.2843	0.3908
		Cd-111		1.00				
		Cd-113	7.7E+15 y	B-	–	0.0926	–	0.0926
		In-113		1.00				
		Cd-113m	14.1 y	B-IT	–	0.1847	<E-04	0.1848
		Cd-113		1.4000E-03				
		In-113		9.9860E-01				
		Cd-115	53.46 h	B-	–	0.3182	0.1926	0.5107
		In-115m		1.00				
		Cd-115m	44.6 d	B-	–	0.6045	0.0329	0.6374
		In-115		9.9989E-01				
		In-115m		1.0578E-04				
		Cd-117	2.49 h	B-	–	0.4379	1.0796	1.5175
		In-117m		9.1507E-01				
		In-117		8.4933E-02				
		Cd-117m	3.36 h	B-	–	0.2279	2.0437	2.2716
		In-117		9.9002E-01				
		In-117m		9.9831E-03				
		Cd-118	50.3 m	B-	–	0.1614	–	0.1614
		In-118		1.00				
		Cd-119	2.69 m	B-	–	0.8186	1.6380	2.4566
		In-119m		9.0091E-01				
		In-119		9.9092E-02				
		Cd-119m	2.20 m	B-	–	0.6898	2.3021	2.9919
		In-119		9.9787E-01				
		In-119m		2.1328E-03				

Table A.1 continued

Z	Element	Nuclide	Half-life	Decay mode	Emitted energy (MeV/nt)			
					Alpha	Electron	Photon	Total
49	Indium	In-103	60 s	ECB+	—	1.5404	2.7528	4.2932
		Cd-103		1.00				
		In-105	5.07 m	ECB+	—	1.0358	1.9307	2.9666
		Cd-105		1.00				
		In-106m	5.2 m	ECB+	—	1.5889	2.8243	4.4132
		Cd-106		1.00				
		In-106	6.2 m	ECB+	—	1.0844	3.5541	4.6385
		Cd-106		1.00				
		In-107	32.4 m	ECB+	—	0.3263	1.5307	1.8570
		Cd-107		1.00				
		In-108m	39.6 m	ECB+	—	0.7021	2.7655	3.4675
		Cd-108		1.00				
		In-108	58.0 m	ECB+	—	0.1620	3.9164	4.0784
		Cd-108		1.00				
		In-109m	1.34 m	IT	—	0.0416	0.6085	0.6501
		In-109	4.2 h	ECB+	—	0.0334	0.6441	0.6775
		Cd-109		1.00				
		In-110m	69.1 m	ECB+	—	0.6282	1.5782	2.2065
		Cd-110		1.00				
		In-110	4.9 h	ECB+	—	0.0122	3.0972	3.1094
		Cd-110		1.00				
		In-111m	7.7 m	IT	—	0.0675	0.4706	0.5380
		In-111	2.8047 d	EC	—	0.0348	0.4061	0.4409
		Cd-111m		4.9998E-05				
		Cd-111		9.9995E-01				
		In-112m	20.56 m	IT	—	0.1217	0.0347	0.1564
		In-112	14.97 m	ECB+B-	—	0.2452	0.2675	0.5127
		Cd-112		5.6000E-01				
		Sn-112		4.4000E-01				
		In-113m	1.6579 h	IT	—	0.1361	0.2606	0.3967
		In-113		1.00				
		In-114m	49.51 d	ITEC	—	0.1450	0.0804	0.2254
		In-114		9.6750E-01				
		Cd-114		3.2500E-02				
		In-114	71.9 s	B-ECB+	—	0.7740	0.0023	0.7764
		Sn-114		9.9500E-01				
		Cd-114		5.0000E-03				
		In-115m	4.486 h	ITB-	—	0.1748	0.1627	0.3376
		In-115		9.5000E-01				
		Sn-115		5.0000E-02				
		In-115	4.41E+14 y	B-	—	0.1526	—	0.1526
		Sn-115		1.00				
		In-116m	54.41 m	B-	—	0.3128	2.4691	2.7819
		Sn-116		1.00				

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Table A.1 continued

Z	Element	Nuclide	Half-life	Decay mode	Emitted energy (MeV/nt)			
					Alpha	Electron	Photon	Total
50	Tin	In-117m	116.2 m	B-IT	–	0.4344	0.0910	0.5254
		In-117	4.7100E-01					
		<i>Sn-117</i>	5.2900E-01					
		In-117	43.2 m	B-	–	0.2673	0.6939	0.9612
		Sn-117m	3.5321E-03					
		<i>Sn-117</i>	9.9647E-01					
		In-118m	4.364 m	B-	–	0.6708	2.7765	3.4473
		<i>Sn-118</i>	1.00					
		In-118	5.0 s	B-	–	1.8792	0.0778	1.9570
		<i>Sn-118</i>	1.00					
		In-119m	18.0 m	B-IT	–	1.0221	0.0660	1.0882
		In-119	5.6000E-02					
		<i>Sn-119</i>	9.4400E-01					
		In-119	2.4 m	B-	–	0.6148	0.7701	1.3849
		Sn-119m	9.4773E-03					
		<i>Sn-119</i>	9.9052E-01					
		In-121m	3.88 m	B-IT	–	1.5288	0.0635	1.5922
		Sn-121	9.8800E-01					
		In-121	1.2000E-02					
		In-121	23.1 s	B-	–	0.9869	0.9266	1.9135
		Sn-121	8.8650E-01					
		Sn-121m	1.1350E-01					
		Sn-106	1.92 m	ECB+	–	0.1279	1.2091	1.3370
		In-106m	1.00					
		Sn-108	10.30 m	ECB+	–	0.0269	0.6847	0.7117
		In-108m	1.00					
		Sn-109	18.0 m	ECB+	–	0.0571	2.2063	2.2634
		In-109	7.1734E-01					
		In-109m	2.8266E-01					
		Sn-110	4.11 h	EC	–	0.0152	0.2918	0.3070
		In-110m	1.00					
		Sn-111	35.3 m	ECB+	–	0.1915	0.4903	0.6818
		In-111	9.9793E-01					
		In-111m	2.0745E-03					
		Sn-113m	21.4 m	ITEC	–	0.0588	0.0137	0.0725
		Sn-113	9.1100E-01					
		<i>In-113</i>	8.9000E-02					
		Sn-113	115.09 d	EC	–	0.0063	0.0237	0.0300
		In-113m	9.9998E-01					
		<i>In-113</i>	2.2352E-05					
		Sn-117m	13.76 d	IT	–	0.1616	0.1581	0.3197
		<i>Sn-117</i>	1.00					
		Sn-119m	293.1 d	IT	–	0.0781	0.0151	0.0932
		<i>Sn-119</i>	1.00					
		Sn-121m	43.9 y	ITB-	–	0.0354	0.0052	0.0405
		Sn-121	7.7600E-01					
		<i>Sb-121</i>	2.2400E-01					

Table A.1 continued

Z	Element	Nuclide	Half-life	Decay mode	Emitted energy (MeV/nt)			
					Alpha	Electron	Photon	Total
51	Antimony	Sn-121	27.03 h	B- <i>Sb-121</i>	—	0.1156	—	0.1156
		Sn-123m	40.06 m	B- <i>Sb-123</i>	—	0.4788	0.1407	0.6196
		Sn-123	129.2 d	B- <i>Sb-123</i>	—	0.5227	0.0069	0.5296
		Sn-125m	9.52 m	B- <i>Sb-125</i>	—	0.8070	0.3464	1.1534
		Sn-125	9.64 d	B- <i>Sb-125</i>	—	0.8037	0.3346	1.1383
		Sn-126	2.30E+5 y	B- <i>Sb-126m</i>	—	0.1380	0.0569	0.1949
		Sn-127m	4.13 m	B- <i>Sb-127</i>	—	1.1140	0.5686	1.6826
		Sn-127	2.10 h	B- <i>Sb-127</i>	—	0.5199	1.9075	2.4274
		Sn-128	59.07 m	B- <i>Sb-128m</i>	—	0.2457	0.6041	0.8498
		Sn-129	2.23 m	B- <i>Sb-129</i>	—	1.2576	1.0081	2.2657
		Sn-130m	1.7 m	B- <i>Sb-130</i>	—	1.4040	0.8861	2.2901
				<i>Sb-130m</i>				
		Sn-130	3.72 m	B- <i>Sb-130m</i>	—	0.4612	0.9372	1.3984
				<i>Sb-130m</i>				
		Sb-111	75 s	ECB+ <i>Sn-111</i>	—	1.3640	1.4859	2.8499
				<i>Sn-111</i>				
		Sb-113	6.67 m	ECB+ <i>Sn-113</i>	—	0.7203	1.2683	1.9886
				<i>Sn-113m</i>				
		Sb-114	3.49 m	ECB+ <i>Sn-114</i>	—	1.2184	2.6893	3.9077
				<i>Sn-114</i>				
		Sb-115	32.1 m	ECB+ <i>Sn-115</i>	—	0.2342	0.8894	1.1236
				<i>Sn-115</i>				
		Sb-116	15.8 m	ECB+ <i>Sn-116</i>	—	0.5138	2.2788	2.7926
				<i>Sn-116</i>				
		Sb-116m	60.3 m	ECB+ <i>Sn-116</i>	—	0.1408	3.1029	3.2437
				<i>Sn-116</i>				
		Sb-117	2.80 h	ECB+ <i>Sn-117</i>	—	0.0298	0.1864	0.2162
				<i>Sn-117</i>				
		Sb-118	3.6 m	ECB+ <i>Sn-118</i>	—	0.8730	0.8039	1.6769
				<i>Sn-118</i>				
		Sb-118m	5.00 h	ECB+ <i>Sn-118</i>	—	0.0374	2.6130	2.6504
				<i>Sn-118</i>				
		Sb-119	38.19 h	EC <i>Sn-119</i>	—	0.0258	0.0234	0.0492
				<i>Sn-119</i>				

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Table A.1 continued

Z	Element	Nuclide	Half-life	Decay mode	Emitted energy (MeV/nt)			
					Alpha	Electron	Photon	Total
		Sb-120	15.89 m	ECB+	–	0.3077	0.4521	0.7599
		<i>Sn-120</i>		1.00				
		Sb-120m	5.76 d	EC	–	0.0449	2.4663	2.5112
		<i>Sn-120</i>		1.00				
		Sb-122	2.7238 d	B-ECB+	–	0.5618	0.4453	1.0072
		<i>Te-122</i>		9.7590E-01				
		<i>Sn-122</i>		2.4100E-02				
		Sb-122m	4.191 m	IT	–	0.0937	0.0707	0.1644
		Sb-122		1.00				
		Sb-124	60.20 d	B-	–	0.3831	1.8531	2.2362
		<i>Te-124</i>		1.00				
		Sb-124m	93 s	ITB-	–	0.1157	0.4401	0.5558
		Sb-124		7.5000E-01				
		<i>Te-124</i>		2.5000E-01				
		Sb-124n	20.2 m	IT	–	0.0256	0.0004	0.0260
		Sb-124m		1.00				
		Sb-125	2.75856 y	B-	–	0.1010	0.4373	0.5383
		<i>Te-125m</i>		2.3136E-01				
		<i>Te-125</i>		7.6864E-01				
		Sb-126	12.35 d	B-	–	0.3545	2.7552	3.1097
		<i>Te-126</i>		1.00				
		Sb-126m	19.15 m	B-IT	–	0.6322	1.5483	2.1805
		Sb-126		1.4000E-01				
		<i>Te-126</i>		8.6000E-01				
		Sb-127	3.85 d	B-	–	0.3160	0.6934	1.0094
		<i>Te-127</i>		8.2320E-01				
		<i>Te-127m</i>		1.7680E-01				
		Sb-128	9.01 h	B-	–	0.4999	3.0934	3.5934
		<i>Te-128</i>		1.00				
		Sb-128m	10.4 m	B-IT	–	0.9580	1.9063	2.8643
		Sb-128		3.6000E-02				
		<i>Te-128</i>		9.6400E-01				
		Sb-129	4.40 h	B-	–	0.3953	1.4601	1.8553
		<i>Te-129</i>		7.7381E-01				
		<i>Te-129m</i>		2.2619E-01				
		Sb-130	39.5 m	B-	–	0.7579	3.2724	4.0303
		<i>Te-130</i>		1.00				
		Sb-130m	6.3 m	B-	–	1.0290	2.7077	3.7367
		<i>Te-130</i>		1.00				
		Sb-131	23.03 m	B-	–	0.5867	2.0733	2.6601
		<i>Te-131</i>		9.1793E-01				
		<i>Te-131m</i>		8.2067E-02				
		Sb-133	2.5 m	B-	–	0.6785	2.7434	3.4219
		<i>Te-133</i>		8.2662E-01				
		<i>Te-133m</i>		1.7338E-01				

Table A.1 continued

Z	Element	Nuclide	Half-life	Decay mode	Emitted energy (MeV/nt)			
					Alpha	Electron	Photon	Total
52	Tellurium	Te-113	1.7 m	ECB+	—	1.7023	2.2208	3.9231
		Sb-113	1.00					
		Te-114	15.2 m	ECB+	—	0.1551	1.2817	1.4368
		Sb-114	1.00					
		Te-115	5.8 m	ECB+	—	0.8124	2.2449	3.0572
		Sb-115	1.00					
		Te-115m	6.7 m	ECB+	—	0.6939	2.6041	3.2980
		Sb-115	1.00					
		Te-116	2.49 h	ECB+	—	0.0618	0.1122	0.1740
		Sb-116	1.00					
		Te-117	62 m	ECB+	—	0.2139	1.5492	1.7631
		Sb-117	1.00					
		Te-118	6.00 d	EC	—	0.0061	0.0199	0.0260
		Sb-118	1.00					
		Te-119	16.05 h	ECB+	—	0.0143	0.7679	0.7822
		Sb-119	1.00					
		Te-119m	4.70 d	ECB+	—	0.0180	1.5059	1.5239
		Sb-119	1.00					
		Te-121	19.16 d	EC	—	0.0098	0.5775	0.5872
		Sb-121	1.00					
		Te-121m	154 d	ITEC	—	0.0817	0.2176	0.2993
		Te-121		8.8600E-01				
		Sb-121		1.1400E-01				
		Te-123	6.00E+14 y	EC	—	0.0028	0.0004	0.0031
		Sb-123		1.00				
		Te-123m	119.25 d	IT	—	0.0990	0.1477	0.2467
		Te-123		1.00				
		Te-125m	57.40 d	IT	—	0.1091	0.0360	0.1451
		Te-125		1.00				
		Te-127	9.35 h	B-	—	0.2246	0.0049	0.2294
		I-127		1.00				
		Te-127m	109 d	ITB-	—	0.0824	0.0113	0.0937
		Te-127		9.7600E-01				
		I-127		2.4000E-02				
		Te-129	69.6 m	B-	—	0.5436	0.0625	0.6061
		I-129		1.00				
		Te-129m	33.6 d	ITB-	—	0.2709	0.0376	0.3085
		Te-129		6.3000E-01				
		I-129		3.7000E-01				
		Te-131	25.0 m	B-	—	0.7122	0.4200	1.1322
		I-131		1.00				
		Te-131m	30 h	B-IT	—	0.1870	1.4545	1.6415
		I-131		7.7800E-01				
		Te-131		2.2200E-01				
		Te-132	3.204 d	B-	—	0.1108	0.2344	0.3453
		I-132		1.00				

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Table A.1 continued

Z	Element	Nuclide	Half-life	Decay mode	Emitted energy (MeV/nt)			
					Alpha	Electron	Photon	Total
53	Iodine	Te-133	12.5 m	B-	–	0.6897	1.2007	1.8904
		I-133		1.00				
		Te-133m	55.4 m	B-IT	–	0.3880	1.8597	2.2477
		I-133		8.2500E-01				
		Te-133		1.7500E-01				
		Te-134	41.8 m	B-	–	0.2266	0.8714	1.0980
		I-134		1.00				
		I-118	13.7 m	ECB+	–	1.9645	2.0163	3.9809
		Te-118		1.00				
		I-118m	8.5 m	ECB+	–	1.1055	3.7513	4.8567
		Te-118		1.00				
		I-119	19.1 m	ECB+	–	0.5116	0.9110	1.4227
		Te-119		9.9046E-01				
		Te-119m		9.5421E-03				
		I-120	81.6 m	ECB+	–	1.1680	2.6611	3.8291
		Te-120		1.00				
		I-120m	53 m	ECB+	–	0.9070	3.5248	4.4318
		Te-120		1.00				
		I-121	2.12 h	ECB+	–	0.0665	0.3995	0.4660
		Te-121		9.9714E-01				
		Te-121m		2.8631E-03				
		I-122	3.63 m	ECB+	–	1.1055	0.9619	2.0674
		Te-122		1.00				
		I-123	13.27 h	EC	–	0.0282	0.1730	0.2012
		Te-123		9.9996E-01				
		Te-123m		4.4420E-05				
		I-124	4.1760 d	ECB+	–	0.1943	1.1132	1.3075
		Te-124		1.00				
		I-125	59.400 d	EC	–	0.0192	0.0428	0.0621
		Te-125		1.00				
		I-126	12.93 d	ECB+B-	–	0.1606	0.4354	0.5959
		Te-126		5.2700E-01				
		Xe-126		4.7300E-01				
		I-128	24.99 m	B-ECB+	–	0.7463	0.0676	0.8139
		Xe-128		9.3100E-01				
		Te-128		6.9000E-02				
		I-129	1.57E+7 y	B-	–	0.0651	0.0252	0.0902
		Xe-129		1.00				
		I-130	12.36 h	B-	–	0.2786	2.1372	2.4158
		Xe-130		1.00				
		I-130m	8.84 m	ITB-	–	0.1778	0.1097	0.2875
		I-130		8.4000E-01				
		Xe-130		1.6000E-01				
		I-131	8.02070 d	B-	–	0.1918	0.3828	0.5746
		Xe-131m		1.1759E-02				
		Xe-131		9.8824E-01				
		I-132	2.295 h	B-	–	0.4930	2.2645	2.7575
		Xe-132		1.00				

Table A.1 continued

Z	Element	Nuclide	Half-life	Decay mode	Emitted energy (MeV/nt)			
					Alpha	Electron	Photon	Total
54	Xenon	I-132m	1.387 h	ITB-	—	0.1614	0.3403	0.5018
		I-132		8.6000E-01				
		<i>Xe-132</i>		1.4000E-01				
		I-133	20.8 h	B-	—	0.4142	0.6120	1.0262
		Xe-133		9.7115E-01				
		Xe-133m		2.8846E-02				
		I-134	52.5 m	B-	—	0.5776	2.5953	3.1729
		<i>Xe-134</i>		1.00				
		I-134m	3.60 m	ITB-	—	0.0913	0.2887	0.3800
		I-134		9.7700E-01				
		<i>Xe-134</i>		2.3000E-02				
		I-135	6.57 h	B-	—	0.3465	1.5815	1.9280
		Xe-135		8.3432E-01				
		Xe-135m		1.6568E-01				
		Xe-120	40 m	ECB+	—	0.0461	0.4015	0.4476
		I-120		1.00				
		Xe-121	40.1 m	ECB+	—	0.5819	1.4720	2.0539
		I-121		1.00				
		Xe-122	20.1 h	EC	—	0.0100	0.0684	0.0784
		I-122		1.00				
		Xe-123	2.08 h	ECB+	—	0.1875	0.6411	0.8286
		I-123		1.00				
		Xe-125	16.9 h	ECB+	—	0.0350	0.2723	0.3074
		I-125		1.00				
		Xe-127	36.4 d	EC	—	0.0325	0.2806	0.3131
		<i>I-127</i>		1.00				
		Xe-127m	69.2 s	IT	—	0.1293	0.1685	0.2978
		Xe-127		1.00				
		Xe-129m	8.88 d	IT	—	0.1844	0.0517	0.2362
		<i>Xe-129</i>		1.00				
		Xe-131m	11.84 d	IT	—	0.1470	0.0206	0.1676
		<i>Xe-131</i>		1.00				
		Xe-133	5.243 d	B-	—	0.1379	0.0474	0.1854
		<i>Cs-133</i>		1.00				
		Xe-133m	2.19 d	IT	—	0.1924	0.0410	0.2333
		Xe-133		1.00				
		Xe-135	9.14 h	B-	—	0.3208	0.2483	0.5691
		Cs-135		1.00				
		Xe-135m	15.29 m	ITB-	—	0.1008	0.4249	0.5257
		Xe-135		9.9400E-01				
		Cs-135		6.0000E-03				
		Xe-137	3.818 m	B-	—	1.6952	0.1908	1.8859
		Cs-137		1.00				
		Xe-138	14.08 m	B-	—	0.6596	1.1222	1.7818
		Cs-138		1.00				

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Table A.1 continued

Z	Element	Nuclide	Half-life	Decay mode	Emitted energy (MeV/nt)			
					Alpha	Electron	Photon	Total
55	Caesium	Cs-121	155 s	ECB+	–	1.7391	1.1733	2.9123
		Xe-121	1.00					
		Cs-121m	122 s	ECB+IT	–	1.3475	1.1821	2.5296
		Xe-121	8.3000E-01					
		Cs-121	1.7000E-01					
		Cs-123	5.88 m	ECB+	–	0.9509	1.0860	2.0369
		Xe-123	1.00					
		Cs-124	30.8 s	ECB+	–	1.9908	1.1607	3.1515
		Xe-124	1.00					
		Cs-125	45 m	ECB+	–	0.3524	0.7550	1.1075
		Xe-125	1.00					
		Cs-126	1.64 m	ECB+	–	1.3193	1.1545	2.4738
		Xe-126	1.00					
		Cs-127	6.25 h	ECB+	–	0.0293	0.4334	0.4627
		Xe-127	1.00					
		Cs-128	3.640 m	ECB+	–	0.8735	0.8917	1.7651
		Xe-128	1.00					
		Cs-129	32.06 h	ECB+	–	0.0175	0.2802	0.2977
		Xe-129	1.00					
		Cs-130	29.21 m	ECB+B-	–	0.3869	0.5026	0.8894
		Xe-130	9.8400E-01					
		Ba-130	1.6000E-02					
		Cs-130m	3.46 m	ITEC	–	0.0937	0.0737	0.1674
		Cs-130	9.9840E-01					
		Xe-130	1.6000E-03					
		Cs-131	9.689 d	EC	–	0.0064	0.0232	0.0295
		Xe-131	1.00					
		Cs-132	6.479 d	ECB+B-	–	0.0143	0.7151	0.7294
		Xe-132	9.8130E-01					
		Ba-132	1.8700E-02					
		Cs-134	2.0648 y	B-EC	–	0.1639	1.5551	1.7190
		Ba-134	1.00					
		Xe-134	3.0000E-06					
		Cs-134m	2.903 h	IT	–	0.1122	0.0273	0.1394
		Cs-134	1.00					
		Cs-135	2.3E+6 y	B-	–	0.0894	–	0.0894
		Ba-135	1.00					
		Cs-135m	53 m	IT	–	0.0361	1.5972	1.6333
		Cs-135	1.00					
		Cs-136	13.16 d	B-	–	0.1449	2.1283	2.2732
		Ba-136	1.00					
		Cs-137	30.1671 y	B-	–	0.1884	<E-04	0.1884
		Ba-137m	9.4399E-01					
		Ba-137	5.6005E-02					
		Cs-138	33.41 m	B-	–	1.2462	2.3611	3.6073
		Ba-138	1.00					

Table A.1 continued

Z	Element	Nuclide	Half-life	Decay mode	Emitted energy (MeV/nt)			
					Alpha	Electron	Photon	Total
56	Barium	Cs-138m	2.91 m	ITB-	—	0.3201	0.4149	0.7350
		Cs-138		8.1000E-01				
		<i>Ba-138</i>		1.9000E-01				
		Cs-139	9.27 m	B-	—	1.6598	0.3030	1.9629
		Ba-139		1.00				
		Cs-140	63.7 s	B-	—	1.9361	1.7692	3.7054
		Ba-140		1.00				
		Ba-124	11.0 m	ECB+	—	0.1743	0.5728	0.7471
		Cs-124		1.00				
		Ba-126	100 m	ECB+	—	0.0178	0.5803	0.5981
		Cs-126		1.00				
		Ba-127	12.7 m	ECB+	—	0.5971	0.7282	1.3253
		Cs-127		1.00				
		Ba-128	2.43 d	EC	—	0.0086	0.0665	0.0751
		Cs-128		1.00				
		Ba-129	2.23 h	ECB+	—	0.1270	0.3331	0.4601
		Cs-129		1.00				
		Ba-129m	2.16 h	ECB+	—	0.0419	1.5833	1.6252
		Cs-129		1.00				
		Ba-131	11.50 d	EC	—	0.0455	0.4763	0.5219
		Cs-131		1.00				
		Ba-131m	14.6 m	IT	—	0.1102	0.0773	0.1875
		Ba-131		1.00				
		Ba-133	10.52 y	EC	—	0.0553	0.4030	0.4583
		<i>Cs-133</i>		1.00				
		Ba-133m	38.9 h	ITEC	—	0.2259	0.0686	0.2945
		Ba-133		9.9990E-01				
		<i>Cs-133</i>		9.6000E-05				
		Ba-135m	28.7 h	IT	—	0.2080	0.0602	0.2682
		<i>Ba-135</i>		1.00				
		Ba-137m	2.552 m	IT	—	0.0653	0.5963	0.6617
		<i>Ba-137</i>		1.00				
		Ba-139	83.06 m	B-	—	0.9012	0.0457	0.9470
		<i>La-139</i>		1.00				
		Ba-140	12.752 d	B-	—	0.3202	0.1826	0.5029
		La-140		1.00				
		Ba-141	18.27 m	B-	—	0.9624	0.9271	1.8895
		La-141		1.00				
		Ba-142	10.6 m	B-	—	0.4141	1.0469	1.4610
		La-142		1.00				
57	Lanthanum	La-128	5.18 m	ECB+	—	1.3340	2.8283	4.1623
		Ba-128		1.00				
		La-129	11.6 m	ECB+	—	0.6135	0.9212	1.5347
		Ba-129		9.2384E-01				
		Ba-129m		7.6164E-02				
		La-130	8.7 m	ECB+	—	1.1113	2.2324	3.3438
		<i>Ba-130</i>		1.00				

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Table A.1 continued

Z	Element	Nuclide	Half-life	Decay mode	Emitted energy (MeV/nt)			
					Alpha	Electron	Photon	Total
58	Cerium	La-131	59 m	ECB+	–	0.1963	0.6633	0.8596
		Ba-131		1.00				
		La-132	4.8 h	ECB+	–	0.5693	1.9952	2.5645
		Ba-132		1.00				
		La-132m	24.3 m	ITECB+	–	0.1126	0.6706	0.7832
		La-132		7.6000E-01				
		Ba-132		2.4000E-01				
		La-133	3.912 h	ECB+	–	0.0503	0.1605	0.2108
		Ba-133		1.00				
		La-134	6.45 m	ECB+	–	0.7647	0.7196	1.4844
		Ba-134		1.00				
		La-135	19.5 h	ECB+	–	0.0067	0.0361	0.0428
		Ba-135		1.00				
		La-136	9.87 m	ECB+	–	0.2941	0.4070	0.7011
		Ba-136		1.00				
		La-137	6.0E+4 y	EC	–	0.0065	0.0250	0.0315
		Ba-137		1.00				
		La-138	1.02E+11 y	ECB-	–	0.0377	1.2316	1.2693
		Ba-138		6.6400E-01				
		Ce-138		3.3600E-01				
		La-140	1.6781 d	B-	–	0.5346	2.3084	2.8429
		Ce-140		1.00				
		La-141	3.92 h	B-	–	0.9876	0.0268	1.0144
		Ce-141		1.00				
		La-142	91.1 m	B-	–	0.8697	2.3738	3.2435
		Ce-142		1.00				
		La-143	14.2 m	B-	–	1.2995	0.2631	1.5625
		Ce-143		1.00				
		Ce-130	22.9 m	ECB+	–	0.0738	0.5003	0.5741
		La-130		1.00				
		Ce-131	10.2 m	ECB+	–	0.6153	1.6267	2.2420
		La-131		1.00				
		Ce-132	3.51 h	EC	–	0.0180	0.2727	0.2908
		La-132		1.00				
		Ce-133	97 m	ECB+	–	0.3817	0.5430	0.9247
		La-133		1.00				
		Ce-133m	4.9 h	ECB+	–	0.0739	1.7373	1.8112
		La-133		1.00				
		Ce-134	3.16 d	EC	–	0.0072	0.0281	0.0353
		La-134		1.00				
		Ce-135	17.7 h	ECB+	–	0.0296	0.8237	0.8533
		La-135		1.00				
		Ce-137	9.0 h	ECB+	–	0.0165	0.0388	0.0553
		La-137		1.00				
		Ce-137m	34.4 h	ITEC	–	0.2067	0.0558	0.2625
		Ce-137		9.9220E-01				
		La-137		7.8000E-03				

Table A.1 continued

Z	Element	Nuclide	Half-life	Decay mode	Emitted energy (MeV/nt)			
					Alpha	Electron	Photon	Total
59	Praseodymium	Ce-139	137.641 d	EC	—	0.0355	0.1599	0.1954
		<i>La-139</i>		1.00				
		Ce-141	32.508 d	B-	—	0.1710	0.0768	0.2478
		<i>Pr-141</i>		1.00				
		Ce-143	33.039 h	B-	—	0.4364	0.2796	0.7159
		Pr-143		1.00				
		Ce-144	284.91 d	B-	—	0.0916	0.0194	0.1110
		Pr-144		9.9023E-01				
		Pr-144m		9.7699E-03				
		Ce-145	3.01 m	B-	—	0.6794	0.8142	1.4937
		Pr-145		1.00				
		Pr-134	11 m	ECB+	—	1.0801	3.1504	4.2305
		Ce-134		1.00				
		Pr-134m	17 m	ECB+	—	1.5274	2.3101	3.8375
		Ce-134		1.00				
		Pr-135	24 m	ECB+	—	0.5793	0.8741	1.4534
		Ce-135		1.00				
		Pr-136	13.1 m	ECB+	—	0.7545	2.1458	2.9003
		<i>Ce-136</i>		1.00				
		Pr-137	1.28 h	ECB+	—	0.1947	0.3693	0.5640
		Ce-137		1.00				
		Pr-138	1.45 m	ECB+	—	1.1617	0.8153	1.9770
		<i>Ce-138</i>		1.00				
		Pr-138m	2.12 h	ECB+	—	0.2208	2.4776	2.6984
		<i>Ce-138</i>		1.00				
		Pr-139	4.41 h	ECB+	—	0.0478	0.1298	0.1777
		Ce-139		1.00				
		Pr-140	3.39 m	ECB+	—	0.5516	0.5467	1.0983
		<i>Ce-140</i>		1.00				
		Pr-142	19.12 h	B-EC	—	0.8098	0.0581	0.8679
		<i>Nd-142</i>		9.9984E-01				
		<i>Ce-142</i>		1.6400E-04				
		Pr-142m	14.6 m	IT	—	0.0036	0.0001	0.0037
		Pr-142		1.00				
		Pr-143	13.57 d	B-	—	0.3150	<E-04	0.3150
		<i>Nd-143</i>		1.00				
		Pr-144	17.28 m	B-	—	1.2084	0.0289	1.2373
		Nd-144		1.00				
		Pr-144m	7.2 m	ITB-	—	0.0475	0.0134	0.0608
		Pr-144		9.9930E-01				
		Nd-144		7.0000E-04				
		Pr-145	5.984 h	B-	—	0.6757	0.0186	0.6943
		<i>Nd-145</i>		1.00				
		Pr-146	24.15 m	B-	—	1.3277	1.0105	2.3382
		<i>Nd-146</i>		1.00				
		Pr-147	13.4 m	B-	—	0.8896	0.4887	1.3784
		Nd-147		1.00				

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Table A.1 continued

Z	Element	Nuclide	Half-life	Decay mode	Emitted energy (MeV/nt)			
					Alpha	Electron	Photon	Total
60	Neodymium	Pr-148	2.29 m	B-	—	1.6637	0.9905	2.6542
		<i>Nd-148</i>		1.00				
		Pr-148m	2.01 m	B-	—	1.6807	0.9337	2.6144
		<i>Nd-148</i>		1.00				
		Nd-134	8.5 m	ECB+	—	0.1728	0.5421	0.7149
		Pr-134m		1.00				
		Nd-135	12.4 m	ECB+	—	1.0551	1.2617	2.3168
		Pr-135		1.00				
		Nd-136	50.65 m	ECB+	—	0.0805	0.2793	0.3598
		Pr-136		1.00				
		Nd-137	38.5 m	ECB+	—	0.3247	1.1789	1.5036
		Pr-137		1.00				
		Nd-138	5.04 h	EC	—	0.0082	0.0438	0.0520
		Pr-138		1.00				
		Nd-139	29.7 m	ECB+	—	0.2084	0.4431	0.6515
		Pr-139		1.00				
		Nd-139m	5.50 h	ECB+IT	—	0.0795	1.5852	1.6647
		Pr-139		8.8200E-01				
		Nd-139		1.1800E-01				
		Nd-140	3.37 d	EC	—	0.0069	0.0287	0.0357
		Pr-140		1.00				
		Nd-141	2.49 h	ECB+	—	0.0165	0.0765	0.0930
		<i>Pr-141</i>		1.00				
		Nd-141m	62.0 s	ITECB+	—	0.0622	0.6947	0.7569
		Nd-141		9.9968E-01				
		<i>Pr-141</i>		3.2000E-04				
61	Promethium	Nd-144	2.29E+15 y	A	1.9052	—	—	1.9052
		<i>Ce-140</i>		1.00				
		Nd-147	10.98 d	B-	—	0.2702	0.1408	0.4109
		Pm-147		1.00				
		Nd-149	1.728 h	B-	—	0.5042	0.3713	0.8756
		Pm-149		1.00				
		Nd-151	12.44 m	B-	—	0.6195	0.8518	1.4712
		Pm-151		1.00				
		Nd-152	11.4 m	B-	—	0.3314	0.1644	0.4957
		Pm-152		1.00				
		Pm-136	107 s	ECB+	—	2.1436	2.7178	4.8614
		Nd-136		1.00				
		Pm-137m	2.4 m	ECB+	—	1.1120	1.7835	2.8955
		Nd-137		1.00				
		Pm-139	4.15 m	ECB+	—	1.0416	0.9390	1.9807
		Nd-139		1.00				
		Pm-140	9.2 s	ECB+	—	2.0432	1.0506	3.0937
		Nd-140		1.00				
		Pm-140m	5.95 m	ECB+	—	0.9932	3.0324	4.0256
		Nd-140		1.00				

Table A.1 continued

Z	Element	Nuclide	Half-life	Decay mode	Emitted energy (MeV/nt)			
					Alpha	Electron	Photon	Total
62	Samarium	Pm-141	20.90 m	ECB+	—	0.6054	0.7349	1.3403
		Nd-141		9.9833E-01				
		Nd-141m		1.6651E-03				
		Pm-142	40.5 s	ECB+	—	1.3122	0.8561	2.1683
		<i>Nd-142</i>		1.00				
		Pm-143	265 d	EC	—	0.0083	0.3157	0.3240
		<i>Nd-143</i>		1.00				
		Pm-144	363 d	EC	—	0.0171	1.5631	1.5802
		Nd-144		1.00				
		Pm-145	17.7 y	ECA	<E-04	0.0126	0.0315	0.0441
		<i>Nd-145</i>		1.00				
		<i>Pr-141</i>		2.8000E-09				
		Pm-146	5.53 y	ECB-	—	0.0941	0.7512	0.8453
		Sm-146		3.4000E-01				
		<i>Nd-146</i>		6.6000E-01				
		Pm-147	2.6234 y	B-	—	0.0619	<E-04	0.0619
		Sm-147		1.00				
		Pm-148	5.368 d	B-	—	0.7284	0.5743	1.3028
		Sm-148		1.00				
		Pm-148m	41.29 d	B-IT	—	0.1699	1.9916	2.1615
		Sm-148		9.5800E-01				
		Pm-148		4.2000E-02				
		Pm-149	53.08 h	B-	—	0.3650	0.0119	0.3769
		<i>Sm-149</i>		1.00				
		Pm-150	2.68 h	B-	—	0.8101	1.4705	2.2806
		<i>Sm-150</i>		1.00				
		Pm-151	28.40 h	B-	—	0.3048	0.3289	0.6337
		Sm-151		1.00				
		Pm-152	4.12 m	B-	—	1.3283	0.2866	1.6149
		<i>Sm-152</i>		1.00				
		Pm-152m	7.52 m	B-	—	0.9055	1.5189	2.4244
		<i>Sm-152</i>		1.00				
		Pm-153	5.25 m	B-	—	0.6881	0.0764	0.7645
		Sm-153		1.00				
		Pm-154	1.73 m	B-	—	0.8706	1.7933	2.6638
		<i>Sm-154</i>		1.00				
		Pm-154m	2.68 m	B-	—	0.9492	1.7996	2.7488
		<i>Sm-154</i>		1.00				
		Sm-139	2.57 m	ECB+	—	1.0871	1.4540	2.5410
		Pm-139		1.00				
		Sm-140	14.82 m	ECB+	—	0.1710	0.5677	0.7387
		Pm-140		1.00				
		Sm-141	10.2 m	ECB+	—	0.7126	1.4089	2.1215
		Pm-141		1.00				
		Sm-141m	22.6 m	ECB+IT	—	0.3995	1.9494	2.3489
		Pm-141		9.9690E-01				
		Sm-141		3.1000E-03				

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Table A.1 continued

Z	Element	Nuclide	Half-life	Decay mode	Emitted energy (MeV/nt)			
					Alpha	Electron	Photon	Total
63	Europium	Sm-142	72.49 m	ECB+	–	0.0446	0.1105	0.1551
		Pm-142	1.00					
		Sm-143	8.75 m	ECB+	–	0.4978	0.5289	1.0267
		Pm-143	1.00					
		Sm-143m	66 s	ITECB+	–	0.0720	0.6844	0.7564
		Sm-143	9.9760E-01					
		Pm-143	2.4000E-03					
		Sm-145	340 d	EC	–	0.0307	0.0642	0.0950
		Pm-145	1.00					
		Sm-146	1.03E+8 y	A	2.5290	–	–	2.5290
		Nd-142	1.00					
		Sm-147	1.060E11 y	A	2.3105	–	–	2.3105
		Nd-143	1.00					
		Sm-148	7E+15 y	A	1.9860	–	–	1.9860
		Nd-144	1.00					
		Sm-151	90 y	B-	–	0.0200	<E-04	0.0200
		Eu-151	1.00					
		Sm-153	46.50 h	B-	–	0.2699	0.0643	0.3341
		Eu-153	1.00					
		Sm-155	22.3 m	B-	–	0.5674	0.1029	0.6703
		Eu-155	1.00					
		Sm-156	9.4 h	B-	–	0.2093	0.1150	0.3242
		Eu-156	1.00					
		Sm-157	8.03 m	B-	–	0.8750	0.4142	1.2892
		Eu-157	1.00					
		Eu-142	2.34 s	ECB+	–	2.7670	1.1970	3.9641
		Sm-142	1.00					
		Eu-142m	1.223 m	ECB+	–	1.7683	3.4305	5.1987
		Sm-142	1.00					
		Eu-143	2.59 m	ECB+	–	1.3676	1.1227	2.4903
		Sm-143	9.9879E-01					
		Sm-143m	1.2069E-03					
		Eu-144	10.2 s	ECB+	–	2.0761	1.0917	3.1678
		Sm-144	1.00					
		Eu-145	5.93 d	ECB+	–	0.0254	1.2798	1.3052
		Sm-145	1.00					
		Eu-146	4.61 d	ECB+	–	0.0455	2.4007	2.4462
		Sm-146	1.00					
		Eu-147	24.1 d	ECB+A	<E-04	0.0431	0.4721	0.5152
		Sm-147	9.9998E-01					
		Pm-143	2.2000E-05					
		Eu-148	54.5 d	ECB+A	<E-04	0.0224	2.2287	2.2511
		Sm-148	1.00					
		Pm-144	9.4000E-09					
		Eu-149	93.1 d	EC	–	0.0241	0.0661	0.0902
		Sm-149	1.00					
		Eu-150	36.9 y	ECB+	–	0.0285	1.5554	1.5839
		Sm-150	1.00					

Table A.1 continued

Z	Element	Nuclide	Half-life	Decay mode	Emitted energy (MeV/nt)			
					Alpha	Electron	Photon	Total
64	Gadolinium	Eu-150m	12.8 h	B-ECB+	—	0.3122	0.0494	0.3615
		Gd-150		8.9000E-01				
		<i>Sm-150</i>		1.1000E-01				
		Eu-152	13.537 y	ECB+B-	—	0.1286	1.1759	1.3045
		Gd-152		2.7900E-01				
		<i>Sm-152</i>		7.2100E-01				
		Eu-152m	9.3116 h	B-ECB+	—	0.5060	0.2963	0.8023
		Gd-152		7.2000E-01				
		<i>Sm-152</i>		2.8000E-01				
		Eu-152n	96 m	IT	—	0.0666	0.0753	0.1419
		Eu-152		1.00				
		Eu-154	8.593 y	B-EC	—	0.2730	1.2493	1.5223
		<i>Gd-154</i>		9.9980E-01				
		<i>Sm-154</i>		2.0000E-04				
		Eu-154m	46.0 m	IT	—	0.0745	0.0706	0.1451
		Eu-154		1.00				
		Eu-155	4.7611 y	B-	—	0.0647	0.0612	0.1259
		<i>Gd-155</i>		1.00				
		Eu-156	15.19 d	B-	—	0.4579	1.2342	1.6920
		<i>Gd-156</i>		1.00				
		Eu-157	15.18 h	B-	—	0.3961	0.2930	0.6891
		<i>Gd-157</i>		1.00				
		Eu-158	45.9 m	B-	—	0.8920	1.2978	2.1898
		<i>Gd-158</i>		1.00				
		Eu-159	18.1 m	B-	—	0.8923	0.3032	1.1954
		Gd-159		1.00				
		Gd-142	70.2 s	ECB+	—	0.7390	1.0432	1.7822
		Eu-142		1.00				
		Gd-143m	110.0 s	ECB+	—	1.2854	2.1211	3.4065
		Eu-143		1.00				
		Gd-144	4.47 m	ECB+	—	0.5858	0.9086	1.4944
		Eu-144		1.00				
		Gd-145	23.0 m	ECB+	—	0.3456	2.4236	2.7692
		Eu-145		1.00				
		Gd-145m	85 s	ITECB+	—	0.1927	0.6814	0.8741
		Gd-145		9.4300E-01				
		Eu-145		5.7000E-02				
		Gd-146	48.27 d	EC	—	0.1274	0.2526	0.3800
		Eu-146		1.00				
		Gd-147	38.1 h	ECB+	—	0.0617	1.4030	1.4647
		Eu-147		1.00				
		Gd-148	74.6 y	A	3.2712	—	—	3.2712
		<i>Sm-144</i>		1.00				
		Gd-149	9.28 d	ECB+	—	0.0686	0.5292	0.5978
		Eu-149		1.00				
		Gd-150	1.79E+6 y	A	2.8090	—	—	2.8090
		Sm-146		1.00				

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Table A.1 continued

Z	Element	Nuclide	Half-life	Decay mode	Emitted energy (MeV/nt)			
					Alpha	Electron	Photon	Total
65	Terbium	Gd-151	124 d	ECA	<E-04	0.0394	0.0708	0.1102
		Sm-147		1.0000E-08				
		<i>Eu-151</i>		1.0000E+00				
		Gd-152	1.08E+14 y	A	2.2046	–	–	2.2046
		Sm-148		1.00				
		Gd-153	240.4 d	EC	–	0.0438	0.1057	0.1494
		<i>Eu-153</i>		1.00				
		Gd-159	18.479 h	B-	–	0.3096	0.0539	0.3635
		<i>Tb-159</i>		1.00				
		Gd-162	8.4 m	B-	–	0.3387	0.4170	0.7557
		<i>Tb-162</i>		1.00				
		Tb-146	23 s	ECB+	–	1.4644	3.6256	5.0900
		Gd-146		1.00				
		Tb-147	1.64 h	ECB+	–	0.2810	2.1845	2.4655
		Gd-147		1.00				
		Tb-147m	1.87 m	ECB+	–	0.3196	1.9116	2.2312
		Gd-147		1.00				
		Tb-148	60 m	ECB+	–	0.8411	2.3590	3.2001
		Gd-148		1.00				
		Tb-148m	2.20 m	ECB+	–	0.3106	3.1390	3.4495
		Gd-148		1.00				
		Tb-149	4.118 h	ECB+A	0.6810	0.0871	1.3612	2.1292
		Gd-149		8.3300E-01				
		Eu-145		1.6700E-01				
		Tb-149m	4.16 m	ECB+A	0.0009	0.2091	1.3748	1.5847
		Gd-149		9.9978E-01				
		Eu-145		2.2000E-04				
		Tb-150	3.48 h	ECB+A	<E-04	0.2890	2.4403	2.7293
		Gd-150		1.00				
		Eu-146		7.0000E-06				
		Tb-150m	5.8 m	ECB+	–	0.1267	2.5202	2.6470
		Gd-150		1.00				
		Tb-151	17.609 h	ECB+A	0.0003	0.0800	0.9941	1.0744
		Gd-151		1.00				
		Eu-147		9.5000E-05				
		Tb-151m	25 s	ITECB+	–	0.0793	0.0808	0.1601
		Tb-151		9.3400E-01				
		Gd-151		6.6000E-02				
		Tb-152	17.5 h	ECB+	–	0.2503	1.4932	1.7436
		Gd-152		1.00				
		Tb-152m	4.2 m	ITECB+	–	0.1520	0.7619	0.9139
		Tb-152		7.8800E-01				
		Gd-152		2.1200E-01				
		Tb-153	2.34 d	ECB+	–	0.0481	0.3318	0.3800
		Gd-153		1.00				
		Tb-154	21.5 h	ECB+	–	0.0681	2.2831	2.3512
		<i>Gd-154</i>		1.00				

Table A.1 continued

Z	Element	Nuclide	Half-life	Decay mode	Emitted energy (MeV/nt)			
					Alpha	Electron	Photon	Total
66	Dysprosium	Tb-155	5.32 d	EC	—	0.0434	0.1777	0.2211
		<i>Gd-155</i>		1.00				
		Tb-156	5.35 d	EC	—	0.0835	1.9371	2.0206
		<i>Gd-156</i>		1.00				
		Tb-156m	24.4 h	IT	—	0.0171	0.0370	0.0540
		Tb-156		1.00				
		Tb-156n	5.3 h	IT	—	0.0874	0.0048	0.0922
		Tb-156		1.00				
		Tb-157	71 y	EC	—	0.0057	0.0057	0.0114
		<i>Gd-157</i>		1.00				
		Tb-158	180 y	ECB-	—	0.1117	0.8048	0.9165
		<i>Gd-158</i>		8.3400E-01				
		<i>Dy-158</i>		1.6600E-01				
		Tb-160	72.3 d	B-	—	0.2593	1.1264	1.3856
		<i>Dy-160</i>		1.00				
		Tb-161	6.906 d	B-	—	0.2025	0.0365	0.2390
		<i>Dy-161</i>		1.00				
		Tb-162	7.60 m	B-	—	0.5327	1.1066	1.6393
		<i>Dy-162</i>		1.00				
		Tb-163	19.5 m	B-	—	0.3597	0.7887	1.1484
		<i>Dy-163</i>		1.00				
		Tb-164	3.0 m	B-	—	0.8249	2.4455	3.2704
		<i>Dy-164</i>		1.00				
		Tb-165	2.11 m	B-	—	0.8966	0.8363	1.7330
		<i>Dy-165m</i>		8.9028E-01				
		<i>Dy-165</i>		1.0972E-01				
		Dy-148	3.3 m	ECB+	—	0.0278	0.7170	0.7448
		Tb-148		1.00				
		Dy-149	4.20 m	ECB+	—	0.1120	1.6222	1.7342
		Tb-149		5.6682E-01				
		Tb-149m		4.3318E-01				
		Dy-150	7.17 m	ECB+A	1.5664	0.0070	0.2785	1.8519
		Tb-150		6.4000E-01				
		Gd-146		3.6000E-01				
		Dy-151	17.9 m	ECB+A	0.2341	0.0653	1.3723	1.6717
		Tb-151		5.3377E-01				
		Tb-151m		4.1023E-01				
		Gd-147		5.6000E-02				
		Dy-152	2.38 h	ECA	0.0037	0.0130	0.2867	0.3034
		Tb-152		9.9900E-01				
		Gd-148		1.0000E-03				
		Dy-153	6.4 h	ECB+A	0.0003	0.0901	0.8746	0.9651
		Tb-153		1.00				
		Gd-149		9.4000E-05				
		Dy-154	3.0E+6 y	A	2.9470	—	—	2.9470
		Gd-150		1.00				

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Table A.1 continued

Z	Element	Nuclide	Half-life	Decay mode	Emitted energy (MeV/nt)			
					Alpha	Electron	Photon	Total
67	Holmium	Dy-155	9.9 h	ECB+	–	0.0272	0.6687	0.6959
		Tb-155	1.00					
		Dy-157	8.14 h	EC	–	0.0138	0.3472	0.3610
		Tb-157	1.00					
		Dy-159	144.4 d	EC	–	0.0131	0.0456	0.0587
		Tb-159	1.00					
		Dy-165	2.334 h	B-	–	0.4473	0.0267	0.4740
		Ho-165	1.00					
		Dy-165m	1.257 m	ITB-	–	0.1049	0.0192	0.1240
		Dy-165	9.7760E-01					
		Ho-165	2.2400E-02					
		Dy-166	81.6 h	B-	–	0.1667	0.0433	0.2100
		Ho-166	1.00					
		Dy-167	6.20 m	B-	–	0.7262	0.5326	1.2589
		Ho-167	1.00					
		Dy-168	8.7 m	B-	–	0.4332	0.3948	0.8280
		Ho-168	1.00					
		Ho-150	76.8 s	ECB+	–	1.9857	1.8843	3.8701
		Dy-150	1.00					
		Ho-153	2.01 m	ECB+A	0.0020	0.5387	1.0281	1.5689
		Dy-153	9.9949E-01					
		Tb-149m	5.1000E-04					
		Ho-153m	9.3 m	ECB+A	0.0074	0.6952	1.0618	1.7644
		Dy-153	9.9820E-01					
		Tb-149	1.8000E-03					
		Ho-154	11.76 m	ECB+A	0.0008	1.0927	1.8817	2.9752
		Dy-154	9.9981E-01					
		Tb-150	1.9000E-04					
		Ho-154m	3.10 m	ECB+A	<E-04	0.5434	2.4367	2.9801
		Dy-154	1.00					
		Tb-150m	1.0000E-05					
		Ho-155	48 m	ECB+	–	0.2176	0.6142	0.8318
		Dy-155	1.00					
		Ho-156	56 m	ECB+	–	0.6655	2.1064	2.7719
		Dy-156	1.00					
		Ho-157	12.6 m	ECB+	–	0.0929	0.5835	0.6764
		Dy-157	1.00					
		Ho-159	33.05 m	ECB+	–	0.0576	0.3857	0.4433
		Dy-159	1.00					
		Ho-160	25.6 m	ECB+	–	0.0703	1.6950	1.7653
		Dy-160	1.00					
		Ho-161	2.48 h	EC	–	0.0336	0.0582	0.0918
		Dy-161	1.00					
		Ho-162	15.0 m	ECB+	–	0.0598	0.1640	0.2239
		Dy-162	1.00					
		Ho-162m	67.0 m	ITECB+	–	0.0739	0.5614	0.6353
		Ho-162	6.2000E-01					
		Dy-162	3.8000E-01					

Table A.1 continued

Z	Element	Nuclide	Half-life	Decay mode	Emitted energy (MeV/nt)			
					Alpha	Electron	Photon	Total
68	Erbium	Ho-163	4570 y <i>Dy-163</i>	EC 1.00	—	0.0005	<E-04	0.0006
		Ho-164	29 m <i>Dy-164</i> <i>Er-164</i>	ECB- 6.0000E-01 4.0000E-01	—	0.1470	0.0297	0.1768
		Ho-164m	38.0 m Ho-164	IT 1.00	—	0.0926	0.0472	0.1399
		Ho-166	26.80 h <i>Er-166</i>	B- 1.00	—	0.6963	0.0301	0.7264
		Ho-166m	1.20E+3 y <i>Er-166</i>	B- 1.00	—	0.1497	1.6249	1.7746
		Ho-167	3.1 h <i>Er-167</i>	B- 1.00	—	0.2329	0.3661	0.5990
		Ho-168	2.99 m <i>Er-168</i>	B- 1.00	—	0.8138	0.8762	1.6901
		Ho-168m	132 s Ho-168	IT 1.00	—	0.0518	0.0073	0.0590
		Ho-170	2.76 m <i>Er-170</i>	B- 1.00	—	0.8366	1.7041	2.5407
		Er-154	3.73 m Ho-154 <i>Dy-150</i>	ECB+A 9.9530E-01 4.7000E-03	0.0201	0.0376	0.0759	0.1336
		Er-156	19.5 m Ho-156	EC 1.00	—	0.0943	0.0678	0.1621
		Er-159	36 m Ho-159	ECB+ 1.00	—	0.0738	0.9635	1.0373
		Er-161	3.21 h Ho-161	ECB+ 1.00	—	0.0522	0.9905	1.0427
		Er-163	75.0 m Ho-163	ECB+ 1.00	—	0.0081	0.0403	0.0484
		Er-165	10.36 h <i>Ho-165</i>	EC 1.00	—	0.0080	0.0379	0.0459
		Er-167m	2.269 s <i>Er-167</i>	IT 1.00	—	0.1111	0.0967	0.2078
		Er-169	9.40 d <i>Tm-169</i>	B- 1.00	—	0.1035	<E-04	0.1035
		Er-171	7.516 h Tm-171	B- 1.00	—	0.4205	0.3731	0.7936
		Er-172	49.3 h Tm-172	B- 1.00	—	0.1387	0.5166	0.6552
		Er-173	1.434 m Tm-173	B- 1.00	—	0.7213	0.8313	1.5526
69	Thulium	Tm-161	30.2 m Er-161	ECB+ 1.00	—	0.2308	1.2992	1.5300
		Tm-162	21.70 m <i>Er-162</i>	ECB+ 1.00	—	0.5648	1.9155	2.4803

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Table A.1 continued

Z	Element	Nuclide	Half-life	Decay mode	Emitted energy (MeV/nt)			
					Alpha	Electron	Photon	Total
70	Ytterbium	Tm-163	1.810 h	ECB+	–	0.0716	1.3200	1.3916
		Er-163	1.00					
		Tm-164	2.0 m	ECB+	–	0.5994	0.7760	1.3754
		<i>Er-164</i>	1.00					
		Tm-165	30.06 h	ECB+	–	0.0637	0.5625	0.6262
		Er-165	1.00					
		Tm-166	7.70 h	ECB+	–	0.0892	1.9768	2.0660
		<i>Er-166</i>	1.00					
		Tm-167	9.25 d	EC	–	0.1332	0.1482	0.2814
		<i>Er-167</i>	1.00					
		Tm-168	93.1 d	ECB+B-	–	0.0847	1.2432	1.3279
		<i>Er-168</i>	9.9990E-01					
		<i>Yb-168</i>	1.0000E-04					
		Tm-170	128.6 d	B-EC	–	0.3280	0.0041	0.3321
		<i>Yb-170</i>	9.9869E-01					
		<i>Er-170</i>	1.3100E-03					
		Tm-171	1.92 y	B-	–	0.0255	0.0006	0.0261
		<i>Yb-171</i>	1.00					
		Tm-172	63.6 h	B-	–	0.5327	0.4744	1.0072
		<i>Yb-172</i>	1.00					
		Tm-173	8.24 h	B-	–	0.3103	0.3885	0.6988
		<i>Yb-173</i>	1.00					
		Tm-174	5.4 m	B-	–	0.5129	1.7787	2.2915
		<i>Yb-174</i>	1.00					
		Tm-175	15.2 m	B-	–	0.5196	1.0848	1.6044
		<i>Yb-175</i>	1.00					
		Tm-176	1.85 m	B-	–	0.9972	1.9682	2.9654
		<i>Yb-176</i>	1.00					
		Yb-162	18.87 m	ECB+	–	0.0374	0.2519	0.2893
		Tm-162	1.00					
		Yb-163	11.05 m	ECB+	–	0.2718	0.7277	0.9996
		Tm-163	1.00					
		Yb-164	75.8 m	EC	–	0.0095	0.0542	0.0637
		Tm-164	1.00					
		Yb-165	9.9 m	ECB+	–	0.1518	0.3390	0.4908
		Tm-165	1.00					
		Yb-166	56.7 h	EC	–	0.0417	0.0868	0.1285
		Tm-166	1.00					
		Yb-167	17.5 m	ECB+	–	0.0952	0.2696	0.3648
		Tm-167	1.00					
		Yb-169	32.026 d	EC	–	0.1471	0.3302	0.4773
		<i>Tm-169</i>	1.00					
		Yb-175	4.185 d	B-	–	0.1308	0.0391	0.1699
		<i>Lu-175</i>	1.00					
		Yb-177	1.911 h	B-	–	0.4358	0.1952	0.6309
		<i>Lu-177</i>	1.00					
		Yb-178	74 m	B-	–	0.1925	0.0381	0.2306
		<i>Lu-178</i>	1.00					

Table A.1 continued

Z	Element	Nuclide	Half-life	Decay mode	Emitted energy (MeV/nt)			
					Alpha	Electron	Photon	Total
71	Lutetium	Yb-179	8.0 m	B-	—	0.7093	0.9743	1.6836
		Lu-179	1.00					
		Lu-165	10.74 m	ECB+	—	0.3751	1.1102	1.4853
		Yb-165	1.00					
		Lu-167	51.5 m	ECB+	—	0.1109	1.6920	1.8030
		Yb-167	1.00					
		Lu-169	34.06 h	ECB+	—	0.0477	1.3166	1.3643
		Yb-169	1.00					
		Lu-169m	160 s	IT	—	0.0274	0.0017	0.0290
		Lu-169	1.00					
		Lu-170	2.012 d	ECB+	—	0.0585	2.5636	2.6220
		Yb-170	1.00					
		Lu-171	8.24 d	ECB+	—	0.0928	0.6503	0.7431
		Yb-171	1.00					
		Lu-171m	79 s	IT	—	0.0691	0.0021	0.0711
		Lu-171	1.00					
		Lu-172	6.70 d	ECB+	—	0.1154	1.9565	2.0718
		Yb-172	1.00					
		Lu-172m	3.7 m	IT	—	0.0404	0.0015	0.0419
		Lu-172	1.00					
		Lu-173	1.37 y	EC	—	0.0526	0.1834	0.2360
		Yb-173	1.00					
		Lu-174	3.31 y	ECB+	—	0.0458	0.1163	0.1621
		Yb-174	1.00					
		Lu-174m	142 d	ITEC	—	0.1188	0.0626	0.1814
		Lu-174	9.9380E-01					
		Yb-174	6.2000E-03					
		Lu-176	3.85E+10 y	B-	—	0.3026	0.4799	0.7825
		Hf-176	1.00					
		Lu-176m	3.635 h	B-EC	—	0.4783	0.0146	0.4929
		Hf-176	9.9905E-01					
		Yb-176	9.5000E-04					
		Lu-177	6.647 d	B-	—	0.1479	0.0351	0.1830
		Hf-177	1.00					
		Lu-177m	160.4 d	B-IT	—	0.2687	1.0005	1.2692
		Lu-177	2.1700E-01					
		Hf-177	7.8300E-01					
		Lu-178	28.4 m	B-	—	0.7561	0.1256	0.8817
		Hf-178	1.00					
		Lu-178m	23.1 m	B-	—	0.4907	1.0481	1.5389
		Hf-178	1.00					
		Lu-179	4.59 h	B-	—	0.4869	0.0298	0.5167
		Hf-179	1.00					
		Lu-180	5.7 m	B-	—	0.6352	1.5148	2.1499
		Hf-180	1.00					
		Lu-181	3.5 m	B-	—	0.8512	0.5740	1.4252
		Hf-181	1.00					

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Table A.1 continued

Z	Element	Nuclide	Half-life	Decay mode	Emitted energy (MeV/nt)			
					Alpha	Electron	Photon	Total
72	Hafnium	Hf-167	2.05 m	ECB+	–	0.4952	0.6171	1.1123
		Lu-167	1.00					
		Hf-169	3.24 m	ECB+	–	0.1309	0.6416	0.7725
		Lu-169	9.6904E-01					
		Lu-169m	3.0960E-02					
		Hf-170	16.01 h	EC	–	0.0687	0.4403	0.5090
		Lu-170	1.00					
		Hf-172	1.87 y	EC	–	0.0828	0.1062	0.1890
		Lu-172m	1.00					
		Hf-173	23.6 h	ECB+	–	0.0524	0.3970	0.4494
		Lu-173	1.00					
		Hf-174	2.0E+15 y	A	2.4948	–	–	2.4948
		Yb-170	1.00					
		Hf-175	70 d	EC	–	0.0450	0.3534	0.3984
		Lu-175	1.00					
		Hf-177m	51.4 m	IT	–	0.5078	2.2864	2.7941
		Hf-177	1.00					
		Hf-178m	31 y	IT	–	0.2113	2.2383	2.4496
		Hf-178	1.00					
		Hf-179m	25.05 d	IT	–	0.1897	0.9207	1.1104
		Hf-179	1.00					
		Hf-180m	5.5 h	ITB-	–	0.1437	0.9884	1.1321
		Hf-180	9.9700E-01					
		Ta-180m	3.0000E-03					
		Hf-181	42.39 d	B-	–	0.2052	0.5324	0.7377
		Ta-181	1.00					
		Hf-182	9E+6 y	B-	–	0.0632	0.2397	0.3030
		Ta-182	1.00					
73	Tantalum	Hf-182m	61.5 m	B-IT	–	0.2441	0.9135	1.1575
		Ta-182	4.9070E-01					
		Hf-182	4.2000E-01					
		Ta-182m	8.9280E-02					
		Hf-183	1.067 h	B-	–	0.4478	0.7749	1.2227
		Ta-183	1.00					
		Hf-184	4.12 h	B-	–	0.4787	0.2387	0.7174
		Ta-184	1.00					
		Ta-170	6.76 m	ECB+	–	1.6049	1.0602	2.6652
		Hf-170	1.00					
		Ta-172	36.8 m	ECB+	–	0.5513	1.6967	2.2480
		Hf-172	1.00					
		Ta-173	3.14 h	ECB+	–	0.1686	0.5824	0.7510
		Hf-173	1.00					
		Ta-174	1.14 h	ECB+	–	0.4670	0.9767	1.4438
		Hf-174	1.00					
		Ta-175	10.5 h	ECB+	–	0.0663	1.1126	1.1790
		Hf-175	1.00					
		Ta-176	8.09 h	ECB+	–	0.0849	2.2322	2.3171
		Hf-176	1.00					

Table A.1 continued

Z	Element	Nuclide	Half-life	Decay mode	Emitted energy (MeV/nt)			
					Alpha	Electron	Photon	Total
74	Tungsten	Ta-177	56.56 h	EC	—	0.0240	0.0678	0.0917
		<i>Hf-177</i>		1.00				
		Ta-178	9.31 m	ECB+	—	0.0391	0.1218	0.1609
		<i>Hf-178</i>		1.00				
		Ta-178m	2.36 h	EC	—	0.1616	1.1558	1.3174
		<i>Hf-178</i>		1.00				
		Ta-179	1.82 y	EC	—	0.0078	0.0256	0.0334
		<i>Hf-179</i>		1.00				
		Ta-180	8.152 h	ECB-	—	0.0565	0.0482	0.1048
		<i>Hf-180</i>		8.6000E-01				
		<i>W-180</i>		1.4000E-01				
		Ta-182	114.43 d	B-	—	0.2105	1.2918	1.5023
		<i>W-182</i>		1.00				
		Ta-182m	15.84 m	IT	—	0.2665	0.2659	0.5324
		Ta-182		1.00				
		Ta-183	5.1 d	B-	—	0.3537	0.2963	0.6499
		<i>W-183</i>		1.00				
		Ta-184	8.7 h	B-	—	0.5426	1.5727	2.1152
		<i>W-184</i>		1.00				
		Ta-185	49.4 m	B-	—	0.7416	0.1547	0.8963
		<i>W-185</i>		1.00				
		Ta-186	10.5 m	B-	—	1.0710	1.4202	2.4912
		<i>W-186</i>		1.00				
		W-177	132 m	ECB+	—	0.0970	0.9185	1.0155
		Ta-177		1.00				
		W-178	21.6 d	EC	—	0.0075	0.0164	0.0239
		Ta-178		1.00				
		W-179	37.05 m	EC	—	0.0326	0.0554	0.0880
		Ta-179		1.00				
		W-179m	6.40 m	ITEC	—	0.1661	0.0561	0.2221
		<i>W-179</i>		9.9720E-01				
		Ta-179		2.8000E-03				
		W-181	121.2 d	EC	—	0.0129	0.0404	0.0533
		<i>Ta-181</i>		1.00				
		W-185	75.1 d	B-	—	0.1270	<E-04	0.1270
		<i>Re-185</i>		1.00				
		W-185m	1.597 m	IT	—	0.1738	0.0287	0.2025
		<i>W-185</i>		1.00				
		W-187	23.72 h	B-	—	0.2995	0.4483	0.7478
		<i>Re-187</i>		1.00				
		W-188	69.78 d	B-	—	0.0997	0.0019	0.1016
		<i>Re-188</i>		1.00				
		W-190	30.0 m	B-	—	0.4771	0.1511	0.6282
		<i>Re-190</i>		1.00				
75	Rhenium	Re-178	13.2 m	ECB+	—	0.6192	1.7085	2.3277
		W-178		1.00				

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Table A.1 continued

Z	Element	Nuclide	Half-life	Decay mode	Emitted energy (MeV/nt)			
					Alpha	Electron	Photon	Total
		Re-179	19.5 m	ECB+	–	0.0668	1.0841	1.1509
		W-179		7.6079E-01				
		W-179m		2.3921E-01				
		Re-180	2.44 m	ECB+	–	0.1720	1.2028	1.3748
		W-180		1.00				
		Re-181	19.9 h	ECB+	–	0.1360	0.8044	0.9404
		W-181		1.00				
		Re-182	64.0 h	EC	–	0.2045	1.7982	2.0027
		W-182		1.00				
		Re-182m	12.7 h	ECB+	–	0.0922	1.2214	1.3136
		W-182		1.00				
		Re-183	70.0 d	EC	–	0.1093	0.1575	0.2667
		W-183		1.00				
		Re-184	38.0 d	ECB+	–	0.0562	0.8919	0.9481
		W-184		1.00				
		Re-184m	169 d	ITEC	–	0.1413	0.3836	0.5249
		Re-184		7.5400E-01				
		W-184		2.4600E-01				
		Re-186	3.7183 d	B-EC	–	0.3362	0.0208	0.3570
		Os-186		9.2530E-01				
		W-186		7.4700E-02				
		Re-186m	2.00E+5 y	IT	–	0.1267	0.0207	0.1474
		Re-186		1.00				
		Re-187	4.12E+10 y	B-	–	0.0006	–	0.0006
		Os-187		1.00				
		Re-188	17.0040 h	B-	–	0.7793	0.0613	0.8406
		Os-188		1.00				
		Re-188m	18.59 m	IT	–	0.0976	0.0715	0.1692
		Re-188		1.00				
		Re-189	24.3 h	B-	–	0.3260	0.0556	0.3816
		Os-189m		1.2211E-01				
		Os-189		8.7789E-01				
		Re-190	3.1 m	B-	–	0.6863	1.3376	2.0239
		Os-190		1.00				
		Re-190m	3.2 h	B-IT	–	0.4456	0.9257	1.3714
		Re-190		4.5600E-01				
		Os-190		5.4400E-01				
76	Osmium	Os-180	21.5 m	ECB+	–	0.0298	0.1265	0.1564
		Re-180		1.00				
		Os-181	105 m	ECB+	–	0.0918	1.3835	1.4753
		Re-181		1.00				
		Os-182	22.10 h	EC	–	0.0565	0.4317	0.4882
		Re-182m		1.00				
		Os-183	13.0 h	ECB+	–	0.0792	0.6288	0.7081
		Re-183		1.00				
		Os-183m	9.9 h	ECB+IT	–	0.0416	1.0062	1.0478
		Re-183		8.5000E-01				
		Os-183		1.5000E-01				

Table A.1 continued

Z	Element	Nuclide	Half-life	Decay mode	Emitted energy (MeV/nt)			
					Alpha	Electron	Photon	Total
77	Iridium	Os-185	93.6 d	EC	—	0.0184	0.6917	0.7101
		<i>Re-185</i>		1.00				
		Os-186	2.0E+15 y	A	2.8220	—	—	2.8220
		<i>W-182</i>		1.00				
		Os-189m	5.8 h	IT	—	0.0286	0.0023	0.0309
		<i>Os-189</i>		1.00				
		Os-190m	9.9 m	IT	—	0.1166	1.5894	1.7059
		<i>Os-190</i>		1.00				
		Os-191	15.4 d	B-	—	0.1372	0.0843	0.2215
		<i>Ir-191</i>		1.00				
		Os-191m	13.10 h	IT	—	0.0664	0.0080	0.0744
		Os-191		1.00				
		Os-193	30.11 h	B-	—	0.3797	0.0674	0.4471
		Ir-193m		3.4757E-03				
		<i>Ir-193</i>		9.9652E-01				
		Os-194	6.0 y	B-	—	0.0453	0.0045	0.0498
		Ir-194		1.00				
		Os-196	34.9 m	B-	—	0.3807	0.0805	0.4612
		Ir-196		1.00				
		Ir-180	1.5 m	ECB+	—	1.2439	1.5955	2.8394
		Os-180		1.00				
		Ir-182	15 m	ECB+	—	1.0474	1.4121	2.4595
		Os-182		1.00				
		Ir-183	58 m	ECB+	—	0.1516	1.1889	1.3404
		Os-183m		7.0972E-01				
		Os-183		2.9028E-01				
		Ir-184	3.09 h	ECB+	—	0.3194	1.9644	2.2838
		<i>Os-184</i>		1.00				
		Ir-185	14.4 h	ECB+	—	0.1228	0.8581	0.9809
		Os-185		1.00				
		Ir-186	16.64 h	ECB+	—	0.1453	1.6651	1.8104
		Os-186		1.00				
		Ir-186m	1.92 h	ECB+IT	—	0.1103	1.2523	1.3626
		Os-186		7.5000E-01				
		Ir-186		2.5000E-01				
		Ir-187	10.5 h	ECB+	—	0.0568	0.3325	0.3892
		<i>Os-187</i>		1.00				
		Ir-188	41.5 h	ECB+	—	0.0508	2.0974	2.1483
		<i>Os-188</i>		1.00				
		Ir-189	13.2 d	EC	—	0.0458	0.0793	0.1251
		Os-189m		7.4264E-02				
		<i>Os-189</i>		9.2574E-01				
		Ir-190	11.78 d	EC	—	0.0746	1.4768	1.5514
		<i>Os-190</i>		1.00				
		Ir-190m	1.120 h	IT	—	0.0240	0.0024	0.0264
		Ir-190		1.00				

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Table A.1 continued

Z	Element	Nuclide	Half-life	Decay mode	Emitted energy (MeV/nt)			
					Alpha	Electron	Photon	Total
78	Platinum	Ir-190n	3.087 h	ECIT	–	0.0289	0.0577	0.0867
		Os-190m		9.1400E-01				
		Ir-190		8.6000E-02				
		Ir-191m	4.94 s	IT	–	0.0971	0.0764	0.1735
		<i>Ir-191</i>		1.00				
		Ir-192	73.827 d	B-EC	–	0.2177	0.8165	1.0342
		<i>Pt-192</i>		9.5130E-01				
		<i>Os-192</i>		4.8700E-02				
		Ir-192m	1.45 m	ITB-	–	0.0540	0.0029	0.0569
		Ir-192		9.9983E-01				
		<i>Pt-192</i>		1.7500E-04				
		Ir-192n	241 y	IT	–	0.1617	0.0066	0.1682
		Ir-192		1.00				
		Ir-193m	10.53 d	IT	–	0.0776	0.0027	0.0803
		<i>Ir-193</i>		1.00				
		Ir-194	19.28 h	B-	–	0.8105	0.0911	0.9015
		<i>Pt-194</i>		1.00				
		Ir-194m	171 d	B-	–	0.1422	2.3339	2.4761
		<i>Pt-194</i>		1.00				
		Ir-195	2.5 h	B-	–	0.3803	0.0593	0.4397
		<i>Pt-195</i>		1.00				
		Ir-195m	3.8 h	B-IT	–	0.2606	0.3766	0.6372
		<i>Pt-195m</i>		4.3692E-01				
		Ir-195		5.0000E-02				
		<i>Pt-195</i>		5.1308E-01				
		Ir-196	52 s	B-	–	1.1738	0.2322	1.4060
		<i>Pt-196</i>		1.00				
		Ir-196m	1.40 h	B-	–	0.3844	2.4677	2.8522
		<i>Pt-196</i>		1.00				
		Pt-184	17.3 m	ECB+A	<E-04	0.2004	0.7266	0.9271
		Ir-184		1.00				
		Os-180		1.7000E-05				
		Pt-186	2.08 h	ECA	<E-04	0.0451	0.6844	0.7296
		Ir-186m		8.1942E-01				
		Ir-186		1.8058E-01				
		Os-182		1.0000E-06				
		Pt-187	2.35 h	ECB+	–	0.1549	0.6183	0.7732
		Ir-187		1.00				
		Pt-188	10.2 d	ECA	<E-04	0.0823	0.2055	0.2878
		Ir-188		1.00				
		<i>Os-184</i>		2.9000E-07				
		Pt-189	10.87 h	ECB+	–	0.0996	0.4870	0.5867
		Ir-189		1.00				
		Pt-190	6.50E+11 y	A	3.2490	–	–	3.2490
		Os-186		1.00				
		Pt-191	2.802 d	EC	–	0.0749	0.2960	0.3709
		<i>Ir-191</i>		1.00				

Table A.1 continued

Z	Element	Nuclide	Half-life	Decay mode	Emitted energy (MeV/nt)			
					Alpha	Electron	Photon	Total
79	Gold	Pt-193	50 y	EC	—	0.0071	0.0026	0.0097
		<i>Ir-193</i>		1.00				
		Pt-193m	4.33 d	IT	—	0.1377	0.0132	0.1510
		Pt-193		1.00				
		Pt-195m	4.02 d	IT	—	0.1845	0.0772	0.2617
		<i>Pt-195</i>		1.00				
		Pt-197	19.8915 h	B-	—	0.2552	0.0256	0.2809
		<i>Au-197</i>		1.00				
		Pt-197m	95.41 m	ITB-	—	0.3250	0.0839	0.4089
		Pt-197		9.6700E-01				
		<i>Au-197</i>		3.3000E-02				
		Pt-199	30.80 m	B-	—	0.5455	0.1995	0.7450
		<i>Au-199</i>		1.00				
		Pt-200	12.5 h	B-	—	0.2321	0.0605	0.2926
		<i>Au-200</i>		1.00				
		Pt-202	44 h	B-	—	0.6537	—	0.6537
		<i>Au-202</i>		1.00				
		Au-186	10.7 m	ECB+	—	1.0751	1.4983	2.5735
		Pt-186		1.00				
		Au-187	8.4 m	ECB+A	0.0001	0.1360	1.0652	1.2014
		Pt-187		1.00				
		<i>Ir-183</i>		3.0000E-05				
		Au-190	42.8 m	ECB+	—	0.2129	2.3807	2.5936
		Pt-190		1.00				
		Au-191	3.18 h	ECB+	—	0.0865	0.5946	0.6811
		Pt-191		1.00				
		Au-192	4.94 h	ECB+	—	0.0904	1.9392	2.0296
		<i>Pt-192</i>		1.00				
		Au-193	17.65 h	EC	—	0.0575	0.1673	0.2249
		Pt-193		1.00				
		Au-193m	3.9 s	ITEC	—	0.0904	0.1979	0.2883
		<i>Au-193</i>		9.9970E-01				
		Pt-193m		3.0000E-04				
		Au-194	38.02 h	ECB+	—	0.0421	1.0386	1.0806
		<i>Pt-194</i>		1.00				
		Au-195	186.098 d	EC	—	0.0520	0.0839	0.1359
		<i>Pt-195</i>		1.00				
		Au-195m	30.5 s	IT	—	0.1172	0.2014	0.3186
		<i>Au-195</i>		1.00				
		Au-196	6.183 d	ECB-	—	0.0372	0.4734	0.5105
		<i>Pt-196</i>		9.2800E-01				
		<i>Hg-196</i>		7.2000E-02				
		Au-196m	9.6 h	IT	—	0.3760	0.2473	0.6233
		<i>Au-196</i>		1.00				
		Au-198	2.69517 d	B-	—	0.3277	0.4029	0.7306
		<i>Hg-198</i>		1.00				

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Table A.1 continued

Z	Element	Nuclide	Half-life	Decay mode	Emitted energy (MeV/nt)			
					Alpha	Electron	Photon	Total
80	Mercury	Au-198m	2.27 d	IT	—	0.2748	0.5332	0.8080
		Au-198	1.00	1.00	—	0.2748	0.5332	0.8080
		Au-199	3.139 d	B-	—	0.1451	0.0961	0.2412
		<i>Hg-199</i>	1.00	1.00	—	0.1451	0.0961	0.2412
		Au-200	48.4 m	B-	—	0.7303	0.2737	1.0041
		<i>Hg-200</i>	1.00	1.00	—	0.7303	0.2737	1.0041
		Au-200m	18.7 h	B-IT	—	0.2433	1.9843	2.2276
		Au-200	1.8000E-01	1.8000E-01	—	0.2433	1.9843	2.2276
		<i>Hg-200</i>	8.2000E-01	8.2000E-01	—	0.2433	1.9843	2.2276
		Au-201	26 m	B-	—	0.4259	0.0346	0.4605
		<i>Hg-201</i>	1.00	1.00	—	0.4259	0.0346	0.4605
		Au-202	28.8 s	B-	—	1.0749	0.1720	1.2469
		<i>Hg-202</i>	1.00	1.00	—	1.0749	0.1720	1.2469
		Hg-190	20.0 m	ECB+	—	0.0539	0.2018	0.2557
		Au-190	1.00	1.00	—	0.0539	0.2018	0.2557
		Hg-191m	50.8 m	ECB+	—	0.1378	1.4895	1.6273
		Au-191	1.00	1.00	—	0.1378	1.4895	1.6273
		Hg-192	4.85 h	EC	—	0.0636	0.2749	0.3385
		Au-192	1.00	1.00	—	0.0636	0.2749	0.3385
		Hg-193	3.80 h	ECB+	—	0.0740	0.8377	0.9117
		Au-193	9.6459E-01	9.6459E-01	—	0.0740	0.8377	0.9117
		Au-193m	3.5408E-02	3.5408E-02	—	0.0740	0.8377	0.9117
		Hg-193m	11.8 h	ECB+IT	—	0.0470	1.0239	1.0709
		Au-193m	8.9197E-01	8.9197E-01	—	0.0470	1.0239	1.0709
		Hg-193	7.1000E-02	7.1000E-02	—	0.0470	1.0239	1.0709
		Au-193	3.7027E-02	3.7027E-02	—	0.0470	1.0239	1.0709
		Hg-194	440 y	EC	—	0.0078	0.0027	0.0106
		Au-194	1.00	1.00	—	0.0078	0.0027	0.0106
		Hg-195	10.53 h	ECB+	—	0.0650	0.2008	0.2658
		Au-195	1.00	1.00	—	0.0650	0.2008	0.2658
		Hg-195m	41.6 h	ITECB+	—	0.1480	0.2047	0.3527
		Hg-195	5.4200E-01	5.4200E-01	—	0.1480	0.2047	0.3527
		Au-195	4.5800E-01	4.5800E-01	—	0.1480	0.2047	0.3527
		Hg-197	64.94 h	EC	—	0.0702	0.0740	0.1442
		<i>Au-197</i>	1.00	1.00	—	0.0702	0.0740	0.1442
		Hg-197m	23.8 h	ITEC	—	0.2170	0.0977	0.3148
		Hg-197	9.1400E-01	9.1400E-01	—	0.2170	0.0977	0.3148
		<i>Au-197</i>	8.6000E-02	8.6000E-02	—	0.2170	0.0977	0.3148
		Hg-199m	42.66 m	IT	—	0.3487	0.1836	0.5323
		<i>Hg-199</i>	1.00	1.00	—	0.3487	0.1836	0.5323
		Hg-203	46.612 d	B-	—	0.0990	0.2380	0.3370
		<i>Tl-203</i>	1.00	1.00	—	0.0990	0.2380	0.3370
		Hg-205	5.2 m	B-	—	0.5390	0.0053	0.5444
		<i>Tl-205</i>	1.00	1.00	—	0.5390	0.0053	0.5444
		Hg-206	8.15 m	B-	—	0.4211	0.1215	0.5426
		Tl-206	1.00	1.00	—	0.4211	0.1215	0.5426
		Hg-207	2.9 m	B-	—	0.8262	2.6614	3.4876
		Tl-207	1.00	1.00	—	0.8262	2.6614	3.4876

Table A.1 continued

Z	Element	Nuclide	Half-life	Decay mode	Emitted energy (MeV/nt)			
					Alpha	Electron	Photon	Total
81	Thallium	Tl-190	2.6 m	ECB+	—	1.5288	1.2914	2.8202
		Hg-190	1.00					
		Tl-190m	3.7 m	ECB+	—	0.7985	2.4550	3.2535
		Hg-190	1.00					
		Tl-194	33.0 m	ECB+	—	0.5972	0.9143	1.5115
		Hg-194	1.00					
		Tl-194m	32.8 m	ECB+	—	0.3037	2.5218	2.8255
		Hg-194	1.00					
		Tl-195	1.16 h	ECB+	—	0.0740	1.2250	1.2991
		Hg-195	9.9656E-01					
		Hg-195m	3.4358E-03					
		Tl-196	1.84 h	ECB+	—	0.1782	1.8734	2.0516
		Hg-196	1.00					
		Tl-197	2.84 h	ECB+	—	0.0544	0.4587	0.5130
		Hg-197	1.00					
		Tl-198	5.3 h	ECB+	—	0.0414	2.0108	2.0522
		Hg-198	1.00					
		Tl-198m	1.87 h	ECB+IT	—	0.2006	1.2168	1.4175
		Tl-198	4.6000E-01					
		Hg-198	5.4000E-01					
		Tl-199	7.42 h	ECB+	—	0.0600	0.2520	0.3120
		Hg-199	1.00					
		Tl-200	26.1 h	ECB+	—	0.0408	1.3106	1.3514
		Hg-200	1.00					
		Tl-201	72.912 h	EC	—	0.0447	0.0938	0.1385
		Hg-201	1.00					
		Tl-202	12.23 d	EC	—	0.0233	0.4658	0.4891
		Hg-202	1.00					
		Tl-204	3.78 y	B-EC	—	0.2372	0.0013	0.2385
		Pb-204	9.7100E-01					
		Hg-204	2.9000E-02					
		Tl-206	4.200 m	B-	—	0.5398	0.0001	0.5399
		Pb-206	1.00					
		Tl-206m	3.74 m	IT	—	0.2003	2.4192	2.6195
		Tl-206	1.00					
		Tl-207	4.77 m	B-	—	0.4952	0.0024	0.4975
		Pb-207	1.00					
		Tl-208	3.053 m	B-	—	0.6113	3.3603	3.9716
		Pb-208	1.00					
		Tl-209	2.161 m	B-	—	0.6875	2.1426	2.8302
		Pb-209	1.00					
		Tl-210	1.30 m	B-	—	1.2699	2.7632	4.0331
		Pb-210	1.00					
82	Lead	Pb-194	12.0 m	ECB+A	<E-04	0.0844	1.0832	1.1677
		Tl-194	1.00					
		Hg-190	7.3000E-08					

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Table A.1 continued

Z	Element	Nuclide	Half-life	Decay mode	Emitted energy (MeV/nt)			
					Alpha	Electron	Photon	Total
83	Bismuth	Pb-195m	15 m	ECB+	–	0.3167	1.6547	1.9714
		Tl-195	1.00					
		Pb-196	37 m	ECB+	–	0.0969	0.4940	0.5909
		Tl-196	1.00					
		Pb-197	8 m	ECB+	–	0.0818	1.5319	1.6137
		Tl-197	1.00					
		Pb-197m	43 m	ECB+IT	–	0.2477	1.1732	1.4209
		Tl-197	8.1000E-01					
		Pb-197	1.9000E-01					
		Pb-198	2.4 h	EC	–	0.0781	0.4385	0.5166
		Tl-198	1.00					
		Pb-199	90 m	ECB+	–	0.0584	1.0394	1.0978
		Tl-199	1.00					
		Pb-200	21.5 h	EC	–	0.0997	0.2086	0.3083
		Tl-200	1.00					
		Pb-201	9.33 h	ECB+	–	0.0594	0.7562	0.8156
		Tl-201	1.00					
		Pb-201m	61 s	IT	–	0.2633	0.3658	0.6291
		Pb-201	1.00					
		Pb-202	5.25E+4 y	ECA	0.0260	0.0061	0.0025	0.0346
		Tl-202	9.9000E-01					
		Hg-198	1.0000E-02					
		Pb-202m	3.53 h	ITEC	–	0.1321	1.9925	2.1246
		Pb-202	9.0500E-01					
		Tl-202	9.5000E-02					
		Pb-203	51.873 h	EC	–	0.0530	0.3143	0.3672
		Tl-203	1.00					
		Pb-204m	67.2 m	IT	–	0.1030	2.0634	2.1664
		Pb-204	1.00					
		Pb-205	1.53E+7 y	EC	–	0.0062	0.0025	0.0087
		Tl-205	1.00					
		Pb-209	3.253 h	B-	–	0.1974	–	0.1974
		Bi-209	1.00					
		Pb-210	22.20 y	B-A	<E-04	0.0404	0.0053	0.0457
		Bi-210	1.00					
		Hg-206	1.9000E-08					
		Pb-211	36.1 m	B-	–	0.4543	0.0644	0.5187
		Bi-211	1.00					
		Pb-212	10.64 h	B-	–	0.1766	0.1450	0.3217
		Bi-212	1.00					
		Pb-214	26.8 m	B-	–	0.2948	0.2533	0.5481
		Bi-214	1.00					
		Bi-197	9.3 m	ECB+	–	0.2887	1.6992	1.9879
		Pb-197	5.6103E-01					
		Pb-197m	4.3897E-01					
		Bi-200	36.4 m	ECB+	–	0.2469	2.4355	2.6825
		Pb-200	1.00					

Table A.1 continued

Z	Element	Nuclide	Half-life	Decay mode	Emitted energy (MeV/nt)			
					Alpha	Electron	Photon	Total
		Bi-201	108 m	ECB+	—	0.0612	1.7304	1.7916
		Pb-201		5.4833E-01				
		Pb-201m		4.5167E-01				
		Bi-202	1.72 h	ECB+	—	0.1515	2.7561	2.9076
		Pb-202		1.00				
		Bi-203	11.76 h	ECB+	—	0.0809	2.3850	2.4659
		Pb-203		1.00				
		Bi-204	11.22 h	ECB+	—	0.0807	2.9162	2.9968
		Pb-204m		9.8525E-02				
		Pb-204		9.0148E-01				
		Bi-205	15.31 d	ECB+	—	0.0346	1.6913	1.7259
		Pb-205		1.00				
		Bi-206	6.243 d	ECB+	—	0.1379	3.2796	3.4175
		Pb-206		1.00				
		Bi-207	32.9 y	ECB+	—	0.1193	1.5370	1.6563
		Pb-207		1.00				
		Bi-208	3.68E+5 y	EC	—	0.0144	2.6460	2.6604
		Pb-208		1.00				
		Bi-210	5.013 d	B-A	<E-04	0.3889	<E-04	0.3889
		Po-210		1.00				
		Tl-206		1.3200E-06				
		Bi-210m	3.04E+6 y	A	5.0064	0.0475	0.2607	5.3146
		Tl-206		1.00				
		Bi-211	2.14 m	A B-	6.6757	0.0100	0.0473	6.7330
		Tl-207		9.9724E-01				
		Po-211		2.7600E-03				
		Bi-212	60.55 m	B-A	2.2164	0.5046	0.1038	2.8247
		Po-212		6.4060E-01				
		Tl-208		3.5940E-01				
		Bi-212n	7.0 m	B-	—	0.5351	—	0.5351
		Po-212m		1.00				
		Bi-213	45.59 m	B-A	0.1245	0.4440	0.1277	0.6963
		Po-213		9.7910E-01				
		Tl-209		2.0900E-02				
		Bi-214	19.9 m	B-A	0.0012	0.6631	1.4793	2.1436
		Po-214		9.9979E-01				
		Tl-210		2.1000E-04				
		Bi-215	7.6 m	B-	—	0.6694	0.2534	0.9228
		Po-215		1.00				
		Bi-216	2.17 m	B-	—	1.3295	0.7385	2.0680
		Po-216		1.00				
84	Polonium	Po-203	36.7 m	ECB+A	0.0060	0.1672	1.6348	1.8080
		Bi-203		9.9890E-01				
		Pb-199		1.1000E-03				
		Po-204	3.53 h	ECA	0.0362	0.1839	1.1650	1.3850
		Bi-204		9.9340E-01				
		Pb-200		6.6000E-03				

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Table A.1 continued

Z	Element	Nuclide	Half-life	Decay mode	Emitted energy (MeV/nt)			
					Alpha	Electron	Photon	Total
84	Polonium	Po-205	1.66 h	ECB+A	0.0021	0.0660	1.5846	1.6527
		Bi-205		9.9900E-01				
		Pb-201		4.0000E-04				
		Po-206	8.8 d	ECA	0.2903	0.1655	1.1928	1.6486
		Bi-206		9.4550E-01				
		Pb-202		5.4500E-02				
		Po-207	5.80 h	ECB+A	0.0011	0.0490	1.2847	1.3347
		Bi-207		9.9979E-01				
		Pb-203		2.1000E-04				
		Po-208	2.898 y	AEC	5.2154	<E-04	<E-04	5.2154
		Bi-208		2.2300E-05				
		Pb-204		9.9998E-01				
		Po-209	102 y	AEC	4.9529	0.0030	0.0063	4.9622
		Pb-205		9.9520E-01				
		Bi-209		4.8000E-03				
		Po-210	138.376 d	A	5.4075	<E-04	<E-04	5.4075
		Pb-206		1.00				
		Po-211	0.516 s	A	7.5860	0.0002	0.0082	7.5944
		Pb-207		1.00				
		Po-212	2.99E-7 s	A	8.9541	—	—	8.9541
		Pb-208		1.00				
		Po-212m	45.1 s	A	11.7755	0.0004	0.0792	11.8551
		Pb-208		9.9930E-01				
		Po-213	4.2E-6 s	A	8.5370	<E-04	<E-04	8.5370
		Pb-209		1.00				
		Po-214	1.643E-4 s	A	7.8334	<E-04	<E-04	7.8335
		Pb-210		1.00				
		Po-215	1.781E-3 s	A	7.5261	<E-04	0.0002	7.5263
		Pb-211		1.00				
		Po-216	0.145 s	A	6.9064	<E-04	<E-04	6.9064
		Pb-212		1.00				
		Po-218	3.10 m	A B-	6.1134	<E-04	—	6.1135
		Pb-214		9.9980E-01				
		At-218		2.0000E-04				
85	Astatine	At-204	9.2 m	ECB+A	0.2307	0.4485	2.3234	3.0025
		Po-204		9.6200E-01				
		Bi-200		3.8000E-02				
		At-205	26.2 m	ECB+A	0.6021	0.2575	1.1444	2.0039
		Po-205		9.0000E-01				
		Bi-201		1.0000E-01				
		At-206	30.6 m	ECB+A	0.0517	0.3324	2.4807	2.8647
		Po-206		9.9110E-01				
		Bi-202		8.9000E-03				
		At-207	1.80 h	ECB+A	0.5049	0.1298	2.0136	2.6483
		Po-207		9.1400E-01				
		Bi-203		8.6000E-02				

Table A.1 continued

Z	Element	Nuclide	Half-life	Decay mode	Emitted energy (MeV/nt)			
					Alpha	Electron	Photon	Total
86	Radon	At-208	1.63 h	ECB+A	0.0316	0.1598	3.0408	3.2322
		Po-208		9.9450E-01				
		Bi-204		5.5000E-03				
		At-209	5.41 h	ECB+A	0.2360	0.1172	2.2846	2.6378
		Po-209		9.5900E-01				
		Bi-205		4.1000E-02				
		At-210	8.1 h	ECB+A	0.0097	0.0796	2.9622	3.0515
		Po-210		9.9825E-01				
		Bi-206		1.7500E-03				
		At-211	7.214 h	ECA	2.4998	0.0059	0.0367	2.5424
		Po-211		5.8200E-01				
		Bi-207		4.1800E-01				
		At-215	1.00E-4 s	A	8.1778	<E-04	0.0002	8.1780
		Bi-211		1.00				
		At-216	3.00E-4 s	A	7.9407	0.0013	0.0025	7.9446
		Bi-212		1.00				
		At-217	3.23E-2 s	A	7.2008	<E-04	0.0002	7.2011
		Bi-213		9.9988E-01				
		At-218	1.5 s	A B-	6.8042	0.0011	—	6.8053
		Bi-214		9.9900E-01				
		Rn-218		1.0000E-03				
		At-219	56 s	A	6.1343	—	—	6.1343
		Bi-215		9.7000E-01				
		At-220	3.71 m	B-A	0.4842	1.2130	0.4497	2.1469
		Rn-220		9.2000E-01				
		Bi-216		8.0000E-02				
		Rn-207	9.25 m	ECB+A	1.3126	0.2197	0.9854	2.5177
		At-207		7.9000E-01				
		Po-203		2.1000E-01				
		Rn-209	28.5 m	ECB+A	1.0466	0.1167	1.1953	2.3586
		At-209		8.3000E-01				
		Po-205		1.7000E-01				
		Rn-210	2.4 h	AEC	5.9121	0.0091	0.0610	5.9821
		Po-206		9.6000E-01				
		At-210		4.0000E-02				
		Rn-211	14.6 h	ECB+A	1.6205	0.0663	1.8727	3.5595
		At-211		7.2600E-01				
		Po-207		2.7400E-01				
		Rn-212	23.9 m	A	6.3847	<E-04	0.0003	6.3850
		Po-208		1.00				
		Rn-215	2.30 us	A	8.8390	—	—	8.8390
		Po-211		1.00				
		Rn-216	4.5E-5 s	A	8.2000	—	—	8.2000
		Po-212		1.00				
		Rn-217	5.40E-4 s	A	7.8856	—	—	7.8856
		Po-213		1.00				

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Table A.1 continued

Z	Element	Nuclide	Half-life	Decay mode	Emitted energy (MeV/nt)			
					Alpha	Electron	Photon	Total
87	Francium	Rn-218	3.5E-2 s	A	7.2618	<E-04	0.0008	7.2626
		Po-214	1.00					
		Rn-219	3.96 s	A	6.8801	0.0068	0.0586	6.9456
		Po-215	1.00					
		Rn-220	55.6 s	A	6.4040	<E-04	0.0006	6.4047
		Po-216	1.00					
		Rn-222	3.8235 d	A	5.5899	<E-04	0.0004	5.5903
		Po-218	1.00					
		Rn-223	24.3 m	B-	—	0.6282	0.3444	0.9726
		Fr-223	1.00					
		Fr-212	20.0 m	ECB+A	2.7731	0.1294	1.1415	4.0440
		Rn-212	5.7000E-01					
		At-208	4.3000E-01					
		Fr-219	2.0E-2 s	A	7.4437	0.0004	0.0036	7.4477
		At-215	1.00					
		Fr-220	27.4 s	A B-	6.7413	0.0163	0.0105	6.7680
		At-216	9.9650E-01					
		Ra-220	3.5000E-03					
		Fr-221	4.9 m	A	6.4199	0.0089	0.0294	6.4582
		At-217	1.00					
88	Radium	Fr-222	14.2 m	B-	—	0.7145	0.1806	0.8951
		Ra-222	1.00					
		Fr-223	22.00 m	B-A	0.0003	0.3829	0.0583	0.4415
		Ra-223	1.00					
		At-219	6.0000E-05					
		Fr-224	3.33 m	B-	—	0.8751	0.5523	1.4274
		Ra-224	1.00					
		Fr-227	2.47 m	B-	—	0.7967	0.4499	1.2466
		Ra-227	1.00					
		Ra-219	10 ms	A	7.9071	0.0600	0.1702	8.1372
		Rn-215	1.00					
		Ra-220	1.79E-2 s	A	7.5904	0.0002	0.0047	7.5952
		Rn-216	1.00					
		Ra-221	28 s	A	6.7915	0.0690	0.0390	6.8995
		Rn-217	1.00					
		Ra-222	38.0 s	A	6.6710	0.0009	0.0092	6.6811
		Rn-218	1.00					
		Ra-223	11.43 d	A	5.7702	0.0781	0.1413	5.9895
		Rn-219	1.00					
		Ra-224	3.66 d	A	5.7766	0.0023	0.0104	5.7893
		Rn-220	1.00					
		Ra-225	14.9 d	B-	—	0.1050	0.0145	0.1194
		Ac-225	1.00					
		Ra-226	1600 y	A	4.8603	0.0039	0.0074	4.8716
		Rn-222	1.00					
		Ra-227	42.2 m	B-	—	0.4511	0.1508	0.6019
		Ac-227	1.00					

Table A.1 continued

Z	Element	Nuclide	Half-life	Decay mode	Emitted energy (MeV/nt)			
					Alpha	Electron	Photon	Total
89	Actinium	Ra-228	5.75 y	B-	—	0.0132	0.0031	0.0163
		Ac-228	1.00	1.00	—	0.2201	0.0794	0.2995
		Ra-230	93 m	B-	—	0.2201	0.0794	0.2995
		Ac-230	1.00	1.00	—	0.2201	0.0794	0.2995
		Ac-223	2.10 m	A	6.6721	0.0254	0.0190	6.7165
		Fr-219	9.9000E-01	9.9000E-01	6.6721	0.0254	0.0190	6.7165
		Ac-224	2.78 h	ECA	0.5662	0.0490	0.2325	0.8477
		Ra-224	9.0900E-01	9.0900E-01	0.5662	0.0490	0.2325	0.8477
		Fr-220	9.1000E-02	9.1000E-02	0.5662	0.0490	0.2325	0.8477
		Ac-225	10.0 d	A	5.8920	0.0248	0.0171	5.9338
		Fr-221	1.00	1.00	5.8920	0.0248	0.0171	5.9338
		Ac-226	29.37 h	B-ECA	0.0003	0.2914	0.1327	0.4245
		Th-226	8.3000E-01	8.3000E-01	0.0003	0.2914	0.1327	0.4245
		Ra-226	1.7000E-01	1.7000E-01	0.0003	0.2914	0.1327	0.4245
		Fr-222	6.0000E-05	6.0000E-05	0.0003	0.2914	0.1327	0.4245
		Ac-227	21.772 y	B-A	0.0693	0.0150	0.0011	0.0853
		Th-227	9.8620E-01	9.8620E-01	0.0693	0.0150	0.0011	0.0853
		Fr-223	1.3800E-02	1.3800E-02	0.0693	0.0150	0.0011	0.0853
		Ac-228	6.15 h	B-	—	0.4495	0.8671	1.3166
		Th-228	1.00	1.00	—	0.4495	0.8671	1.3166
90	Thorium	Ac-230	122 s	B-	—	0.9229	0.5440	1.4668
		Th-230	1.00	1.00	—	0.9229	0.5440	1.4668
		Ac-231	7.5 m	B-	—	0.6361	0.4190	1.0550
		Th-231	1.00	1.00	—	0.6361	0.4190	1.0550
		Ac-232	119 s	B-	—	0.9707	1.1528	2.1235
		Th-232	1.00	1.00	—	0.9707	1.1528	2.1235
		Ac-233	145 s	B-	—	0.8355	0.4993	1.3348
		Th-233	1.00	1.00	—	0.8355	0.4993	1.3348
		Th-223	0.60 s	A	7.4134	0.0575	0.0754	7.5463
		Ra-219	1.00	1.00	7.4134	0.0575	0.0754	7.5463
		Th-224	1.05 s	A	7.2635	0.0129	0.0232	7.2996
		Ra-220	1.00	1.00	7.2635	0.0129	0.0232	7.2996
		Th-226	30.57 m	A	6.4219	0.0211	0.0089	6.4519
		Ra-222	1.00	1.00	6.4219	0.0211	0.0089	6.4519
		Th-227	18.68 d	A	5.9883	0.0755	0.1317	6.1955
		Ra-223	1.00	1.00	5.9883	0.0755	0.1317	6.1955
		Th-228	1.9116 y	A	5.4956	0.0210	0.0036	5.5202
		Ra-224	1.00	1.00	5.4956	0.0210	0.0036	5.5202
		Th-229	7.34E+3 y	A	4.9584	0.1217	0.0971	5.1772
		Ra-225	1.00	1.00	4.9584	0.1217	0.0971	5.1772
		Th-230	7.538E+4 y	A	4.7538	0.0146	0.0018	4.7702
		Ra-226	1.00	1.00	4.7538	0.0146	0.0018	4.7702
		Th-231	25.52 h	B-	—	0.1622	0.0269	0.1891
		Pa-231	1.00	1.00	—	0.1622	0.0269	0.1891
		Th-232	1.405E10 y	A	4.0688	0.0126	0.0015	4.0829
		Ra-228	1.00	1.00	4.0688	0.0126	0.0015	4.0829

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Table A.1 continued

Z	Element	Nuclide	Half-life	Decay mode	Emitted energy (MeV/nt)			
					Alpha	Electron	Photon	Total
91	Protactinium	Th-233	22.3 m	B-	—	0.4140	0.0375	0.4515
		Pa-233		1.00				
		Th-234	24.10 d	B-	—	0.0622	0.0105	0.0728
		Pa-234m		1.00				
		Th-235	7.1 m	B-	—	0.6713	0.0537	0.7251
		Pa-235		1.00				
		Th-236	37.5 m	B-	—	0.3671	0.0346	0.4017
		Pa-236		1.00				
		Pa-227	38.3 m	AEC	5.5650	0.0226	0.0234	5.6110
		Ac-223		8.5000E-01				
		Th-227		1.5000E-01				
		Pa-228	22 h	ECB+A	0.1217	0.1320	1.3694	1.6230
		Th-228		9.8000E-01				
		Ac-224		2.0000E-02				
		Pa-229	1.50 d	ECA	0.0274	0.0131	0.0666	0.1071
		Th-229		9.9520E-01				
		Ac-225		4.8000E-03				
		Pa-230	17.4 d	ECB-A	0.0002	0.0668	0.6708	0.7377
		Th-230		9.1600E-01				
		U-230		8.4000E-02				
		Ac-226		3.2000E-05				
		Pa-231	3.276E+4 y	A	5.0592	0.0538	0.0450	5.1580
		Ac-227		1.00				
		Pa-232	1.31 d	B-EC	—	0.1738	0.9393	1.1131
		U-232		1.00				
		Th-232		3.0000E-05				
		Pa-233	26.967 d	B-	—	0.2151	0.2229	0.4380
		U-233		1.00				
		Pa-234	6.70 h	B-	—	0.4037	1.4718	1.8755
		U-234		1.00				
		Pa-234m	1.17 m	B-IT	—	0.8171	0.0162	0.8334
		U-234		9.9840E-01				
		Pa-234		1.6000E-03				
		Pa-235	24.5 m	B-	—	0.4886	0.0008	0.4894
		U-235m		9.9990E-01				
		U-235		1.0109E-04				
		Pa-236	9.1 m	B-	—	0.8049	0.9148	1.7197
		U-236		1.00				
		Pa-237	8.7 m	B-	—	0.5780	0.6095	1.1875
		U-237		1.00				
92	Uranium	U-227	1.1 m	A	6.9984	0.0960	0.1199	7.2143
		Th-223		1.00				
		U-228	9.1 m	A	6.6046	0.0231	0.0056	6.6333
		Th-224		9.7500E-01				
		U-230	20.8 d	A	5.9681	0.0216	0.0032	5.9929
		Th-226		1.00				

Table A.1 continued

Z	Element	Nuclide	Half-life	Decay mode	Emitted energy (MeV/nt)			
					Alpha	Electron	Photon	Total
93	Neptunium	U-231	4.2 d	ECA	0.0002	0.0847	0.0896	0.1745
		Pa-231	1.00					
		Th-227	4.0000E-05					
		U-232	68.9 y	A	5.3948	0.0164	0.0023	5.4135
		Th-228	1.00					
		U-233	1.592E+5 y	A	4.9013	0.0059	0.0013	4.9085
		Th-229	1.00					
		U-234	2.455E+5 y	A	4.8430	0.0137	0.0020	4.8587
		Th-230	1.00					
		U-235	7.04E+8 y	A	4.4693	0.0530	0.1669	4.6891
		Th-231	1.00					
		U-235m	26 m	IT	—	<E-04	<E-04	<E-04
		U-235	1.00					
		U-236	2.342E+7 y	A	4.5592	0.0114	0.0018	4.5723
		Th-232	1.00					
		U-237	6.75 d	B-	—	0.1991	0.1442	0.3433
		Np-237	1.00					
		U-238	4.468E+9 y	ASF	4.2584	0.0092	0.0014	4.2691
		Th-234	1.00					
		SF	5.4500E-07					
		U-239	23.45 m	B-	—	0.4108	0.0519	0.4626
		Np-239	1.00					
		U-240	14.1 h	B-	—	0.1276	0.0099	0.1374
		Np-240m	1.00					
		U-242	16.8 m	B-	—	0.3859	0.0413	0.4272
		Np-242	1.00					
		Np-232	14.7 m	ECB+	—	0.1073	1.1969	1.3042
		U-232	1.00					
		Np-233	36.2 m	ECA	<E-04	0.0144	0.0911	0.1055
		U-233	1.00					
		Pa-229	1.0000E-05					
		Np-234	4.4 d	ECB+	—	0.0574	1.1086	1.1660
		U-234	1.00					
		Np-235	396.1 d	ECA	0.0001	0.0105	0.0071	0.0178
		U-235	9.9598E-01					
		U-235m	3.9933E-03					
		Pa-231	2.6000E-05					
		Np-236	1.54E+5 y	ECB-A	0.0074	0.2372	0.1594	0.4039
		U-236	8.7300E-01					
		Pu-236	1.2500E-01					
		Pa-232	1.6000E-03					
		Np-236m	22.5 h	ECB-	—	0.0880	0.0507	0.1387
		U-236	5.2000E-01					
		Pu-236	4.8000E-01					
		Np-237	2.144E+6 y	A	4.8499	0.0681	0.0350	4.9529
		Pa-233	1.00					

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Table A.1 continued

Z	Element	Nuclide	Half-life	Decay mode	Emitted energy (MeV/nt)			
					Alpha	Electron	Photon	Total
94	Plutonium	Np-238	2.117 d	B-	–	0.2519	0.5879	0.8398
		Pu-238		1.00				
		Np-239	2.3565 d	B-	–	0.2623	0.1846	0.4469
		Pu-239		1.00				
		Np-240	61.9 m	B-	–	0.5095	1.0538	1.5632
		Pu-240		1.00				
		Np-240m	7.22 m	B-IT	–	0.6779	0.3225	1.0004
		Pu-240		9.9890E-01				
		Np-240		1.1000E-03				
		Np-241	13.9 m	B-	–	0.4341	0.0395	0.4736
		Pu-241		1.00				
		Np-242	2.2 m	B-	–	0.9027	0.2654	1.1681
		Pu-242		1.00				
		Np-242m	5.5 m	B-	–	0.7551	0.9194	1.6744
		Pu-242		1.00				
		Pu-232	33.7 m	ECA	1.5402	0.0087	0.0633	1.6122
		Np-232		7.7000E-01				
		U-228		2.3000E-01				
		Pu-234	8.8 h	ECA	0.3776	0.0114	0.0693	0.4583
		Np-234		9.4000E-01				
		U-230		6.0000E-02				
		Pu-235	25.3 m	ECA	0.0002	0.0228	0.0957	0.1187
		Np-235		9.9997E-01				
		U-231		2.7000E-05				
		Pu-236	2.858 y	ASF	5.8524	0.0128	0.0022	5.8674
		U-232		1.00				
		SF		1.3700E-09				
		Pu-237	45.2 d	ECA	0.0002	0.0171	0.0537	0.0710
		Np-237		1.00				
		U-233		4.2000E-05				
		Pu-238	87.7 y	ASF	5.5803	0.0107	0.0021	5.5930
		U-234		1.00				
		SF		1.8500E-09				
		Pu-239	2.411E+4 y	A	5.2357	0.0075	0.0011	5.2442
		U-235m		9.9940E-01				
		U-235		6.0000E-04				
		Pu-240	6564 y	ASF	5.2434	0.0105	0.0019	5.2559
		U-236		1.00				
		SF		5.7500E-08				
		Pu-241	14.35 y	B-A	0.0001	0.0052	<E-04	0.0054
		Am-241		9.9998E-01				
		U-237		2.4500E-05				
		Pu-242	3.75E+5 y	ASF	4.9738	0.0090	0.0017	4.9855
		U-238		1.00				
		SF		5.5400E-06				
		Pu-243	4.956 h	B-	–	0.1729	0.0259	0.1988
		Am-243		1.00				

Table A.1 continued

Z	Element	Nuclide	Half-life	Decay mode	Emitted energy (MeV/nt)			
					Alpha	Electron	Photon	Total
95	Americium	Pu-244	8.00E+7 y	ASF	4.6513	0.0197	0.0211	4.9094
		U-240		9.9879E-01				
		SF		1.2100E-03				
		Pu-245	10.5 h	B-	—	0.3190	0.4027	0.7217
		Am-245		1.00				
		Pu-246	10.84 d	B-	—	0.1159	0.1431	0.2590
		Am-246m		1.00				
		Am-237	73.0 m	ECA	0.0015	0.0802	0.3714	0.4531
		Pu-237		9.9975E-01				
		Np-233		2.5000E-04				
		Am-238	98 m	ECB+A	<E-04	0.0485	0.9023	0.9509
		Pu-238		1.00				
		Np-234		1.0000E-06				
		Am-239	11.9 h	ECA	0.0006	0.1709	0.2436	0.4150
		Pu-239		9.9990E-01				
		Np-235		1.0000E-04				
		Am-240	50.8 h	ECA	<E-04	0.0758	1.0355	1.1112
		Pu-240		1.00				
		Np-236		1.9000E-06				
		Am-241	432.2 y	A	5.5712	0.0373	0.0293	5.6379
		Np-237		1.00				
		Am-242	16.02 h	B-EC	—	0.1806	0.0188	0.1994
		Cm-242		8.2700E-01				
		Pu-242		1.7300E-01				
		Am-242m	141 y	ITA	0.0238	0.0439	0.0056	0.0734
		Am-242		9.9550E-01				
		Np-238		4.5000E-03				
		Am-243	7.37E+3 y	A	5.3583	0.0234	0.0585	5.4402
		Np-239		1.00				
		Am-244	10.1 h	B-	—	0.3330	0.8052	1.1382
		Cm-244		1.00				
		Am-244m	26 m	B-	—	0.5187	0.0172	0.5359
		Cm-244		9.9960E-01				
		Am-245	2.05 h	B-	—	0.2874	0.0324	0.3198
		Cm-245		1.00				
		Am-246	39 m	B-	—	0.7241	0.7498	1.4739
		Cm-246		1.00				
		Am-246m	25.0 m	B-	—	0.5033	0.9799	1.4832
		Cm-246		1.00				
		Am-247	23.0 m	B-	—	0.5683	0.1348	0.7031
		Cm-247		1.00				
96	Curium	Cm-238	2.4 h	ECA	0.2537	0.0117	0.0829	0.3482
		Am-238		9.6160E-01				
		Pu-234		3.8400E-02				
		Cm-239	2.9 h	ECB+	—	0.0291	0.2593	0.2883
		Am-239		1.00				

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Table A.1 continued

Z	Element	Nuclide	Half-life	Decay mode	Emitted energy (MeV/nt)			
					Alpha	Electron	Photon	Total
97	Berkelium	Cm-240	27 d	ASF	6.3650	0.0108	0.0022	6.3781
		Pu-236		9.9700E-01				
		SF		3.9000E-08				
		Cm-241	32.8 d	ECA	0.0603	0.1342	0.5034	0.6979
		Am-241		9.9000E-01				
		Pu-237		1.0000E-02				
		Cm-242	162.8 d	ASF	6.2041	0.0096	0.0020	6.2156
		Pu-238		1.00				
		SF		6.3700E-08				
		Cm-243	29.1 y	AEC	5.8929	0.1342	0.1353	6.1624
		Pu-239		9.9760E-01				
		Am-243		2.4000E-03				
		Cm-244	18.10 y	ASF	5.8915	0.0079	0.0017	5.9014
		Pu-240		1.00				
		SF		1.3710E-06				
		Cm-245	8.5E+3 y	ASF	5.4474	0.0824	0.1084	5.6382
		Pu-241		1.00				
		SF		6.1000E-09				
		Cm-246	4.76E+3 y	ASF	5.4654	0.0085	0.0050	5.5285
		Pu-242		9.9974E-01				
		SF		2.6300E-04				
		Cm-247	1.56E+7 y	A	5.0292	0.0114	0.3138	5.3544
		Pu-243		1.00				
		Cm-248	3.48E+5 y	ASF	4.7212	0.7716	1.3127	22.5608
		Pu-244		9.1610E-01				
		SF		8.3900E-02				
		Cm-249	64.15 m	B-	—	0.2835	0.0200	0.3035
		Bk-249		1.00				
		Cm-250	8300 y	AB-SF	0.9283	8.4270	13.3167	161.287
		Pu-246		1.8000E-01				
		Bk-250		8.0000E-02				
		SF		7.4000E-01				
		Cm-251	16.8 m	B-	—	0.4545	0.1112	0.5657
		Bk-251		1.00				
		Bk-245	4.94 d	ECA	0.0075	0.1326	0.2352	0.3752
		Cm-245		9.9880E-01				
		Am-241		1.2000E-03				
		Bk-246	1.80 d	EC	—	0.0554	0.8546	0.9101
		Cm-246		1.00				
		Bk-247	1.38E+3 y	A	5.7031	0.0691	0.1468	5.9190
		Am-243		1.00				
		Bk-248m	23.7 h	B-EC	—	0.1910	0.0559	0.2470
		Cf-248		7.0000E-01				
		Cm-248		3.0000E-01				
		Bk-249	330 d	B-A	<E-04	0.0324	<E-04	0.0325
		Cf-249		1.00				
		Am-245		1.4500E-05				

Table A.1 continued

Z	Element	Nuclide	Half-life	Decay mode	Emitted energy (MeV/nt)			
					Alpha	Electron	Photon	Total
98	Californium	Bk-250	3.212 h	B-	—	0.2949	0.8983	1.1932
		Cf-250	1.00	1.00	—	0.3693	0.0915	0.4608
		Bk-251	55.6 m	B-	—	0.3693	0.0915	0.4608
		Cf-251	1.00	1.00	—	0.3693	0.0915	0.4608
		Cf-244	19.4 m	A	7.3196	0.0075	0.0020	7.3291
		Cm-240	1.00	1.00	—	0.3693	0.0915	0.4608
		Cf-246	35.7 h	ASF	6.8527	0.0060	0.0014	6.8606
		Cm-242	1.00	1.00	—	0.3693	0.0915	0.4608
		SF	2.5000E-06	2.5000E-06	—	0.3693	0.0915	0.4608
		Cf-247	3.11 h	ECA	0.0022	0.0463	0.1050	0.1536
		Bk-247	9.9965E-01	9.9965E-01	—	0.3693	0.0915	0.4608
		Cm-243	3.5000E-04	3.5000E-04	—	0.3693	0.0915	0.4608
		Cf-248	334 d	ASF	6.3518	0.0075	0.0020	6.3670
		Cm-244	9.9997E-01	9.9997E-01	—	0.3693	0.0915	0.4608
		SF	2.9000E-05	2.9000E-05	—	0.3693	0.0915	0.4608
		Cf-249	351 y	ASF	5.9262	0.0399	0.3282	6.2944
		Cm-245	1.00	1.00	—	0.3693	0.0915	0.4608
		SF	5.0200E-09	5.0200E-09	—	0.3693	0.0915	0.4608
		Cf-250	13.08 y	ASF	6.1168	0.0103	0.0111	6.2897
		Cm-246	9.9923E-01	9.9923E-01	—	0.3693	0.0915	0.4608
		SF	7.7000E-04	7.7000E-04	—	0.3693	0.0915	0.4608
		Cf-251	900 y	A	5.8803	0.1705	0.1245	6.1754
		Cm-247	1.00	1.00	—	0.3693	0.0915	0.4608
		Cf-252	2.645 y	ASF	6.0177	0.2516	0.4572	12.8107
		Cm-248	9.6908E-01	9.6908E-01	—	0.3693	0.0915	0.4608
		SF	3.0920E-02	3.0920E-02	—	0.3693	0.0915	0.4608
99	Einsteinium	Cf-253	17.81 d	B-A	0.0188	0.0908	0.0048	0.1144
		Es-253	9.9690E-01	9.9690E-01	—	0.3693	0.0915	0.4608
		Cm-249	3.1000E-03	3.1000E-03	—	0.3693	0.0915	0.4608
		Cf-254	60.5 d	ASF	0.0183	10.0442	16.8399	222.890
		Cm-250	3.1000E-03	3.1000E-03	—	0.3693	0.0915	0.4608
		SF	9.9690E-01	9.9690E-01	—	0.3693	0.0915	0.4608
		Cf-255	85 m	B-	—	0.2178	—	0.2178
		Es-255	1.00	1.00	—	0.2178	—	0.2178
		Es-249	102.2 m	ECB+A	0.0392	0.0437	0.4129	0.4958
		Cf-249	9.9430E-01	9.9430E-01	—	0.3693	0.0915	0.4608
		Bk-245	5.7000E-03	5.7000E-03	—	0.3693	0.0915	0.4608
		Es-250	8.6 h	EC	—	0.3281	1.2235	1.5516
		Cf-250	9.8500E-01	9.8500E-01	—	0.3693	0.0915	0.4608
		Es-250m	2.22 h	ECB+	—	0.0343	0.5549	0.5892
		Cf-250	1.00	1.00	—	0.3693	0.0915	0.4608
		Es-251	33 h	ECA	0.0329	0.0522	0.1016	0.1868
		Cf-251	9.9500E-01	9.9500E-01	—	0.3693	0.0915	0.4608
		Bk-247	5.0000E-03	5.0000E-03	—	0.3693	0.0915	0.4608
		Es-253	20.47 d	ASF	6.7335	0.0022	0.0008	6.7366
		Bk-249	1.00	1.00	—	0.3693	0.0915	0.4608
		SF	8.9000E-08	8.9000E-08	—	0.3693	0.0915	0.4608

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Table A.1 continued

Z	Element	Nuclide	Half-life	Decay mode	Emitted energy (MeV/nt)			
					Alpha	Electron	Photon	Total
100	Fermium	Es-254	275.7 d	A B-SF	6.5244	0.0727	0.0208	6.6179
		Bk-250	1.00					
		Fm-254	1.7400E-06					
		SF	3.0000E-08					
		Es-254m	39.3 h	B-AECSF	0.0208	0.2408	0.4757	0.8269
		Fm-254	9.8000E-01					
		Bk-250	3.2000E-03					
		Cf-254	7.6000E-04					
		SF	4.5000E-04					
		Es-255	39.8 d	B-ASF	0.5117	0.0737	0.0007	0.5950
		Fm-255	9.2000E-01					
		Bk-251	8.0000E-02					
		SF	4.5000E-05					
		Es-256	25.4 m	B-	—	0.5822	0.0032	0.5854
		Fm-256	1.00					
		Fm-251	5.30 h	ECB+A	0.1252	0.0337	0.1588	0.3176
		Es-251	9.8200E-01					
		Cf-247	1.8000E-02					
		Fm-252	25.39 h	ASF	7.1449	0.0064	0.0018	7.1578
		Cf-248	9.9998E-01					
		SF	2.3000E-05					
		Fm-253	3.00 d	ECA	0.8352	0.1084	0.0713	1.0150
		Es-253	8.8000E-01					
		Cf-249	1.2000E-01					
		Fm-254	3.240 h	ASF	7.2956	0.0095	0.0086	7.4339
		Cf-250	9.9941E-01					
		SF	5.9200E-04					
		Fm-255	20.07 h	ASF	7.1292	0.0954	0.0170	7.2417
		Cf-251	1.00					
		SF	2.3000E-07					
		Fm-256	157.6 m	ASF	0.5686	6.3116	12.4371	205.496
		Cf-252	8.1000E-02					
		SF	9.1900E-01					
		Fm-257	100.5 d	ASF	6.6154	0.1471	0.1521	7.3388
		Cf-253	9.9790E-01					
		SF	2.1000E-03					

USER GUIDE TO THE ICRP CD AND THE DECDATA SOFTWARE

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1. Basic Information About the CD and Software Package

The ICRP publication entitled “Decay Data for Dosimetric Calculations” includes a CD that provides electronic files of the unabridged nuclear decay data for 1252 radionuclides of 97 elements. The CD contains the following:

- README.TXT: A file containing information post preparation of the publication
- CONTENT.PDF: A file detailing the content of the CD
- LICENSE.TXT: A file containing the license agreement and a liability statement
- USERGUIDE.PDF: The user guide to the DECDATA software; this document
- ICRP-07 nuclear decay data files:
 - ICRP-07.NDX
 - ICRP-07.RAD
 - ICRP-07.BET
 - ICRP-07.NSF
 - ICRP-07.ACK
- SETUP.EXE: Executable file for installing DECDATA software
- ARCHIVE folder:
 - INPUT folder: EDISTR04 input files for nuclides of the ICRP collection
 - OUTPUT folder: EDISTR04 output files for nuclides of the ICRP collection
 - SOURCE folder: Source code for DECDATA

The software package DECDATA enables viewing the radiological properties of radionuclides of the ICRP collection in a tabular and graphical manner as no detailed tabulations are included in the publication. Data on the yields and energies of the radiations of a user-specified radionuclide can be extracted into ASCII files for subsequent use. DECDATA is intended for use on a personal computer (PC) with Windows operating system.

The CD contains a file, AUTORUN.INF, which invokes SETUP.EXE to install DECDATA when the CD is placed in the CD drive. If this function has been disabled on the computer then, from Explorer, click on SETUP.EXE to install DECDATA. AUTORUN can be bypassed by holding down the shift key when the CD is placed in the drive.

SETUP.EXE is a Windows installation procedure¹ that installs DECDATA and its data files. After the user agrees to the license statement the procedure creates a folder with the default name ICRP07 on the hard drive, installs the software in the folder, and places the DECDATA icon on the desktop. Three additional folders,

¹ The install package was created using Inno Setup compiler available from <http://www.innosetup.com>.

below the main folder, are created. The decay data files reside in the subfolder DATA, decay scheme drawings for selected files are in the FIGS subfolder, and all output files generated by DECDATA are written to the subfolder OUTPUT. The subfolder REPORT contains documents of potential interest to the user, including this document.

DECDATA is a console application developed using PowerBASICTM Console Compiler² with Perfect Sync, Inc³ modules Console ToolsTM and Graphics ToolsTM providing additional functionality. Graphical support is provided by the DPlot Jr⁴ module. Documents in the REPORT subfolder can be selected for viewing within DECDATA, as can some of DECDATA output files. This capability depends on the user having associated files with file extensions *pdf* and *txt* with Acrobat Reader⁵ and an ASCII editor (such as Microsoft's Notepad), respectively.

² PowerBASIC, Inc.; 1978 Tamiami Trail S. #200; Venice, FL 34293. See <http://www.powerbasic.com/>.

³ Perfect Sync, Inc. 6511 Franklin Woods Dr., Traverse City, MI. See <http://perfectsync.com/>.

⁴ HydeSoft Computing, Inc., 110 Roseland Drive, Vicksburg, MS 39180. See <http://www.dplot.com/index.htm>.

⁵ Adobe Acrobat Reader of Adobe Inc. is available from <http://www.pdf-2007.com/index.asp>.

2. Information on the Nuclear Decay Data Files

The nuclear decay data are embodied in five formatted (hence readable with an ASCII editor) direct-access files, each with the root name ICRP-07. The file ICRP-07.RAD contains the data on the absolute yields (or intensities) and mean⁶ or discrete energies of the radiations emitted by the 1252 nuclides of the ICRP-07 collection. This file contains data for all emitted radiations; i.e., no cutoff on the number of radiations has been applied to the data of the electronic files as would be required in printed tabulations.

The file ICRP-07.BET contains the beta spectra for all beta emitters in the collection. EDISTR04 (Dillman, 1980; Endo *et al.* 2005), the software used to compute the energies and yields of the emitted radiations, computes the spectrum for each beta transition to determine the mean beta energy of the transitions and tabulates the composite spectra for all beta transitions of the radionuclide. Only the composite spectra are included in this file.

The file ICRP-07.ACK contains the Auger and Coster-Kronig (CK) electron spectra for selected radionuclides. These discrete emissions were collected into no more than 15 groups for each decay mode in the ICRP-07.RAD file. Hereafter Auger and Coster-Kronig (CK) is denoted as Auger-CK.

The spectra of neutrons emitted during spontaneous fission are contained in the file ICRP-07.NSF. In the ICRP-07 collection, 28 radionuclides decay by spontaneous fission.

To facilitate access to the data in these files, an additional file ICRP-07.NDX was constructed. A brief description of each of these five files is presented below. For convenience, the files are referred to by their file extension, i.e., NDX, RAD, BET, ACK, and NSF.

Index File: ICRP-07.NDX

The NDX file serves as the entrance into the larger radiation (RAD) and spectral (BET, ACK, and NSF) files. The NDX file contains one record for each nuclide of the collection. Fields of the nuclide record specify the location of the nuclide's records in the RAD, BET, ACK, and NSF files. In addition to these so-called pointers, the record contains fields giving the nuclide's physical half-life, decay mode (e.g., alpha or beta), identity of progeny (daughters), fraction of the nuclear transformations forming each daughter (so-called branching fraction), the total energies emitted by alpha, electron, and photon emissions (including electrons and photons accompanying spontaneous fission), and other supportive data. A full description of the NDX records is given in Table 1.

The records of the index file have been sorted by the nuclide field, making it possible for user-developed software to locate the record for a radionuclide of interest using a binary search algorithm. While the purpose of the NDX file is to provide entrance into the other data files, it is of considerable utility in its own right. For

⁶ Mean values are reported for beta particles, Auger and Coster-Kronig electron groups, and the radiations emitted during spontaneous fission – fission fragments, neutrons, and delayed beta emission.

Table 1. Structure of Records in ICRP-07.NDX File.

Field	Format ^a	Description
Record 1^b		
First	I4	Record number of first data record
Last	I4	Record number of last data record
Data Records (First,..., Last)^c		
Nuclide	A7	Name of nuclide (parent); e.g., Am-241, Tc-99m
Half-life	A8	Half-life of nuclide
Units	A2	Half-life units: μ s - microsecond, ms - millisecond, s - second, m - minute, d - day, and y - year
Decay Mode	A8	A denotes alpha, B- beta minus, B+ beta plus, EC electron capture, IT internal transition, and SF spontaneous fission
Pointer-1	I7	Location of nuclide in ICRP-07.RAD file
Pointer-2	I7	Location of nuclide in ICRP-07.BET file
Pointer-3	I7	Location of nuclide in ICRP-07.ACK file
Pointer-4	I6	Location of nuclide in ICRP-07.NSF file
The next block of three fields is repeated for radioactive daughter $i, i = 1$ to 4.		
Daughter- i	A7	Name of daughter nuclide i
Record- i	I6	Location of daughter i in ICRP-07.NDX file
Branch- i	E11.0	Branching fraction to daughter i
E-alpha	E7.0	Energy of alpha emissions (MeV/nt)
E-electron	E8.0	Energy of electrons, including beta (MeV/nt)
E-photon	E8.0	Energy of photon emission (MeV/nt)
Number-1	I4	Number of photon of energy less than 10 keV per nt
Number-2	I4	Number of photons of energy greater than 10 keV per nt
Number-3	I4	Number of beta transitions per nt
Number-4	I5	Number of monoenergetic electrons per nt
Number-5	I4	Number of alpha transitions per nt
AMU	E11.0	Atomic mass of radionuclide (Audi <i>et al.</i> 2003)
Γ_{10}	E10.0	Air kerma-rate constant ($\text{Gy}\cdot\text{m}^2 (\text{Bq s})^{-1}$)
K_{air}	E9.0	Air kerma coefficient ($\text{Gy}\cdot\text{m}^2 (\text{Bq s})^{-1}$)

^a The format is expressed in FORTRAN notation, e.g., A8 denotes an alphanumeric field of length 8, I5 an integer field of length 5, and E11.0 a real numeric field of length 11.

^b FORTRAN Format(2I4)

^c FORTRAN Format (A7,A8,A2,A8,3I7,I6,1X,3(A7,I6,E11.0,1X),A7,I6,E11.0,F7.0,2F8.0,3I4,I5,I4,E11.0,E10.0,E9.0)

example, DECDATA constructs the decay chain headed by a radionuclide from information in the NDX file - this is discussed further below.

Both the radioactive and stable daughter nuclei and their branching fractions are included in the file. The Tantalum-180 m daughter, formed in 0.3% of the decays of Hf-180 m is indicated here as stable as in NUBASE2003 (Audi *et al.* 2003). In the NDX file, daughter nuclei with a zero record pointer (Record- i of Table 1) are stable. A list of the radionuclides of the ICRP-07 collection is contained in the REPORT folder (ICRP_Nuclides.TXT).

Radiation File: ICRP-07.RAD

The records of the RAD file contain the data on the energy and yield of each radiation emitted in nuclear transformations of the radionuclide. No cutoff is applied to the number of radiations in this file. The fields of the RAD records are described in

Table 2. Structure of Records in ICRP-07.RAD File.

Field	Format	Description
<u>Nuclide Record</u>		
Nuclide	A7	Nuclide name; e.g., Tc-99m
T _{1/2}	E11.0	Physical half-life of nuclide
Time unit	A2	Units of T _{1/2} , see Table 1
N	I9	Number of data records
<u>Data Record (1, ..., N)</u>		
ICODE	A2	Radiation type (see Table 3)
Yield	E12.0	Yield of the radiation (/nt)
Energy	E12.0	Energy of the radiation (MeV)
JCODE	A3	Mnemonic (see Table 3)

Table 2. The first record (header record) of a nuclide in the file contains the name of the nuclide, its half-life, and the number of data records for each radiation that follow. The fields of the data record are: (1) an integer code (ICODE) that identifies the radiation type; (2) the absolute yield of the radiation (i.e., number per nuclear transformation); (3) the unique or average energy of the radiation; and (4) a two-character mnemonic denoting the radiation type. The ICODE and the mnemonic are defined in Table 3. The nuclide record in the NDX file includes a field, Pointer-1 of Table 1, which contains the record number of header record of the nuclide in the RAD file, a so-called pointer.

The data records for the nuclides are ordered by radiation type, the order of ICODE, and by increasing energy. The radiations group are:

- Photons- ICODE 1-3: x ray, γ ray, annihilation photons, and prompt and delayed photons of spontaneous fission
- Beta particles- ICODE 4-5: mean energy of each beta transition

Table 3. Description of ICODE Variable.

ICODE	Mnemonic For ICODE	Description
1	G	Gamma rays
	PG	Prompt gamma rays*
	DG	Delayed gamma rays*
2	X	X rays
3	AQ	Annihilation photons
4	B+	Beta + particles
5	B –	Beta – particles
	BD	Delayed beta particles*
6	IE	IC electrons
7	AE	Auger electrons
8	A	Alpha particles
9	AR	Alpha recoil nuclei
10	FF	Fission fragments
11	N	Neutrons

* Prompt and delayed radiations of spontaneous fission.

- Monoenergetic electrons- ICODE 6-7: internal conversion and Auger-CK electrons
- Alpha particles- ICODE 8
- Alpha recoil nuclei- ICODE 9
- Fission fragments- ICODE 10
- Neutrons- ICODE 11

The records of each group are sorted by increasing energy. This sorting facilitates interpolation in an energy-dependent function, e.g., the energy-dependent specific absorbed fraction data for the radiation type. The mnemonic can be used to identify a particular radiation type within its group (for example, annihilation photons, delayed gamma rays, etc.).

Beta Spectra File: ICRP-07.BET

The BET file contains the beta spectrum for each beta emitter of the ICRP-07 collection. The spectral data are tabulated on a fixed logarithmic-type energy grid. For each nuclide, the header record gives the name of the nuclide and the number of data records that follow. The fields of the data records contain the electron energy E (MeV) and the number of electrons emitted per nuclear transformation with energy between E and $E + dE$. The structure of the records of the BET file is given in Table 4. A field of the nuclide record in the NDX file, Pointer-2 of Table 1, contains the record number of the nuclide's header record in the BET file.

Auger-CK Electron Spectra File: ICRP-07.ACK

For 136 selected nuclides, detailed Auger-CK electron spectra are contained in the ACK file. These emissions were collapsed into no more than 15 groups for each decay mode in the RAD file. The maximum number of discrete electrons in the spectrum of a radionuclide is 3015 in the case of Hg-195 m and Hg-197 m. The fields of the header record contain the name of the nuclide and the number of data records that follow. The fields of the data records contain the electron energy E (eV), the yield (number of electrons of that energy per nuclear transformation), and an identification of the orbital transition. The data records have been sorted by increasing electron energy. The format of the records is given in Table 5. A list of the nuclides, for which detailed spectra are included in the ACK file, is contained in the REPORT folder (AugerList.TXT). A field of the nuclide record in the NDX file, Pointer-3 of Table 1, contains the record number of the nuclide's header record in the ACK file.

Table 4. Structure of Records in ICRP-07.BET File.

Field	Format	Description
<u>Nuclide Record</u>		
Nuclide	A7	Nuclide name; e.g., Tc-99m
N	I10	Number of data records
<u>Data Record (1, ..., N)</u>		
Energy	F7.0	Energy grid point (MeV)
Number	E10.0	Number of beta particles per MeV per nuclear transformation at this energy

Table 5. Structure of Records in ICRP-07.ACK File.

Field	Format	Description
<u>Nuclide Record</u>		
Nuclide	A7	Nuclide name; e.g., I-123
T _{1/2}	A17	Half-life with its units
N	18	Number of discrete electron lines
<u>Data Record (1, ..., N)</u>		
Yield	E11.0	Number of electrons per nt at this energy
Energy	E12.0	Electron energy (eV)
Transition	A9	Identification of atomic transition; e.g., L1 M2 M3

Table 6. Structure of Records in ICRP-07.NSF File.

Field	Format	Description
<u>Nuclide Record</u>		
Nuclide	A7	Nuclide name; e.g., Cf-252
BF	E13.0	SF branching fraction
N	I9	Number of neutron energy bins
<u>Data Record (1, ..., N)</u>		
Energy-1	E8.0	Lower energy (MeV)
Energy-2	E9.0	Upper energy (MeV)
Yield	E12.0	Number of neutrons per nt in bin

Neutron Spectra File: ICRP-07.NSF

The NSF file contains the spectra of neutrons emitted by radionuclides undergoing spontaneous fission. In the ICRP-07 collection, 28 radionuclides decay by spontaneous fission. For each nuclide, the header record gives the name of the nuclide, the branching fraction for spontaneous fission, and the number of data records that follow. The fields of data records define a neutron energy bin (a lower and upper energy are given) and the number of neutrons per nuclear transformation in that energy bin. The format of the records of the NSF file is given in Table 6. A field of the nuclide record in the NDX file, Pointer-4 of Table 1, contains the record number of the nuclide's header record in the NSF file.

3. Air Kerma for Ideal Point Source

A field of the nuclide record in the NDX file contains the quantity air kerma as a measure of the radiation field in the vicinity of a point source of the radionuclides. Kerma, an acronym for the kinetic energy released per unit mass, is the kinetic energy of charged particles liberated by photon and neutron interactions per unit mass.

3.1. Air Kerma-Rate Constant

The nuclide record in the NDX file has a field containing the air-kerma rate constant. This constant, a characteristic of a radionuclide, is defined in terms of an ideal point source. The International Commission on Radiation Units and Measurements (ICRU 1998) defined the constant as $\Gamma_\delta = l^2 \dot{K}_\delta / A$ where $\dot{K}_\delta$ is the air kerma rate due to photons of energy greater than δ at a distance l in vacuum from a point source of the radionuclide of activity A . The photons referred to include gamma rays, characteristic x rays, and inner bremsstrahlung; the latter is not addressed in the ICRP-07 collection because of its low yield and energy. Annihilation photons, included in the tabulated emissions, are not included in the definition of the constant as they would not be present for an ideal point source in a vacuum. In a source of finite size, attenuation and scattering occur, and external bremsstrahlung can be produced. In addition, any medium between the source and the point of measurement will give rise to absorption and scattering, external bremsstrahlung, and annihilation photons. In many cases, these processes can substantially influence the observed kerma rate. The constant, as defined, is given by

$$\Gamma_\delta = \frac{1}{4\pi} \sum_i (\mu_k/\rho)_i Y_i E_i \quad (1)$$

where $(\mu_k/\rho)_i$ is the mass energy-transfer coefficient in air for photons of energy E_i emitted by the nuclide with yield Y_i . The mass energy-transfer coefficients used to derive the tabulated values are from Shultis and Faw (1999). The value of δ was 10 keV. The air kerma-rate constant is the SI equivalent of the quantity referred to as the specific gamma constant.

3.2. Point Source Air-Kerma Coefficient

As noted above, the air kerma-rate constant does not consider annihilation photons associated with positron emission and the neutrons and photons accompanying spontaneous fission. To address these radiations, the air-kerma coefficient $K_{air,\delta}$ for a hypothetical point source is defined as

$$K_{air,\delta} = \frac{1}{4\pi} \left[\sum_i (\mu_k/\rho)_i Y_i E_i + \sum_i Y(E_i, E_{i+1}) \bar{k}(E_i, E_{i+1}) \right] \quad (2)$$

and is included on the nuclide record in the NDX file. The first summation in the right hand side Eq. (2) extends over all photons of energy greater than δ , including annihilation photons arising from positron emission and the delayed and prompt

gamma-rays accompanying spontaneous fission. The second summation in Eq. (2) is the contribution to air kerma of neutrons accompanying spontaneous fission with yield $Y(E_i, E_{i+1})$ per nuclear transformation and $\bar{k}(E_i, E_{i+1})$ denotes the value of the neutron air kerma coefficient (Chadwick *et al.* 1999) averaged over the energies E_i to E_{i+1} . Only neutrons of energy greater than δ (10 keV) are considered. The constant and the coefficient are numerically equal except in the cases of positron emission or spontaneous fission decay. The cautionary statements above regarding the air kerma-rate constant and the observed kerma rate for a real source in air are applicable to the coefficient.

4. Basic Operation and Features of the DECDATA Software

DECDATA can be invoked by clicking on its desktop icon. To access the records for a radionuclide of interest the user clicks on the chemical symbol of the element in the periodic table. Displayed in the lower right corner of the screen is the name of the element corresponding to the chemical symbol over which the mouse cursor is currently located (the cursor is not shown in Fig. 1). If the ICRP-07 collection contains no radioisotopes of the element (as for He, Li, B, Md, No, and Lr) then the phrase “No Data” appears in the lower right corner of the display. If the mouse is clicked (left click) while over an element then a list box of the radioisotopes of the element in the collection (see Fig. 2) is displayed. If only a single radioisotope of the element is included in the collection, then clicking on that element results in the display of the summary information for the radioisotope – hydrogen is the only such case. After a radioisotope of the element is selected from the list box of multiple isotopes, e.g., Cf-252, the software displays the physical half-life, decay mode, specific activity, and radioactive progeny of the radionuclide followed by a summary tabulation of its emissions as shown in Fig. 3. Graphical display of the decay scheme of 333 radionuclides was published earlier (Eckerman and Endo 2008) and these graphics are included here in the FIGS folder. A message is displayed as seen in Fig. 3 when the graphic exists for the selected radionuclide.

The summary reports the number of records of each radiation type in the RAD file, the yield or number per nuclear transformation (nt), the total emitted energy (MeV per nt), the mean (average) emitted energy (MeV), and the equilibrium

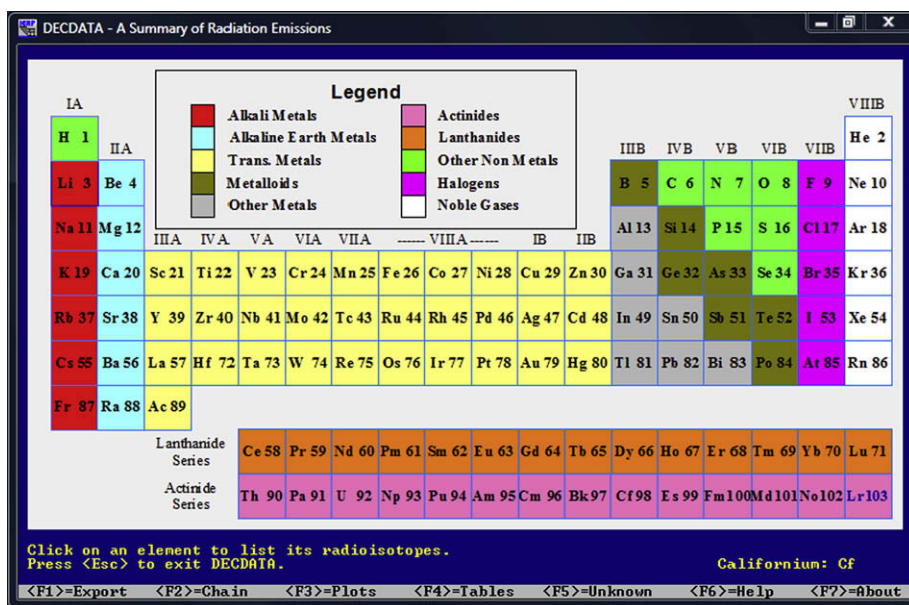


Fig. 1. DECDATA's interface for selecting radioisotopes of the elements in ICRP-07.

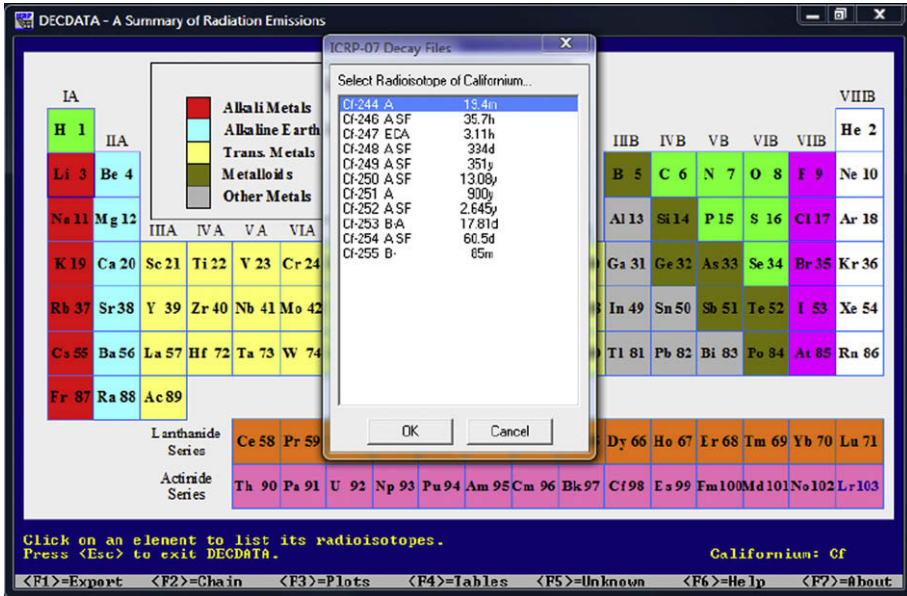


Fig. 2. Menu to select a radioisotope of the selected element.

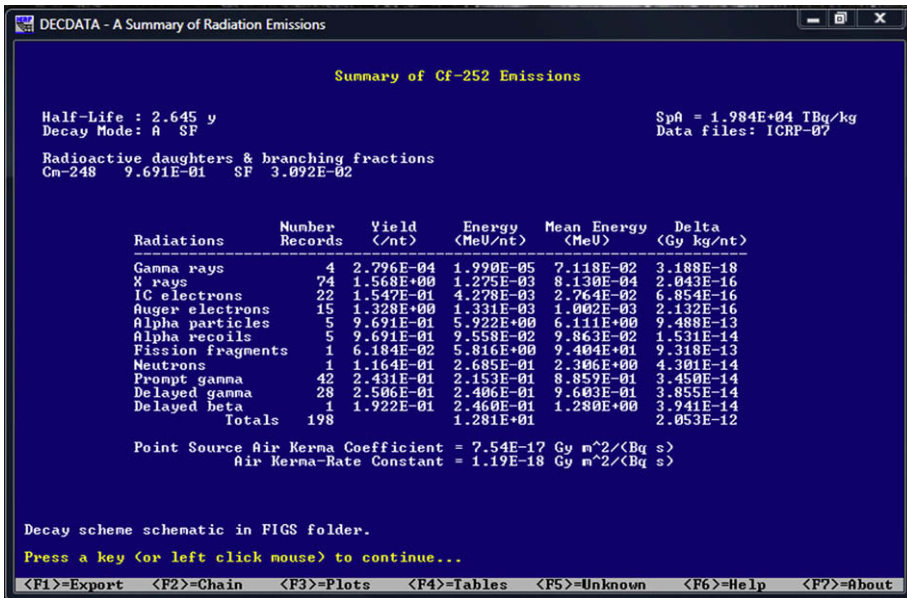


Fig. 3. Summary display of the Cf-252 emissions.

absorbed dose quantity Δ (Gy kg nt⁻¹). For brevity, nuclear transformation (Bq s) is abbreviated as nt in the display. The 14 radiation groups of the summary table are

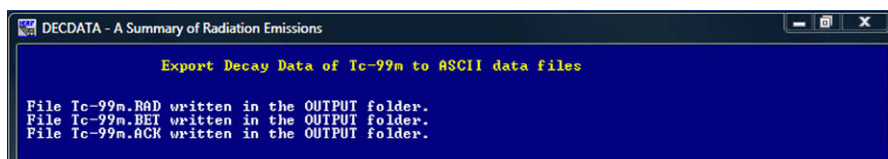
listed as: Gamma rays, X rays, Annh photons, Beta+, Beta-, IC electrons, Auger electrons, Alpha particles, Alpha recoil, Fission fragments, Neutrons, Prompt gamma, Delayed gamma, and Delayed beta. In the event photons of energy greater than 10 keV are emitted, the air kerma-rate constant and air kerma coefficient for a point source are displayed. The latter quantity includes for positron emitters the contributions from annihilation photons and for spontaneous fission the contributions from the neutron and the prompt and delayed gamma radiations.

The last line of the display delineates actions assigned to the function keys F1 through F7. These keys can be pressed at any time. The first three keys act on the displayed nuclide while the other keys are not specific to a nuclide. The key assignments are summarized below and described in more detail in the paragraphs that follow.

Function Key	Action
F1	Export to ASCII files the nuclear decay data for the displayed nuclide
F2	Show the decay chain headed by the displayed nuclide and tabulate the cumulative energy associated with the chain members
F3	Graphically display the beta and neutron spectra (continuous) and line spectra for the discrete energies of alpha particles, IC electrons, Auger-CK electrons, and photon radiations
F4	List the radionuclides in the collection sorted by: a) atomic number, b) half-life, c) total emitted energy, d) alpha emitters, e) beta emitters, f) internal transition (IT), g) spontaneous fission, h) principal alpha emission, i) principal beta transition, j) principal photon emission, k) nuclides with detailed Auger-CK spectra
F5	Given an alpha or photon emission of known energy, a search of the RAD file is carried out in an attempt to identify a radionuclide in the ICRP-07 which emits such a radiation
F6	Display documents in the REPORT folder
F7	Display the usual software credit screen

F1 key: Export nuclide data

If the F1 key is pressed while the summary information for a radionuclide is displayed, then the nuclear decay data for the nuclide is extracted from the data files and written to ASCII files in the OUTPUT folder. In the case where the data from Tc-99m is to be extracted the following files are created:



The names of the output files are constructed from the nuclide name with the file extension being the extension of the ICRP-07 file from which the data were extracted. In each output file, the second line of the file is the header record of the nuclide in the file from which the data are extracted. A partial listing of the Tc-99m.RAD file follows.

```

File: Tc-99m.RAD for Tc-99m
Tc-99m      6.015h    143
Tl/2 = 6.015h Decay Mode: ITB-
Radiations of each type listed in increasing energy
Number of photon radiations: 86
Number of beta radiations: 3
Number of monoenergetic electron radiations: 54
ICODE Y (/nt) E(MeV) Mnemonic
START RADIATION RECORDS
2 5.49690E+00 7.87678E-06 X
  // // // // // // // // // //
  ::::::::::::::::::::::::::::
  // // // // // // // // // //
  1 6.65489E-11 2.17260E-03 G
  2 1.69925E-06 2.23236E-03 X
  // // // // // // // // // //
  ::::::::::::::::::::::::::::::
  // // // // // // // // // //
  1 9.69400E-07 3.22400E-01 G
  5 1.07767E-06 3.01202E-02 B-
  5 2.59439E-05 1.02026E-01 B-
  // // // // // // // // // //
  ::::::::::::::::::::::::::::::
  // // // // // // // // // //
  6 5.99387E-11 3.22400E-01 IE
END RADIATION RECORDS

```

As seen in above listing, the file begins with a few brief comments and then lists, for each radiation type, the number of records in the file. The decay data are bracketed between a start and end delimiter (“START RADIATION RECORDS” and “END RADIATION RECORDS”, respectively). The radiation records are in the format of the RAD file. See [Table 2](#) for the definition of ICODE and the mnemonic.

The records of the Tc-99m.BET file have the same structure as the record in the BET file but are bracketed by start and end delimiters. A few lines from the Tc-99m.bet file are listed below.

```

File: Tc-99m.BET for Tc-99m
Tc-99m      100
Beta Spectrum for Tc-99m
Spectrum is normalized to 1 nt (Bq s)
To normalize to 1 beta, divide by 3.7000E-05
Number of energy points: 100
E(MeV)      P(E)
START RADIATION RECORDS
0.00000 2.319E-04
0.00010 2.318E-04

```

```

::      ::
0.36000 7.143E-06
0.40000 1.990E-06
0.43618 0.000E+00
END RADIATION RECORDS

```

For selected radionuclides, Tc-99m being one, detailed spectra of the Auger-CK electron emissions are contained in the ACK file. While this level of detail is not necessary in calculations of mean absorbed dose, it is of interest in microdosimetric calculations, e.g., the dose to the cell nucleus from such an emitter incorporated in the cell nucleus. Below is an excerpt from the Tc-99m. ACK file.

```

File: Tc-99m.ACK for Tc-99m
Tc-99m          968
Auger/Coster-Kronig Spectrum for Tc-99m
Number of electrons: 968
Y(/nt)      E(eV)      transition
START RADIATION RECORDS
6.57103E-05 3.34000E+00 M1 M4 M5
1.66012E-04 7.28000E+00 M1 M5 M5
7.71672E-08 9.83000E+00 M1 M2 N1
3.19642E-03 1.22200E+01 M2 M3 N4
6.44567E-04 1.23300E+01 M1 M2 N1
7.26819E-03 1.25100E+01 M2 M3 N5
::      ::      ::      ::      ::
1.90107E-11 2.20272E+04 K N3 N4
1.90107E-11 2.20275E+04 K N3 N5
1.26734E-11 2.20300E+04 K N3 01
END RADIATION RECORDS

```

In the case of Tc-99m, the emissions of 968 discrete electrons are tabulated. The unit of the electron energy in this file is eV. In this case the electron energy ranges over four decades.

If decay data for a radionuclide which decays by spontaneous fission, e.g., Cf-252, is requested, a file of the neutron spectra associated with the spontaneous fission process is created. Below is an excerpt from the Cf-252.NSF file.

```

Output File Cf-252.NSF for Cf-252
Cf-252      3.092E-02      52
Neutron Spectra for Cf-252
Number of neutrons per fission - 3.767
Spectrum is normalized to 1 nt (Bq-s)
To normalize to 1 fission, divide by 3.09E-02
To normalize to 1 neutron, divide by 1.1641E-01
Number of energy bins: 52
E1 (MeV) E2 (MeV) Yield (/nt)
START RADIATION RECORDS

```



```

4.14E-07 1.00E-06 1.85105E-11
1.00E-06 1.00E-05 1.15581E-09
1.00E-05 5.00E-05 1.27176E-08
5.00E-05 1.00E-04 2.56849E-08
  ::  ::  ::  ::  ::  ::
1.20E+01 1.30E+01 3.82221E-05
1.30E+01 1.40E+01 1.82774E-05
1.40E+01 1.50E+01 8.66434E-06
END RADIATION RECORDS

```

F2 key: Show the decay chain

The decay of some radionuclides results in formation of radioactive nuclei (daughters) and hence a serial decay chain. The F2 key displays the decay chain and is illustrated below for the case of the parent radionuclide Es-254m, the longest chain in the ICRP-07 collection. After selecting Es-254m, its summary display is shown and then pressing F2 results in the display of Fig. 4.a. The values under the columns headed by *f1*, *f2*, *f3*, and *f4* denote the fractions of the nuclear transformations of the nuclide in the second column which form the daughter nuclei shown immediately to the right of those fractions. Spontaneous fission, which is a decay mode of several chain members, is denoted as SF in the table. Note that stable nuclei are shown in yellow.

The decay chain information is presented in two screens when the length of the chain is greater than 7 members, as in this case. Upon continuing the display, the additional information provides an evaluation of the potential dosimetric impor-

Nuclide	Half-life	f1	Nuclide	f2	Daughter Nuclide	f3	Nuclide	f4	Nuclide
1 Es-254m	39.3h	9.800E-01	Fm-254	7.600E-04	Cf-254	3.200E-03	Bk-250	4.500E-04	SF
2 Fm-254	3.240h	9.994E-01	Cf-250	5.920E-04	SF				
3 Cf-254	60.5d	3.100E-03	Cm-250	9.969E-01	SF				
4 Cm-250	8300y	8.000E-02	Bk-250	1.800E-01	Pu-246	7.400E-01	SF		
5 Bk-250	3.212h	1.000E+00	Cf-250						
6 Cf-250	13.08y	9.992E-01	Cm-246	7.700E-04	SF				
7 Pu-246	10.84d	1.000E+00	Am-246m						
8 Am-246m	25.0m	1.000E+00	Cm-246						
9 Cm-246	4.76E+3y	9.997E-01	Pu-242	2.630E-04	SF				
10 Pu-242	3.75E+5y	1.000E+00	U-238	5.540E-06	SF				
11 U-238	4.468E+9y	1.000E+00	Th-234	5.450E-07	SF				
12 Th-234	24.10d	1.000E+00	Pa-234m						
13 Pa-234m	1.17m	1.600E-03	Pa-234	9.984E-01	U-234				
14 Pa-234	6.70h	1.000E+00	U-234						
15 U-234	2.455E+5y	1.000E+00	Th-230						
16 Th-230	7.538E+4y	1.000E+00	Ra-226						
17 Ra-226	1600y	1.000E+00	Rn-222						
18 Rn-222	3.8235d	1.000E+00	Po-218						
19 Po-218	3.10m	9.993E-01	Pb-214	2.000E-04	At-218				
20 Pb-214	26.8m	1.000E+00	Bi-214						
21 At-218	1.5s	9.990E-01	Bi-214	1.000E-03	Rn-218				
22 Bi-214	19.9m	9.998E-01	Po-214	2.100E-04	Tl-210				
23 Rn-218	3.5E-2s	1.000E+00	Po-214						
24 Po-214	1.643E-4s	1.000E+00	Pb-210						
25 Tl-210	1.30m	1.000E+00	Pb-210						
26 Pb-210	22.20y	1.000E+00	Bi-210	1.900E-08	Hg-206				
27 Bi-210	5.013d	1.000E+00	Po-210	1.320E-06	Tl-206				
28 Hg-206	8.15m	1.000E+00	Tl-206						
29 Po-210	138.376d	1.000E+00	Pb-206						
30 Tl-206	4.200m	1.000E+00	Pb-206						

Stable isotope(s) shown in yellow.
Press a key (or left click mouse) to continue...

Fig. 4.a. Display of the Es-254m decay chain.

DECDATA - A Summary of Radiation Emissions

Es-254m: Activity, Transformations, and Cumulative Energies (MeV) at 100y

Nuclide	T1/2	$\lambda(t)/\lambda_0$	nt/ $\lambda_0(s)$	Alpha	Electron	Photon
1 Es-254m	39.3h	0.000E+00	2.041E+05	4.246E+03	4.914E+04	9.709E+04
2 Fr-254	3.240h	0.000E+00	2.000E+05	1.464E+06	5.104E+04	9.881E+04
3 Cf-254	60.5d	0.000E+00	1.551E+02	1.464E+06	5.260E+04	1.014E+05
4 Cm-250	8300y	1.262E-12	3.990E-03	1.464E+06	5.260E+04	1.014E+05
5 Bk-250	3.212h	1.010E-13	6.532E+02	1.464E+06	5.279E+04	1.020E+05
6 Cf-250	13.08y	1.683E-06	1.996E+05	2.684E+06	5.484E+04	1.042E+05
7 Pu-246	10.84d	2.272E-13	7.178E-04	2.684E+06	5.484E+04	1.042E+05
8 Am-246m	25.0m	2.272E-13	7.178E-04	2.684E+06	5.484E+04	1.042E+05
9 Cm-246	4.76E+3y	9.093E-07	2.356E+03	2.697E+06	5.486E+04	1.042E+05
10 Pu-242	3.75E+5y	1.379E-10	1.862E-01	2.697E+06	5.486E+04	1.042E+05
11 U-238	4.468E+9y	9.152E-19	8.441E-10	2.697E+06	5.486E+04	1.042E+05
12 Th-234	24.10d	9.132E-19	8.414E-10	2.697E+06	5.486E+04	1.042E+05
13 Pa-234m	1.17m	9.132E-19	8.414E-10	2.697E+06	5.486E+04	1.042E+05
14 Pa-234	6.70h	1.461E-21	1.346E-12	2.697E+06	5.486E+04	1.042E+05
15 U-234	2.455E+5y	7.527E-23	5.693E-14	2.697E+06	5.486E+04	1.042E+05
16 Th-230	7.538E+4y	1.540E-23	4.018E-15	2.697E+06	5.486E+04	1.042E+05
17 Ra-226	1600y	1.197E-28	4.010E-15	2.697E+06	5.486E+04	1.042E+05
18 Rn-222	3.8235d	1.189E-28	4.010E-15	2.697E+06	5.486E+04	1.042E+05
19 Po-218	3.10m	1.193E-28	4.010E-15	2.697E+06	5.486E+04	1.042E+05
20 Pb-214	26.8m	1.189E-28	4.009E-15	2.697E+06	5.486E+04	1.042E+05
21 At-218	1.5s	2.381E-32	8.019E-19	2.697E+06	5.486E+04	1.042E+05
22 Bi-214	19.7m	1.193E-28	4.010E-15	2.697E+06	5.486E+04	1.042E+05
23 Rn-218	3.5E-2s	2.386E-35	8.019E-22	2.697E+06	5.486E+04	1.042E+05
24 Po-214	1.643E-4s	1.199E-28	4.009E-15	2.697E+06	5.486E+04	1.042E+05
25 Tl-210	1.30m	2.512E-32	8.420E-19	2.697E+06	5.486E+04	1.042E+05
26 Pb-210	22.20y	3.965E-29	4.010E-15	2.697E+06	5.486E+04	1.042E+05
27 Bi-210	5.013d	4.005E-29	4.010E-15	2.697E+06	5.486E+04	1.042E+05
28 Hg-206	8.15m	7.504E-37	7.618E-23	2.697E+06	5.486E+04	1.042E+05
29 Po-210	138.376d	3.846E-29	4.010E-15	2.697E+06	5.486E+04	1.042E+05
30 Tl-206	4.200m	5.368E-35	5.369E-21	2.697E+06	5.486E+04	1.042E+05

In dosimetric calculations the Es-254m chain can be truncated at member 6.
Press a key (or left click mouse) to continue...

Fig. 4.b. Summary of emitted energies of the Es-254m chain.

tance of the chain members. Fig. 4.b. shows the additional information for the Es-254m chain.

Briefly, DECDATA constructs and displays a table of activities, nuclear transformations, and cumulative energies of emitted alpha, electron, and photon emissions of the chain members over a period of 100 y, assuming a unit activity of the pure parent (e.g., 1 Bq of Es-254m) at time zero. The cumulative energy of alpha, electron (including beta) and photon emissions for the decay series is shown in the right three columns. The length of the chain that accounts for 99% of the total emitted energy over the 100 y time period is calculated for each energy type. For Es-254m the suggestion is made that for dosimetric purposes the chain could be truncated after the sixth member (Cm-250) because the contribution of the rest of the members to the cumulative energy of each radiation type is not significant. Some loss in numerical significance may be evident in these calculations for some decay chains and the guidance concerning truncation is somewhat arbitrary. However, this guidance has proven to be useful when considering the truncation of chains headed by or including long-lived radionuclides. In any event, the user must make the decision regarding truncation of the chain in his application.

F3 key: Plots of radiation emissions

This function key generates, for the displayed radionuclide, a series of plots of the yield vs. energy of the emitted radiations. In the case of beta emission (negatron and positron) and neutrons the plots are continuous function of energy. For radiations of discrete energy a line plots are generated. Fig. 5 show the generated plots displayed in a cascade manner of the emissions of Cf-252. To expand the plot of interest, click

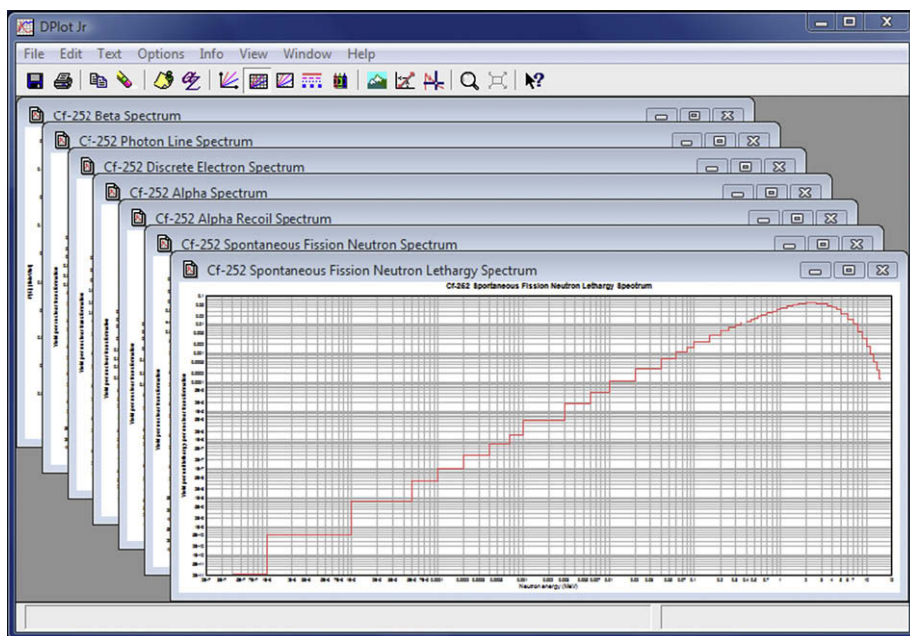


Fig. 5. Cascade plots of Cf-252 emissions.

the “maximize” icon in the upper right corner of the plot. The plots of the other radiations can then be selected for viewing by clicking on the “Window” item on DPlotJr’s tool bar. You can force DECDATA to display the plots in the maximized manner by creating a file named “DPlotJr. INI” in the ICRP07 folder. The type of axis and their numerical extent can be modified by selecting the options item on DPlot Jr menu. Additional information is available under the help menu.

The neutron spectral data in the NSF file are binned and displayed in two plots evident in Fig. 5, the first being simply the binned spectra (y-axis is the number of neutron per nt with energy within the bin) and the second a lethargy plot (y-axis is the number of neutrons in the bin per nt in lethargy units, i.e., number per nt/ $\ln(E_{i+1}/E_i)$ where E_{i+1} and E_i are the upper and lower energy bounds of the i^{th} energy bin).

Upon its very first use, DPlotJr may start with its plot window maximized. After resizing the window in the usual Windows manner (dragging the edge and centering) click on Options on the toolbar, select that last item (General) and remove the check mark on “Always start maximized.” DplotJr will then start as sized in future applications.

If the ICRP-07.ACK file contains the Auger-CK electron spectrum for the selected radionuclide then the spectrum is also plotted in the cascade. The Auger-CK spectrum of Hg-195m is shown in Fig. 6.

F4 key: Nuclide lists

The function key F4 displays a menu providing a number of options for sorting and listing the radionuclides in the ICRP-07 collection. In addition, the last two

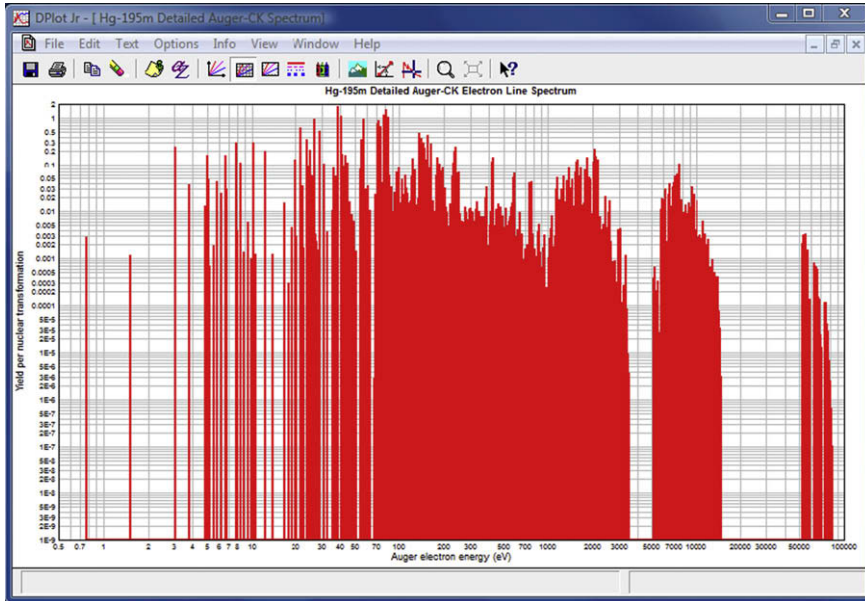


Fig. 6. Detailed Auger and Coster-Kronig electron emissions of Hg-195m.

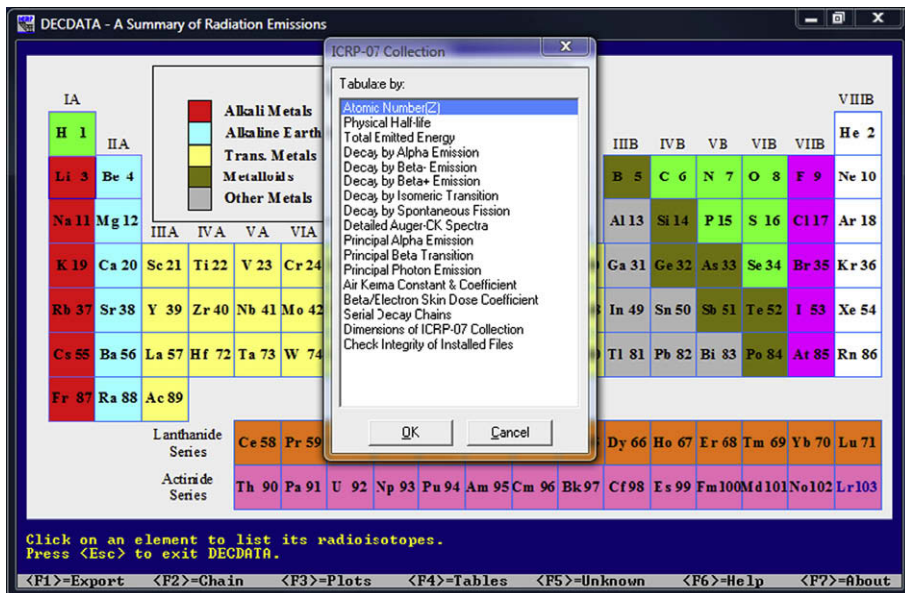


Fig. 7. Menu to characterize the ICRP-07 collection.

menu items characterize the data files. The actions under this menu are displayed and written to the file NucList-X.TXT, where X corresponds to the index of the user-

specified criteria selected from a menu. After the file(s) are constructed DECDATA will open the file using an application registered to open files with the extension TXT. The user is cautioned to be patient while DECDATA constructs the desired information and Windows loads an application to view the file. The menu items are presented below and shown in Fig. 7.

Menu items for sorting and characterizing the radionuclides in ICRP-07

Menu Item	Description
1. Atomic Number(Z)	List nuclides in order of atomic number
2. Physical Half-life	List nuclides by increasing physical half-life
3. Total Emitted Energy	List nuclides by increasing emitted energy
4. Decay by Alpha Emission	List nuclides that undergo alpha decay
5. Decay by Beta- Emission	List nuclides that undergo β^- (negatron) decay
6. Decay by Beta+ Emission	List nuclides that undergo β^+ (positron) decay
7. Decay by Internal Transition	List nuclides that undergo internal transition decay
8. Decay by Spontaneous Fission	List nuclides that undergo spontaneous fission
9. Detailed Auger-CK Spectra	List nuclides in the ACK file
10. Principal Alpha Emission	List nuclides by increasing energy of principal alpha emission
11. Principal Beta Transition	List nuclides by increasing energy of principal beta emission
12. Principal Photon Emission	List nuclides by increasing energy of principal photon emission
13. Air Kerma Constant & Coefficient	Tabulate air kerma-rate constant and coefficient
14. Beta/Electron Skin Dose Coefficient	Compute beta/electron skin dose coefficient
15. Serial Decay Chains	Tabulate serial decay chains
16. Dimensions of ICRP-07 Collection	List dimensions of collection
17. Check Integrity of Installed Files	Verify checksum of ICRP-07 data files.

As the files created by menu items 1 and 9 can be of frequent interest they are included in the REPORT folder as ICRP_Nuclide.TXT and AugerList.TXT, respectively. They can be accessed under the help provisions of the F6 key.

Menu item 14 invokes a calculation of absorbed dose rate to skin ($\text{Gy s}^{-1} \text{cm}^2 \text{Bq}^{-1}$) resulting from contamination of the skin by radionuclides that undergo beta decay or emit energetic conversion electrons. The absorbed dose calculations is based on the paper by Faw (1992) and is calculated as

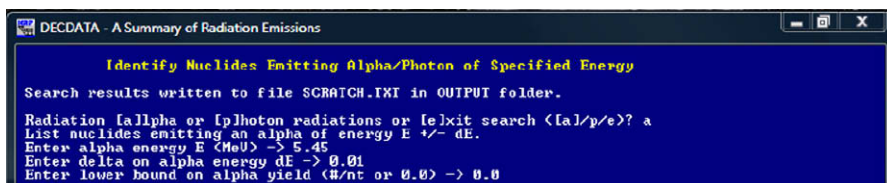
$$D_{\text{skin}} = 1.60210^{-10} \int_0^{\infty} P(E)R(E)dE + \sum_i Y(E_i)R(E_i) \quad (3)$$

where $P(E)$ is the number of electrons of energy between E and $E + dE$ emitted per nt as tabulated in the BET file, $Y(E_i)$ is the number of monoenergetic electrons of energy E_i emitted per nt, and $R(E)$ ($\text{MeV cm}^2 \text{g}^{-1}$) is the absorbed dose to the skin (averaged over a depth of 5 to 10 mg cm^{-2}) as a function of electron energy expressed in functional form by the parameters of Table 3 of Faw (1992). The integral over the beta spectrum is evaluated by fitting a piecewise cubic hermite interpolation polynomial to the integrand using the PCHIP routines of Fritsch and Carlson (1980).

F5 key: Identify unknown radionuclide

This function key provides a means to associate an observed alpha particle or photon with a radionuclide in the ICRP-07 collection. For example, assume the emission

of an alpha particle of energy 5.45 MeV has been observed. Pressing the F5 key opens a screen within which the user first defines whether the observed radiation is an alpha particle or a photon and then enters the observed energy. In addition to the point-value for the energy, a tolerance (or delta) in energy value and a lower bound on its yield (range from zero to 1.0) is requested as shown below.



```

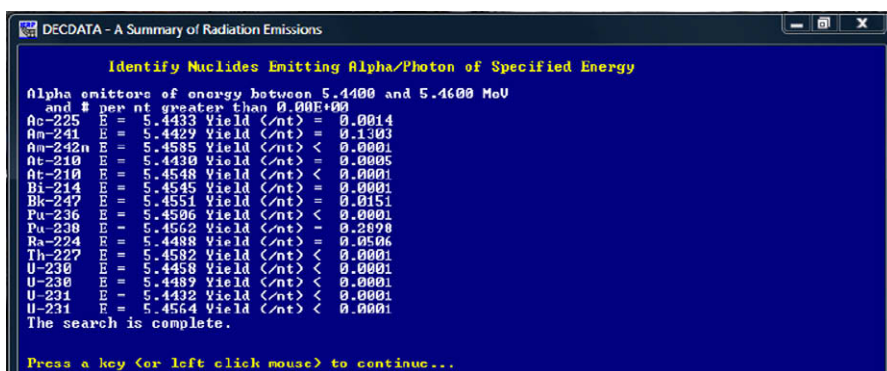
DECDATA - A Summary of Radiation Emissions

Identify Nuclides Emitting Alpha/Photon of Specified Energy

Search results written to file SCRATCH.TXT in OUTPUT folder.

Radiation [alpha or lp] photon radiations or [exit] search [a/p/e]? a
List nuclides emitting an alpha of energy E +/- dE.
Enter alpha energy E (MeV) -> 5.45
Enter delta on alpha energy dE -> 0.01
Enter lower bound on alpha yield (0/nt or 0.0) -> 0.0
  
```

A search of the RAD file is performed, as guided by information in the NDX file. The results appear on the screen and are written to the file SCRATCH.TXT in the OUTPUT folder. In this case the screen display is



```

DECDATA - A Summary of Radiation Emissions

Identify Nuclides Emitting Alpha/Photon of Specified Energy

Alpha emitters of energy between 5.4400 and 5.4600 MeV
and # per nt greater than 0.00E+000
Ac-225 = 5.4433 Yield (nt) = 0.0014
Am-241 = 5.4429 Yield (nt) = 0.1303
Am-242m = 5.4585 Yield (nt) < 0.0001
At-210 = 5.4430 Yield (nt) = 0.0005
At-210 = 5.4548 Yield (nt) < 0.0001
Bi-214 = 5.4545 Yield (nt) = 0.0001
Bk-247 = 5.4551 Yield (nt) = 0.0151
Pu-236 = 5.4506 Yield (nt) < 0.0001
Pu-238 = 5.4562 Yield (nt) = 0.2878
Ra-224 = 5.4488 Yield (nt) = 0.0506
Th-227 = 5.4582 Yield (nt) < 0.0001
U-230 = 5.4458 Yield (nt) < 0.0001
U-230 = 5.4489 Yield (nt) < 0.0001
U-231 = 5.4432 Yield (nt) < 0.0001
U-231 = 5.4564 Yield (nt) < 0.0001
The search is complete.

Press a key (or left click mouse) to continue...
  
```

The search can lead to a substantial number of candidates, particularly for photons, and thus the file SCRATCH.TXT will need to be reviewed. The information on the yield of the radiations within the indicated energy range may be of help in eliminating some candidates. DECDATA will attempt to open the file using the application that has been associated with files with extension TXT, e.g., Microsoft Notepad or an ASCII editor.

F6 key: View files in the REPORT folder

The REPORT folder contains this User Guide, reports produced by the authors (Endo *et al.* 2005, Endo, 2005, Endo and Eckerman 2007) during the course of the work creating the decay database, the original EDISTR report of Dillman (1980), and two files created via the F4 key as discussed above. The user can develop and maintain notes on the operation of DECDATA by creating in the REPORT folder a file with an extension registered by the operating system, e.g., using Microsoft NotePad. The file MyNotes.TXT is provided as a template.

F7 key: About

Provides the usual *About* screen associated with current software. The software license, agree to during the installation of DECADATA, can be viewed at this time by pressing the F8 key.

5. Programmer's Notes

The ICRP-07 data files are formatted direct-access files. Each record in these files is of fixed length and includes a carriage return and line feed so that the files are readable when opened with an ASCII editor. The length of the records, less the carriage return and line feed, for each of the ICRP-07 files is listed below.

Length of Records in the Direct-Access Files		
File	Record Length*	Number of records
ICRP-07.NDX	226	1,253
ICRP-07.RAD	29	455,625
ICRP-07.BET	17	111,325
ICRP-07.ACK	32	131,571
ICRP-07.NSF	29	1,484

* Value does not include the carriage return and line feed.

The number of data records N in the RAD file for a given nuclide is given in the header record for that nuclide in the file but it can also be computed from the nuclide's record in the NDX file as

$$N = \begin{cases} \sum_{i=1}^5 \text{Number} - i + \text{Number} - 5, & SF \notin \text{DecayMode} \\ \sum_{i=1}^5 \text{Number} - i + \text{Number} - 5 + 2, & SF \in \text{DecayMode} \end{cases}$$

where Number- i refers to the fields of the nuclide record so designated in Table 1. Whether a nuclide decays by spontaneous fission can be determined by parsing its decay mode for the characters "SF" or by noting if the pointer into the NSF file (Pointer-4 of Table 1) is nonzero. The number of alpha transitions, Number-5, enters into the above equation twice since records are included in the RAD file for the kinetic energy of the recoiling nuclei formed in alpha decay as well as the alpha particles themselves. The two additional records for those nuclides undergoing spontaneous fission are those of the mean energies of the fission fragments and neutrons accompanying spontaneous fission.

Records in the RAD file for a particular radiation type emitted by a specific radio-nuclide can be read as tabulated below. In the table below the fields Pointer- i and Number- i defined in Table 1 of the nuclide's record in the NDX file are denoted by P_i and N_i , respectively. The data records for the radiation type in the RAD file lie between the lower and upper record numbers specified in the last two columns below.

Radiation Type	Record Number of Radiations in RAD File		
	Number	Location of Data Records for Radiation Type	
	Records	From	To
Photons, $E < 10$ keV	N_1	$P_1 + 1$	$P_1 + N_1$
Photons, $E \geq 10$ keV	N_2	$P_1 + N_1 + 1$	$P_1 + N_1 + N_2$
Photons, all	$N_1 + N_2$	$P_1 + 1$	$P_1 + N_1 + N_2$
Beta particles	N_3	$P_1 + N_1 + N_2 + 1$	$P_1 + N_1 + N_2 + N_3$
Monoenergetic electrons	N_4	$P_1 + N_1 + N_2 + N_3 + 1$	$P_1 + N_1 + N_2 + N_3 + N_4$
Alpha particles	N_5	$P_1 + N_1 + N_2 + N_3 + N_4 + 1$	$P_1 + N_1 + N_2 + N_3 + N_4 + N_5$
Alpha recoil	N_5	$P_1 + N_1 + N_2 + N_3 + N_4 + N_5 + 1$	$P_1 + N_1 + N_2 + N_3 + N_4 + 2N_5$
Fission fragments ^a	I	$P_1 + N_1 + N_2 + N_3 + N_4 + 2N_5 + 1$	$P_1 + N_1 + N_2 + N_3 + N_4 + 2N_5$ + 1
Neutrons ^a	I	$P_1 + N_1 + N_2 + N_3 + N_4 + 2N_5 + 1$	$P_1 + N_1 + N_2 + N_3 + N_4 + 2N_5$ + 2

^a Records present only if nuclide undergo spontaneous fission, i.e., $P_4 > 0$.

The dimensions of the ICRP-07 collection can be tabulated by selecting the penultimate item on the menu displayed via the F4 key. The listing created gives the maximum number of radiations of the various types in the RAD file and the number of discrete energy points in the continuous beta and neutron spectra. For example, the maximum number of records in the RAD file for radiations emitted by a single radionuclide is 4121 in the case of Lu-170. The resultant list is shown below.

Dimensions of the ICRP-07 Collection

Radionuclides	-	1252
Radiation records	-	4121 (Lu-170)
Beta transitions	-	79 (Gd-145)
Alpha transitions	-	52 (Pu-239)
Photons: $E < 10$ keV	-	83 (Sm-141m)
Photons: $E > 10$ keV	-	664 (Lu-170)
Total	-	699 (Lu-170)
Discrete electrons	-	3409 (Lu-170)

Energy Range: Discrete Radiations

Betas*	-	6.2E-04 to 4.9806 MeV (Re-187, N-16)
Photons	-	3.1E-07 to 9.9100 MeV (Na-24, Cf-246)
Electrons	-	8.8E-06 to 8.8693 MeV (In-114m, N-16)
Alphas	-	1.85224 to 11.6512 MeV (Nd-144, Po-212m)
Alpha recoils	-	0.05148 to 0.2242 MeV (Pb-202, Po-212m)
Fission fragments\$	-	84.84610 to 97.7707 MeV (U-238, Fm-252)
Fission neutrons\$	-	1.68813 to 2.3823 MeV (U-238, Cm-240)

*Average energy of beta transitions

\$Average energy in spontaneous fission

Spectral Dimensions

Beta spectrum	-	138 (N-16)
Auger-CK spectrum	-	3015 (Hg-195m)
Neutron spectrum	-	52 (Cf-246)

Energy Range: Spectra

Beta	-	0.00 to 10.4207 MeV (N-16)
Neutron (bins)	-	0.41 eV to 15.0000 MeV (U-238, Cm-240)
Auger-CK electrons	-	0.15 eV to 88.1763 keV (Tb-157, Bi-204)

Decay Chains

Chain length	-	30 (Es-254m)
Elements Number	-	97
Radioisotopes	-	29 (Indium)

Data File		Record length	# records
ICRP-O7.NDX	-	226	1253
ICRP-O7.RAD	-	29	455625
ICRP-O7.BET	-	17	111325
ICRP-O7.ACK	-	32	131571
ICRP-O7.NSF	-	29	1484

The DECDATA source code is included in the folder SOURCE of the ARCHIVE folder on the CD. The functions and subroutines of the code can be readily translated to other programming languages.

SUPPLEMENTARY DATA

Supplementary data associated with this article can be found, in the online version, at [doi:10.1016/j.icrp.2008.10.001](https://doi.org/10.1016/j.icrp.2008.10.001).

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